Who Reaches for Yield?*

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Abstract

Using Norwegian administrative data, we show that households at every level of liquid wealth reduce their risky share after a contractionary monetary policy shock, consistent with reaching for yield. The response is convex in wealth and flattens at the top. Portfolio flow inherits this gradient but moves differently: over the tightening phase, the wealthiest households buy equities while less affluent households sell. A model-free accounting identity that decomposes net purchases of risky assets shows why. On impact, capital losses offset the rebalancing withdrawal of the wealthiest and hold their flow near zero while households below them sell. By the trough of the contraction, net saving at the top turns the wealthiest into net buyers. We develop a theory of reaching for yield based on certainty-equivalent portfolio adjustment, in which monetary policy changes households' marginal utility but not their sensitivity to risk. Higher interest income raises effective risk aversion and lowers the risky share. The model generates the convex wealth gradient with countercyclical idiosyncratic disaster risk because the marginal utility of the wealthy is less sensitive to resource changes. Together, these forces explain why monetary tightening reduces risky exposure throughout the wealth distribution while reallocating equity ownership toward the top.

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1 Introduction

It has been a recurring concern for central bankers that loose monetary policy encourages risk-taking in financial markets. The phenomenon, referred to as *reaching for yield*, has been documented and studied as an important channel of monetary policy transmission to the macroeconomy (Bauer et al., 2023). At the individual level, the choice to take on or reduce risk reflects how much a household is exposed to aggregate risk and predicts its consumption volatility. In the aggregate, equity transactions move asset prices when the demand curves are steep (Gabaix and Koijen, 2021), and they contribute to wealth inequality through the redistribution of risky asset holdings. Because asset prices and risk appetite drive aggregate demand (Caballero and Simsek, 2020), it matters all the more to the real economy when marginal propensities to consume differ across the wealth distribution. Understanding household portfolio adjustment thus matters for both the distributional and aggregate effects of monetary policy.

Yet the risk-taking debate has centered on banks and institutions. On the household side, while the income and consumption effects of monetary policy are by now well studied (Kaplan et al., 2018; Holm et al., 2021), far less is known about how household risky asset allocation responds, and in particular which households across the wealth distribution take on risk and which pull back from it as interest rates move. There are only a handful of studies examining reaching for yield among households, in a lab experiment that ignores cross-sectional heterogeneity (Lian et al., 2019), with self-reported surveys that embed measurement errors (Bonaparte et al., 2024), and using brokerage data that covers selected portfolios (Gomes et al., 2025), leaving our understanding limited and the concerns far from resolved.

This paper studies how household risk-taking behaviors respond to monetary policy across the liquid wealth distribution and what economic theory explains them. Specifically, we focus on two measures. The first is the *portfolio risky share*, the fraction of liquid wealth invested in risky assets, which reflects the household's risk appetite. The second is the *portfolio net flow*, the net purchases of risky assets, which captures the household's actual trading behavior. We make three contributions, one methodological, one empirical, and one theoretical.

We start with the methodology, which connects the portfolio risky share to the net flow. A household's exposure to risky assets and its demand for risky assets need not move in the same direction. After a monetary tightening, falling equity prices mechanically reduce the risky share of household portfolios. A household may therefore purchase equities even while reducing its overall exposure to risk. This distinction turns out to matter systematically across the wealth distribution: while households at all wealth levels reduce their risky share after a tightening, the wealthiest households become net buyers of equities as less wealthy households sell. To address this, we propose a model-free

accounting identity that decomposes a household's net purchases of risky assets into three channels: First, portfolio rebalancing as the deliberate shift between safe and risky assets, which ties closely to the portfolio risky share. The second channel corresponds to cash flow, the increment of wealth that an investor saves out of his income. An individual would actively buy risky assets to balance his portfolio when there is more money in the current account. Third, asset valuation is the flow induced by movements in equity prices: investors who earn capital gains on risky wealth would withdraw from risky investment to avoid over-exposure, if their risk appetites do not change. Because it follows as an exact identity from the household's balance sheet, the decomposition holds regardless of preferences, information, or market structure. The responses of each channel to monetary policy are separately estimable, and the three together suggest the driving forces of changes in the net flow.

We then provide empirical evidence using Norwegian administrative data that covers the entire resident population from 1993 to 2019. Norway's tax and wealth registry data offer several advantages for studying household portfolio responses: universal coverage without attrition or selection; detailed balance sheet and income information at the individual level, which together let us construct all three channels; and the absence of top-coding, which is particularly important for the upper tail of the wealth distribution where equity holdings concentrate. We link them to the Norwegian Shareholder Registry, which reports individual stock-level holdings from 2005 onward, and to the identified series of monetary policy shocks from Holm et al. (2021). We estimate local projections across liquid wealth groups and trace the impulse response of each variable to a contractionary monetary policy shock.

To the best of our knowledge, we are the first to document how household risky share reacts to monetary policy across the wealth distribution. We find that, after a monetary tightening, households on average reduce the share of liquid wealth held in risky assets at every wealth level, the universal reaching for yield regardless of changes in portfolio size. On impact, the decline becomes steeper through the middle of the distribution, then flattens at the top, showing a convex gradient. The convexity remains at longer horizons until the trough of monetary contraction, after which the risky share recovers and the gradient becomes U-shaped. Our findings suggest that reaching for yield is a persistent, expectations-driven behavior that spans the whole distribution but weakens toward the top.

The flow response, estimated directly, emerges clearly at longer horizons: as the contraction deepens, a positive wealth gradient builds, with middle-wealth households significant net sellers of risky assets and the wealthiest net buyers. On impact, however, the direct estimate is insignificant across the distribution, because the flow is the small difference between two large and noisy quantities.

To see what drives the flow and its reversal, we decompose the direct estimates through

the accounting identity at every horizon. The wealth gradient of the flow is inherited from the risky share: through the tightening phase, the flow tracks the rebalancing channel closely, so the convexity of the risky-share response is what makes the flow flatten and then reverse toward the top. The reversal itself has two sources at two horizons. On impact, capital losses scale with equity exposure, so the valuation channel offsets the rebalancing outflow of the wealthiest and holds their flow near zero while households below them, whose rebalancing outflow is stronger, turn to selling. By the trough of the contraction, net saving at the top builds up, consistent with the higher interest income the wealthy earn on their deposits, and turns the wealthiest into net buyers. The three channels also run on different clocks: valuation moves with equity prices and is strongest when the policy rate moves, rebalancing follows the risky share and unwinds with the contraction, and cash flow builds slowly and dominates the lower half of the distribution only in the longer run, when households across the distribution turn to buying. The reversal is consistent with the high-frequency brokerage evidence of Gomes et al. (2025); the administrative data see it build over two years rather than within weeks, since the annual frequency misses within-year rebalancing and capital gains must be imputed rather than read from transactions, adding measurement error that widens confidence bands at some horizons.

Third, we develop a portfolio choice theory that gives reaching for yield a foundation, and shows how the convexity of its wealth gradient, passed through the accounting identity, keeps the wealthiest from selling as the rest of the distribution does and turns them into net buyers. In a standard incomplete-markets economy with CRRA preferences, we assume certainty-equivalent portfolio adjustment: monetary policy moves households' expected marginal utility but not their sensitivity to aggregate risk, which matters for their welfare only at second order. In other words, households adjust their portfolios using only the first-order responses that characterize their consumption and saving. Under this assumption, a higher interest rate raises interest income and makes the household safely richer in the next period, lowering its expected marginal utility. As expected marginal utility is negatively correlated with the household's effective risk aversion, higher interest income makes it more effectively risk averse. The household therefore optimally reduces its risky share to avoid over-exposure to the aggregate risk, which is reaching for yield. Formally, the optimal risky share is obtained by a second-order perturbation of the household's Euler equation with respect to aggregate risk, and its response by a first-order perturbation with respect to the monetary policy shock.

It has long been established that rational portfolio theory rules out reaching for yield: in the textbook Merton benchmark the risky share is pinned down only by the equity premium, so standard theory predicts no reaching for yield, and the behavior has been explained through reference dependence, salience, or prospect theory (Lian et al., 2019; Gomes et al., 2025). Our theory differs from these in its predictions and its scope. First,

it predicts a response that is symmetric in the shock and independent of the level of the interest rate, which is what we empirically find and where the reference-dependent and prospect-theoretic accounts predict otherwise. Second, because it rests on the same first-order perturbation solutions that heterogeneous-agent models already use, it makes the portfolio choice between safe and risky assets a channel of monetary transmission that can be embedded in those models naturally. Lastly, like these behavioral theories, ours explains reaching for yield by departing from full rationality. Rather than adding complexity to traditional models, our departure rests on a behavioral micro-foundation based on a simplified decision-making process: the bounded rationality of a household who allocates attention by stakes, of which the assumption is the corner solution of a sparse-max problem (Gabaix, 2014).

Beyond explaining reaching for yield, our model is the first to account for the convex gradient in the decline in risky share that flattens at the top after a monetary contraction. We show that the convexity arises from a precautionary motive against tail risk that standard portfolio theory leaves out. Alongside a myopic risk-return demand and a standard income-hedging demand, the optimal share carries a disaster-hedging demand against the countercyclical probability of rare idiosyncratic disasters, the tendency for large labor income losses to become more likely in recessions (Guvenen et al., 2014b; Halvorsen et al., 2024). Under our certainty-equivalent portfolio adjustment assumption, a larger disaster risk raised by a contractionary shock results in a higher expected marginal utility, which needs to be offset by agents' portfolio choice for optimality. However, offsetting it requires a portfolio adjustment whose size depends on the sensitivity of expected marginal utility to a marginal change in the expected wealth. For the wealthy who are too rich to be touched by getting richer, this sensitivity is low. So answering the same rise in tail risk takes a larger increase in expected wealth from the equity premium, thus a larger offsetting increase in the risky share. That larger increase dampens their reaching-for-yield response, and the profile flattens at the top. This flattening is what opens the door to the reversal: as the rebalancing outflow of the wealthy levels off while their capital losses keep deepening with equity exposure, the cash flow effect comes to dominate at the very top and turns the flow positive, the reversal in net flows we document.

Related Literature Our paper contributes to an empirical and a theoretical literature, which we take in turn.

Empirical literature. We contribute first to the empirical literature on *reaching for yield*. The behavior was first documented among institutional investors, who tilt toward higher-yielding securities within a risk class (Becker and Ivashina, 2015; Choi and Kronlund, 2018). Evidence for households is thinner and less conclusive. Lian et al. (2019) find it in experiments and surveys, Agarwal et al. (2020) in Indian bank records

around term-deposit expirations, and Daniel et al. (2021) in a US discount brokerage, where investors tilt toward high-dividend stocks and high-income funds as rates fall, while evidence from US household surveys is mixed (Alzuabi et al., 2020; Bonaparte et al., 2024).

A more recent literature turns to heterogeneity: Gomes et al. (2025) find in Chinese brokerage accounts that reaching for yield weakens with wealth, with the wealthiest reversing it in their net purchases of equities, and Gabaix et al. (2024) find that ultra-high-net-worth US households buy equities in downturns while others sell. These studies observe trades within a single institution, part of the portfolios of a selected set of investors, or self-reported holdings. Our registry data cover the complete liquid portfolios of the entire population, participants and non-participants alike, without top-coding, which lets us measure the risky share at every point of the liquid wealth distribution and document the reversal found in brokerage accounts in population data. Closest to us, Bernardino et al. (2025) use the same Norwegian registry to study how monetary tightening changes the value and composition of household wealth across net wealth quartiles. We differ in object and in resolution. Our object is the allocation and the trades behind it: the risky share, and the net flow into risky assets measured in value. We rank households by liquid wealth, the base of the safe-risky allocation, at percentile resolution in the upper tail, where the reversal we document is concentrated. And where they separate price from quantity, our identity decomposes the flow into rebalancing, net saving, and valuation, linking the risky-share response to the trades it implies.

More broadly, we contribute to the literature using microdata to trace monetary transmission to households, which has focused on consumption, income, and debt (Di Maggio et al., 2017; Coibion et al., 2017; Beraja et al., 2019; Cloyne et al., 2020; Flodén et al., 2021; Amberg et al., 2022; Andersen et al., 2023). We build on Holm et al. (2021), whose Norwegian shock series and framework we use. Our results add a distributional effect this literature has not isolated: because the wealthiest buy risky assets after a tightening while others sell, monetary policy redistributes risk itself, moving equity ownership up the wealth distribution and reshaping who gains when prices recover.

Finally, we build on household finance research that uses registry data to study risk-taking: participation and portfolio choice over the life cycle (Fagereng et al., 2017), the determinants of risky shares (Calvet and Sodini, 2014), heterogeneous returns to wealth (Fagereng et al., 2020a), responses to the equity premium (Fagereng et al., 2023), and consumption out of capital gains (Di Maggio et al., 2020), alongside evidence of inertia and contrarian trading among individual investors (Kaniel et al., 2008; Bilias et al., 2010). Closest in method is Calvet et al. (2009), who split the change in the risky share into a passive component driven by returns and an active component driven by trades. Our identity decomposes the flow rather than the share, and separates a channel their split absorbs into the active component, the saving out of income that flows into risky assets.

Theoretical literature. On the theory side, we contribute first to explanations of reaching for yield. Standard portfolio theory with full rationality rules it out: in the Merton (1969) benchmark the risky share depends on the equity premium alone. Existing explanations therefore add a behavioral factor, reference dependence and salience (Lian et al., 2019) or prospect theory (Gomes et al., 2025), or a constraint on the objective, a sustainable-spending mandate (Campbell and Sigalov, 2022) or a rule that funds consumption from capital income such as dividends and coupons (Daniel et al., 2021). We obtain it instead for unconstrained households with CRRA preferences, from a simplification in how they evaluate shocks: households respond to the first-order effects of policy on marginal utility without revising their sensitivity to aggregate risk, the corner solution of a sparse-max attention problem (Gabaix, 2014). With countercyclical disaster risk, the same mechanism delivers the wealth gradient that none of these accounts addresses.

Our model builds on portfolio choice theory with non-tradable labor income, from Merton (1969) to its life-cycle extensions (Viceira, 2001; Cocco et al., 2005), and characterizes the optimal portfolio and its response to monetary policy by perturbation (Bhandari et al., 2023; Auclert et al., 2024b). Methodologically, we extend the first-order marginal-utility responses of Auclert et al. (2024b) from the determination of the steady-state portfolio to its adjustment after an aggregate shock.

The ingredient we add, counter-cyclical idiosyncratic disaster risk, comes from incomplete-markets asset pricing. Since Mankiw (1986) and Constantinides and Duffie (1996), that literature has asked whether state dependence in uninsurable idiosyncratic risk is large enough to matter (see Panageas, 2020, for a survey). Recent administrative earnings data show that business-cycle variation in income risk takes the form of left-skewness rather than higher variance (Guvenen et al., 2014b; Halvorsen et al., 2024). Models in which households face a time-varying probability of rare idiosyncratic disasters, by analogy with aggregate rare disasters (Rietz, 1988; Barro, 2006; Gabaix, 2012; Gourio, 2012; Wachter, 2013), match the level and dynamics of the equity premium (Constantinides and Ghosh, 2017; Schmidt, 2025). On the cross-section, this tail risk shapes portfolio choice: Catherine (2022) finds in a life-cycle model that it lowers the equity share, with an effect that attenuates at high wealth, where marginal utility is less sensitive to resources, and Catherine et al. (2024) find that exposure to it explains stock market shares across Swedish households (see also Meeuwis, 2022; Choi and Robertson, 2020). These papers study the level of risk premia or of the equity share. We further study how the risky share adjusts to a monetary policy shock, and show that the same attenuation at high wealth, applied to the response rather than the steady state level, produces the convex gradient and, through the identity, the reversal of flows.

Finally, our results bear on models of monetary transmission through heterogeneous household balance sheets. Kaplan et al. (2018) show that most transmission runs through

indirect general-equilibrium effects, and later work isolates particular margins: unhedged interest-rate exposure (Auclert, 2019), which reappears in our cash-flow channel; rebalancing between liquid and illiquid assets (Luetticke, 2021), the counterpart of our rebalancing term; and redistribution toward households more willing to bear risk (Kekre and Lenel, 2022; Bilbiie et al., 2022), which operates through the valuation margin we measure. Melcangi and Sterk (2025) show that stock market participation shapes transmission and that the resulting stock investment channel dominates the consumption channels usually emphasized. The margin they emphasize is extensive; ours is mixed, since we measure the allocation of participants and non-participants together and the participation gap enters our identity explicitly. We add to this literature channels that correspond to margins these models contain, estimated rather than assumed; cross-sectional responses that quantitative models with portfolio choice can be asked to match; and, because our theory uses the same first-order objects, a natural way to embed the safe-risky allocation in them. Our model is deliberately partial equilibrium, aimed at analytical results linking the wealth gradient to primitives of the household problem. A companion paper embeds these mechanisms in general equilibrium to quantify the macroeconomic consequences of heterogeneous portfolio responses (Zhang, 2026a).

Structure The remainder of this paper is organized as follows. Section 2 introduces the accounting identity and its three transmission channels. Section 3 describes the Norwegian administrative data. Section 4 presents the empirical analysis, including the estimation strategy, the channel-by-channel impulse responses, and the implied portfolio flow decomposition. Section 5 develops the portfolio choice model and derives the theoretical predictions for the wealth gradient. Section 6 concludes.

2 Accounting Identity

When investors reduce the portfolio risky share after a monetary contraction, they are not necessarily the net sellers of risky assets. This is because the portfolio risky share is a ratio of risky wealth to total liquid wealth, and the latter can change due to cash flow or capital gains. To understand the connection between active compositional adjustment of risky share and the active trading behavior of flow, we propose a model-free accounting framework that decomposes household net purchases of risky assets into three distinct transmission channels. The decomposition follows as an exact identity from the household's balance sheet, requiring no behavioral assumptions about preferences, information, or market structure.

We organize this decomposition around a two-account representation of the household liquid portfolio, consisting of a *current account* holding safe assets such as cash, deposits and bonds; and a *investment account* holding risky assets such as equities and stock

mutual funds, with net flows transferring wealth between the two.

$$F_{i,t} \equiv a_{i,t+1}^{risky} - a_{i,t}^{risky} - CG_{i,t} \tag{1}$$

where $F_{i,t}$ is the net flow of household i into risky assets in period t, $a_{i,t+1}^{risky}$ is the market value of risky assets held in the investment account at the end of period t that will carry over to period $t+1$, and $CG_{i,t}$ is the capital gain earned in period t on risky asset holdings carried over from the last period, due to variations of market prices. The sum of last-period risky wealth $a_{i,t}^{risky}$ and current-period capital gains $CG_{i,t}$ is the period t market value of risky assets carried over from period $t-1$. The net flow $F_{i,t}$ thus measures the active trading behavior: it is positive when the household purchases risky assets and negative when it sells.

As the riskiness of investments lies in the return fluctuations and it is independent of the size of the portfolio, we can rewrite the net flow in terms of the portfolio risky share $\theta_{i,t}$ and total liquid wealth $a_{i,t+1}$:

$$F_{i,t} = \theta_{i,t+1} \cdot \underbrace{(a_{i,t+1}^{risky} + a_{i,t+1}^{safe})}_{a_{i,t+1}} - a_{i,t}^{risky} \; - \; CG_{i,t} \tag{2}$$

where the portfolio size of household i, the sum of both current account ($a_{i,t+1}^{safe}$) and investment account ($a_{i,t+1}^{risky}$), is denoted by $a_{i,t+1}$, and the portfolio risky share $\theta_{i,t+1} \equiv \frac{a_{i,t+1}^{risky}}{a_{i,t+1}}$ measures the fraction of total liquid wealth held in risky assets of household i at the end of period t, which is the chosen exposure to risk that usually receives the most attention in academia and in practice.

This structure reflects the economic logic of what drives net flows into risky assets. Flows arise from three fundamentally distinct sources. The first is *structural portfolio rebalancing*: the household deliberately adjusts the composition of its existing wealth ($\theta_{i,t+1}$), shifting between safe and risky assets. The second is *portfolio expansion through net saving*: unexpected changes reshape household budget constraints through interest income, labor income, consumption, etc. that alter the pool of resources available for saving in the current account ($a_{i,t+1}^{safe}$). The third is *revaluation of risky assets*: exogenous shocks revalue existing equity holdings through equity price movements and induce fluctuations in the risky account balance ($CG_{i,t}$).

These three sources operate through different economic mechanisms, respond to different features of the monetary shock, and interact differently with household characteristics such as income exposure, wealth level and portfolio composition. Conflating them in a single reduced-form flow regression obscures which force drives observed portfolio adjustments. The two-account framework separates them cleanly, and the resulting decomposition makes each channel separately identifiable and empirically estimable.

We further define the *Net saving* $NS_{i,t}$ as the change in total liquid portfolio size net of capital gains. Equivalently, it is the household's active contribution to its portfolio from current disposable income and consumption decisions.

Proposition 1 (Accounting Identity)**.** *The impulse response of a household i's net flows into risky assets $F_{i,t}$, facing a monetary policy shock ε^{MP} on risk-free interest rate R_t^b, satisfies the first-order decomposition:*

$$\frac{dF_{i,t}}{dR_t^b} = a_{i,t+1}\frac{\partial\theta_{i,t+1}}{\partial R_t^b} + MPR_{i,t}\frac{dNS_{i,t}}{dR_t^b} - (1-MPR_{i,t})\frac{dCG_{i,t}}{dR_t^b} \tag{3}$$

$$= \underbrace{a_{i,t+1}\frac{d\theta_{i,t+1}}{dR_t^b}}_{Portfolio\ Rebalancing} + \underbrace{\theta_{i,t+1}\frac{dNS_{i,t}}{dR_t^b}}_{Cash\ Flow} - \underbrace{(1-\theta_{i,t+1})\frac{dCG_{i,t}}{dR_t^b}}_{Asset\ Valuation} \tag{4}$$

where $a_{i,t+1}$ denotes total liquid wealth at the end of period t and $\theta_{i,t+1}$ is the associated ratio of risky wealth to total liquid wealth. The marginal propensity to take risk $MPR_{i,t} \equiv \frac{d(\theta_i a_i)}{da_i}$ serves as a sufficient statistic that measures the fraction of an additional unit of liquid wealth that the household allocates to risky assets. $\frac{\partial\theta_{i,t+1}}{\partial R_t^b}$ denotes the response of portfolio risk exposure conditional on portfolio size $a_{i,t+1}$.

Proof. See Appendix D.1. □

The identity follows from differentiating the household balance sheet with respect to the monetary policy shock and rearranging. No assumptions about preferences, asset structure, or portfolio adjustment costs are required. The decomposition isolates three economically distinct forces that jointly determine how monetary policy transmits to household portfolio flows and can be directly estimated in the data.

Portfolio Rebalancing Channel The first term in equation (4), $a_{i,t+1}\frac{d\theta_{i,t+1}}{dR_t^b}$, isolates the effect of the monetary shock on portfolio composition. It separates deliberate rebalancing decisions from mechanical net flows due to the expansion of portfolio size. Under monetary tightening, households who reduce their desired risky share generate net outflows through this channel as they withdraw from their equity investment. Its magnitude scales with total liquid wealth $a_{i,t+1}$, so even modest changes in risky share translate into large absolute flows for wealthy households.

Cash Flow Channel The second term, $\theta_{i,t+1}\frac{dNS_{i,t}}{dR_t^b}$, captures how cash flow that enters the portfolio in the form of net saving translates into risky asset purchases. Intuitively, households who enjoy positive cash flow in their current account after monetary tightening increase absolute risky asset holdings, allocating new saving according to the marginal propensity to take risk $MPR_{i,t}$, which combines the direct share $\theta_{i,t+1}$ and the indirect

wealth effect $a_{i,t+1}\frac{\partial\theta_{i,t+1}}{\partial a_{i,t+1}}$. Portfolio risky share rises with liquid wealth: wealthier households on average invest a larger fraction of savings into risky assets, implying the exposure of flow to marginal net saving interacts systematically with wealth.

Asset Valuation Channel The third term, $(1-\theta_{i,t+1})\frac{dCG_{i,t}}{dR_t^b}$, captures the effect of equity price movements on flows. Capital gains $CG_{i,t}$ also affect the flow response through marginal change in portfolio size, but it fundamentally differs from net saving as it inflates the balance of the investment account. When the stock price reacts to the monetary policy shock, for example, a price drop, stockholders would target a stable risky share by directing positive flow into equities, in order for the active flow to counteract passive risky share reduction. The scaling by $(1-\theta_{i,t+1})$ corresponds to the complement of portfolio risky share.

3 Norwegian Data

In this section we introduce the Norwegian data and provide descriptive statistics for an overview of the portfolios of Norwegian households. In the end we discuss the relative strength and limitations of the data for our empirical portfolio analysis.

3.1 Household Income and Wealth

Our study draws on Norwegian administrative data covering the full population of residents from 1993 to 2019. The primary data source is the Income and Wealth Tax Records, which provide information on income sources, asset holdings, and liabilities for all residents. Norway levies both income and wealth taxes on its inhabitants, so the tax authority collects detailed annual data on household balance sheets down to individual asset categories. Income is recorded as the annual flow over the calendar year, while asset values are measured at the end of each year. The data are largely third-party reported (employers report labor income directly to the tax authority, and banks report deposit balances and interest income), which substantially limits the scope for measurement error relative to self-reported survey data.

We combine the tax records with the Norwegian Shareholder Registry, which records year-end share holdings at the individual stock level for all residents. This allows us to track the price and quantity of each stock held on an annual frequency. We use the Shareholder Registry data from 2005 onward, when individual-level holdings become available. For the earlier period 1993–2004, stock holdings are observed only at the aggregate level. We also draw on national accounts data for mutual fund returns. All registries are linked via unique personal identifiers at the individual level.

Asset Classification We focus on liquid portfolios and classify assets according to riskiness. Liquid safe assets consist of bank deposits and bonds; liquid risky assets consist of directly held stocks and stock mutual funds. This classification captures the assets most relevant for active portfolio rebalancing in response to monetary policy shocks, while abstracting from housing and private business wealth that are subject to substantially larger adjustment costs and longer rebalancing horizons.

Sample Restrictions We restrict the estimation sample to residents aged 18 and older whose liquid wealth is positive, and whose net income and consumption exceed that year's *Grunnbeløp* (G), the base amount in the Norwegian National Insurance Scheme. Because the portfolio outcomes are constructed from changes between adjacent year-end balance sheets, we also require positive liquid asset holdings and a well-defined risky share in both the current and previous year. Finally, because the portfolio flow is backed out as a residual from imputed capital gains and is normalized by liquid wealth, its tails are dominated by imputation error and by households with a small liquidity base. We therefore trim the top and bottom 1 percent of the normalized flow distribution in each year when estimating the flow responses.

3.2 Descriptive Statistics

Table 1 presents summary statistics for the full sample averaged over 1993–2019. The mean after-tax income is USD 46,556 and mean liquid wealth is USD 40,137, substantially less than mean total wealth of USD 312,607, reflecting the dominant role of housing and other illiquid assets in Norwegian household balance sheets. Capital income net of capital expenses is nearly zero on average (USD −743) but highly dispersed, ranging from −9,211 at the 10th percentile to 2,679 at the 90th, suggesting a sign reversal across the wealth distribution. Within liquid portfolios, deposits account for the overwhelming majority of holdings with a mean of USD 32,348, compared to USD 6,294 for risky assets. Both directly held stocks and mutual funds have a median of zero, confirming that risky asset holdings are concentrated among a minority of households. Among risky assets, mutual funds (mean USD 3,396) are more widely held than directly held stocks (mean USD 2,898).

Figures 1 and 2 further illustrate how income and portfolio composition vary across the liquid wealth distribution. Figure 1 shows that net income is only mildly increasing in liquid wealth, but its composition shifts remarkably: labor income dominates throughout, while capital income net of capital expenses is negative in the lower deciles, reflecting mortgage interest payments, and turns positive only in the upper deciles, where interest-bearing deposits generate substantial returns. Transfers net of taxes are slightly negative across most of the distribution.

	Mean	S.D.	p10	p50	p90
Age	49.34	18.01	26	48	75
After-Tax Income	46,556	98,861	23,162	41,017	69,633
Labor Income	48,422	50,298	0	47,025	98,370
Capital Income (Net)	−743	94,151	−9,211	−1,184	2,679
Total Wealth	312,607	674,834	−57,487	117,132	900,435
Liquid Wealth	40,137	192,105	714	12,828	96,813
Deposits	32,348	91,607	597	10,371	80,322
Bonds	1,496	35,835	0	0	0
Risky Assets	6,294	146,257	0	0	10,147
– Stocks	2,898	140,703	0	0	1,129
– Mutual Funds	3,396	34,258	0	0	6,230

Table 1: Summary Statistics

Notes: Norwegian data from 1993–2019. Balance-sheet variables are measured at the one-year-lagged date. All values except age are in 2011 U.S. dollars. Capital income is net of capital expenses; transfers are net of taxes. The exported table contains 74,128,688 person-year observations.

Figure 2 shows that deposits account for nearly all liquid wealth in the bottom deciles, while stocks and mutual funds become quantitatively important only in the upper half deciles. Bond holdings are negligible throughout. Together, these patterns imply that the asset valuation channel of monetary policy is concentrated among wealthier households, while poorer households are primarily exposed through the cash flow channel.

Figure 3 documents average portfolio exposure and flow behavior across the liquid wealth distribution. Panel (a) plots the stock market participation rate and the portfolio risky share by liquid wealth percentile. Both measures increase monotonically: participation rises from around 20 percent in the third decile to 78 percent at the top, while the risky share increases from 8 percent to slightly higher than 23 percent. The gap between the two series reflects the fact that even among participants, portfolio risky share increases gradually and risky assets constitute a modest fraction of liquid portfolios.

Panel (b) plots the average portfolio flow rate and the liquid wealth growth rate. The flow rate is monotonically decreasing in liquid wealth, turning negative above the 80th percentile: poorer households are on average net buyers of risky assets relative to their liquid wealth base, while wealthier households are net sellers. The liquid wealth growth rate displays a similar declining, mean-reverting pattern, consistent with poorer households accumulating liquid wealth more rapidly in percentage terms. In contrast to average portfolio share, the gaps between the two series imply that poorer households accumulate liquidity mostly in deposits, while wealthier households reduce liquid asset holdings comparably more from stocks and mutual funds.

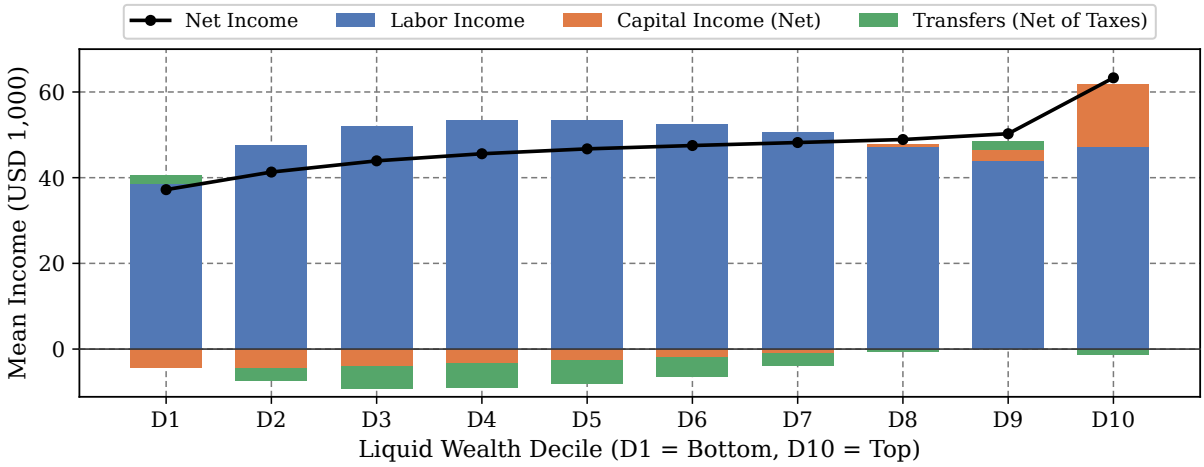


Figure 1: Income Composition Across the Liquid Wealth Distribution

Notes: The figure plots mean income by source across liquid wealth deciles defined using liquid wealth. All values are in 2011 U.S. dollars. The black line denotes net income. Bars above zero decompose gross income into labor income, capital income net of capital expenses, and transfers net of taxes; bars below zero show negative components.

3.3 Comparison with Brokerage Data

Our empirical setting is closely related to Gomes et al. (2025), who use detailed brokerage data from a major Chinese brokerage to examine how retail investors respond to interest rate changes. The two approaches are complementary, each with distinct strengths and limitations.

The Norwegian administrative data's strengths are precisely what a brokerage sample lacks. First, coverage is universal: the dataset includes the entire resident population without attrition, apart from death and emigration. This coverage of the full population rather than of active participants alone, avoids the selection bias inherent in brokerage samples. Second, the data are not top-coded, which is particularly important for studying portfolio behavior at the top of the wealth distribution where equity holdings are concentrated. Third, the data record the complete balance sheet alongside income, consumption, and demographic records, which enables us to study the full composition of household liquid portfolios and to conduct a comprehensive analysis of the driving forces of portfolio choices within a single unified framework, neither feasible with holdings data alone. Lastly, the panel longevity allows us to estimate dynamic responses to monetary policy shocks over multiple years and across multiple episodes of policy tightening and easing. Moreover, we pair them with the well-identified series of monetary policy shocks from Holm et al. (2021), rather than general interest rate movements, which strengthens the causal interpretation of the decomposition.

These advantages come at a cost. The brokerage data in Gomes et al. (2025) record individual stock transactions, allowing them to study the timing and intensity of trading

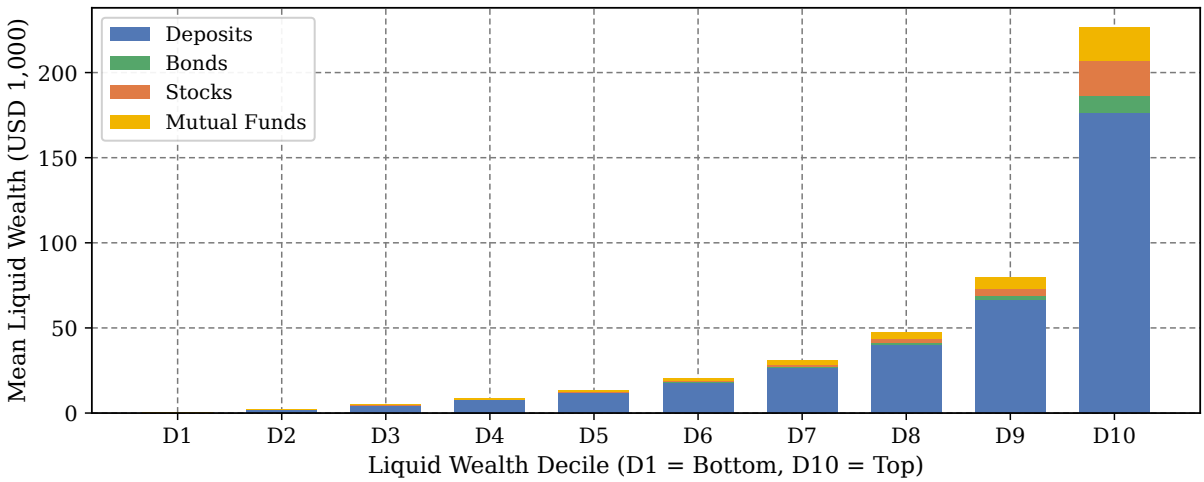


Figure 2: Liquid Portfolio Composition Across the Liquid Wealth Distribution

Notes: The figure plots mean liquid asset holdings by asset class across liquid wealth deciles defined using liquid wealth. All values are in 2011 U.S. dollars.

responses with precision that annual administrative data cannot match. Our annual frequency limits the ability to capture within-year rebalancing dynamics and introduces temporal aggregation that may attenuate short-lived responses. In addition, because we observe only end-of-year portfolio positions, active portfolio flows must be backed out as the residual change in risky wealth net of imputed capital gains, rather than observed directly from transaction records. This imputation introduces measurement error, particularly in the pre-2005 period when individual stock holdings are unavailable and capital gains are approximated using the aggregate Oslo Stock Exchange index.

In sum, the two datasets are best viewed as addressing different aspects of the same question. The brokerage data are well suited to documenting the high-frequency trading response to interest rate movements, while the administrative data are uniquely suited to decomposing the channels through which monetary policy reshapes household portfolios across the wealth distribution and to identifying which channels drive these responses.

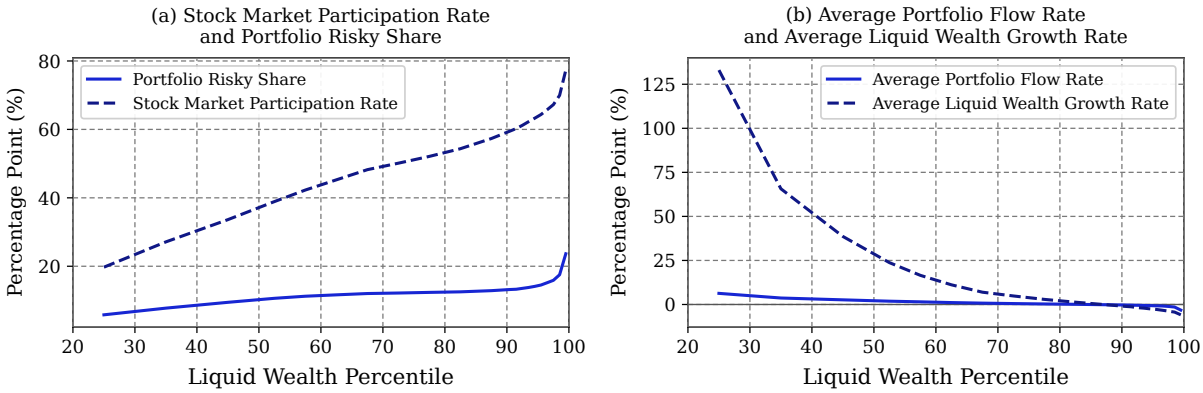


Figure 3: Portfolio Exposure and Flow Behavior Across the Liquid Wealth Distribution

Notes: Panel (a) plots the average stock market participation rate and portfolio risky share by liquid wealth percentile group. Panel (b) plots the average portfolio flow rate and liquid wealth growth rate by liquid wealth percentile group. Groups below the 20th percentile are omitted because the near-zero liquidity denominator makes normalized ratios unstable. All series are averaged across 1993–2019.

4 Empirical Analysis

This section presents the empirical analysis. We first describe the construction of key variables that are not directly observed in the administrative data, including risky share, net saving, and capital gains. We then lay out the empirical strategy, which combines narrative monetary policy shocks with local projections estimated both at the aggregate level and across the liquid wealth distribution. We present impulse responses for aggregate and household-level variables to establish the broad effects of monetary policy, before turning to the three channels identified by the accounting identity. Finally, we decompose the flow response and the relative contributions of the channels.

4.1 Variable Construction

Our empirical analysis requires constructing several variables that are not directly observed in the administrative data. We describe each in turn.

Capital Gains For directly held stocks, we use the Norwegian Stockholder Registry to compute capital gains on individual stocks from 2005 onward. The registry records each household's holdings of individual stocks at the end of the year. For each stock, we calculate the annual capital gain as the change in its real price multiplied by the number of shares held at the beginning of the year. We adopt an end-of-year trading convention, treating the observed net change in stock holdings as if it occurred at the end-of-year price. Capital gains are therefore calculated on beginning-of-year holdings.

For 1993–2004, we cannot calculate capital gains stock by stock. We instead use

average capital gains for stocks traded on the Oslo Stock Exchange and apply them to each household's reported stock portfolio, following Holm et al. (2021). For stock mutual funds, we do not observe holdings of individual funds. We therefore impute capital gains using the same annual aggregate stock-fund price change from the Financial Accounts for all households, combined with each household's reported mutual-fund holdings.

Net Saving and Net Liquid Saving We distinguish between household net saving and net liquid saving. Household net saving is the change in net wealth after removing capital gains:

$$S_{i,t} = w_{i,t+1} - w_{i,t} - CG_{i,t}^{\text{tot}}, \tag{5}$$

where $w_{i,t+1}$ denotes household net wealth, calculated by subtracting outstanding debt from the value of housing, liquid assets, outstanding claims, cars, and boats at the end of year t. The term $CG_{i,t}^{\text{tot}}$ includes capital gains on housing, directly held stocks, and stock mutual funds. Private business wealth is excluded from this measure. Household net saving is used to impute consumption.

Net liquid saving is the corresponding measure for the household's liquid portfolio:

$$NS_{i,t} = a_{i,t+1} - a_{i,t} - CG_{i,t}, \tag{6}$$

where $a_{i,t}$ denotes total liquid assets, consisting of deposits, bonds, directly held stocks, and stock mutual funds, and $CG_{i,t}$ denotes capital gains on directly held stocks and stock mutual funds. Net liquid saving measures the household's contribution to its liquid portfolio after removing changes in market value. This is the saving measure that enters the portfolio accounting identity.

Consumption Consumption expenditures excluding housing services are imputed from the household budget constraint:

$$c_{i,t} = y_{i,t} - S_{i,t}, \tag{7}$$

where $y_{i,t}$ denotes disposable income and $S_{i,t}$ denotes household net saving as defined above. This measure follows Holm et al. (2021).

Risky Share The portfolio risky share is the fraction of total liquid assets held in directly held stocks and stock mutual funds:

$$\theta_{i,t+1} = \frac{a_{i,t+1}^{\text{risky}}}{a_{i,t+1}}, \tag{8}$$

where $a_{i,t+1}^{\text{risky}}$ denotes the household's holdings of directly held stocks and stock mutual funds at the end of year t.

Flow Following (1), the net flow into risky assets is imputed as the corresponding measure for the household's liquid risky wealth:

$$F_{i,t} = a_{i,t+1}^{risky} - a_{i,t}^{risky} - CG_{i,t}. \tag{9}$$

4.2 Empirical Strategy

We use the series of monetary policy shocks identified by the narrative approach of Romer and Romer (2004) as implemented for Norway by Holm et al. (2021). The shock series is constructed by regressing policy rate changes on Norges Bank and Consensus Economics forecasts of inflation and output growth, as well as the lagged policy rate and the exchange rate. The residuals from this regression capture the component of policy rate movements not predicted by the central bank's information set at the time of each decision. The resulting series is aggregated to annual frequency by summing monthly shocks within each calendar year. Figure B.1 in Appendix B plots annual interest rates and the monetary policy shock series over the sample period.

We estimate impulse response functions using local projections (Jordà, 2005). For aggregate variables, the specification takes the form

$$y_{t+h} - y_{t-1} = \alpha^h + \varepsilon^h \vartheta_t^{MP} + \sum_{n=1}^{3} \varpi_n^h \vartheta_{t-n}^{MP} + u_t^h \tag{10}$$

where $y_{t+h} - y_{t-1}$ is the cumulative log-change in the macro variable from the pre-shock period to horizon h (expressed in level-differences for the policy rate and unemployment rate), ϑ_t^{MP} is the monetary policy shock, and ϑ_{t-n}^{MP} for $n = 1, 2, 3$ are its lags. For the aggregate specification, we do not include lags of the dependent variable as controls, following Holm et al. (2021).

For household-level variables, we estimate the responses in a panel local projection framework. When pooling across all households, the outcome variable is normalized by the sample mean of the pre-shock level:

$$\frac{Y_{i,t+h} - Y_{i,t-1}}{\bar{Y}_{t-1}} = \omega_i^h + \varepsilon^h \vartheta_t^{MP} + \sum_{n=1}^{3} \varpi_n^h \vartheta_{t-n}^{MP} + \sum_{n=1}^{2} \phi_n^h X_{i,t-n} + u_{i,t}^h \tag{11}$$

where ω_i^h denotes individual-by-horizon fixed effects and $X_{i,t-n}$ are two lags of the one-year growth rate of the dependent variable. The coefficient ε^h captures the average h-year response of the variable, as a percentage of its pre-shock mean, to a one percentage point contractionary monetary policy shock.

When estimating heterogeneous responses across the liquid wealth distribution, we normalize all variables by the individual's pre-shock liquid wealth:

$$\frac{Y_{i,t+h} - Y_{i,t-1}}{A_{i,t-1}} = \omega_i^h + \varepsilon_g^h \vartheta_t^{MP} + \sum_{n=1}^{3} \varpi_{g,n}^h \vartheta_{t-n}^{MP} + \sum_{n=1}^{2} \phi_{g,n}^h X_{i,t-n} + u_{i,t}^h \quad \forall\, i \in g \tag{12}$$

where g indexes liquid wealth groups defined by year-specific percentile cutpoints. This normalization ensures that all channel responses are expressed in the same units as a fraction of pre-shock liquidity, and can therefore be aggregated through the accounting identity derived in proposition 1. We define wealth groups of increasing granularity: deciles from the 1st to the 50th percentile, followed by five-percentile bins from p50 to p90, and individual percentiles from p90 to p100. This grouping provides finer resolution at the top of the distribution, where portfolio heterogeneity is most pronounced. For the risky share $\theta_{i,t}$, which is already a ratio, the outcome variable is the level change $\theta_{i,t+h} - \theta_{i,t-1}$ without further normalization.

All panel specifications include individual-by-horizon fixed effects and are estimated using the Driscoll and Kraay (1998) covariance estimator with a lag truncation of two years, which accounts for both cross-sectional dependence arising from the common monetary policy shock, and serial correlation in the projection residuals. We report 95 and 68 percent confidence bands throughout.

4.3 Aggregate and Household Responses

Figure 4 presents impulse responses of aggregate variables to a one percentage point contractionary monetary policy shock. The responses are broadly consistent with those documented in Holm et al. (2021). The policy rate rises on impact, remains elevated for two to three years, then becomes expansionary and reverts toward zero over the following years. Similarly, GDP declines by around two percent at its trough after four years, and unemployment rate rises by 0.7 percentage points at the peak in three to four years. Consumption expenditures gradually decline by slightly over one percent and trace the GDP responses closely.

Figure 5 shows the corresponding responses at the household level. These micro-level responses mirror the aggregate patterns but provide additional detail on the composition of income and balance sheet adjustments. Consumption expenditures decline by roughly 1 percent in the short run, consistent with the aggregate response. Disposable income falls on impact by slightly less than 1 percent, driven mostly by financial income. Capital revenues and capital expenditures both exhibit large swings on impact and show some decline at longer horizons.

The balance sheet responses are particularly relevant for the portfolio analysis that follows. Total wealth declines sharply, falling by roughly 8 percent after three years. This

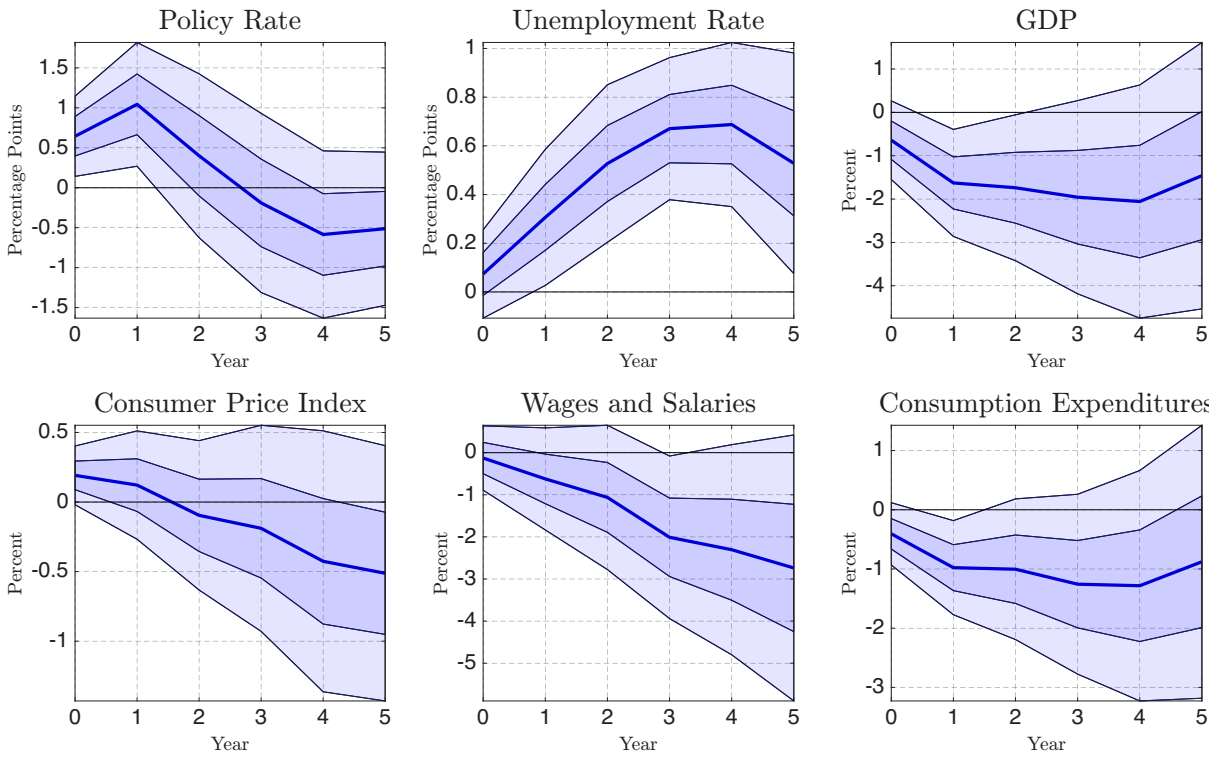


Figure 4: Impulse Responses of Aggregate Variables

Notes: Impulse responses to a one percentage point contractionary monetary policy shock, estimated by local projections at annual frequency. Shaded areas denote 68 and 95 percent confidence bands based on Newey and West (1987) standard errors.

aggregate decline masks a striking compositional shift: risky assets fall by up to 12 percent, while safe assets increase by roughly 2 percent. The divergent response of risky and safe assets (the contraction in equity holdings accompanied by an expansion in deposits) is the reduced-form counterpart of the portfolio responses we examine across the distribution next. On the other hand, wide confidence intervals of safe assets and capital incomes indicate large wealth heterogeneity.

4.4 The Risky Share: Reaching for Yield

For the rest of the analysis, we exclude the bottom two wealth groups (below p20) from the figures for two reasons: First, the risky share, portfolio net flow, and risky asset holdings of the bottom 20% of the liquidity distribution are not significant compared with the others. Second, the near-zero liquidity base of these households produces extreme normalizations and uninformative estimates.

Figure 6 presents the impulse response of the risky share across the liquid wealth distribution. The risky share falls across the entire distribution, and the decline persists across the horizons: households cut the share of their liquid portfolio held in risky assets and hold it down until the contraction passes. This is the risky-share signature of reaching

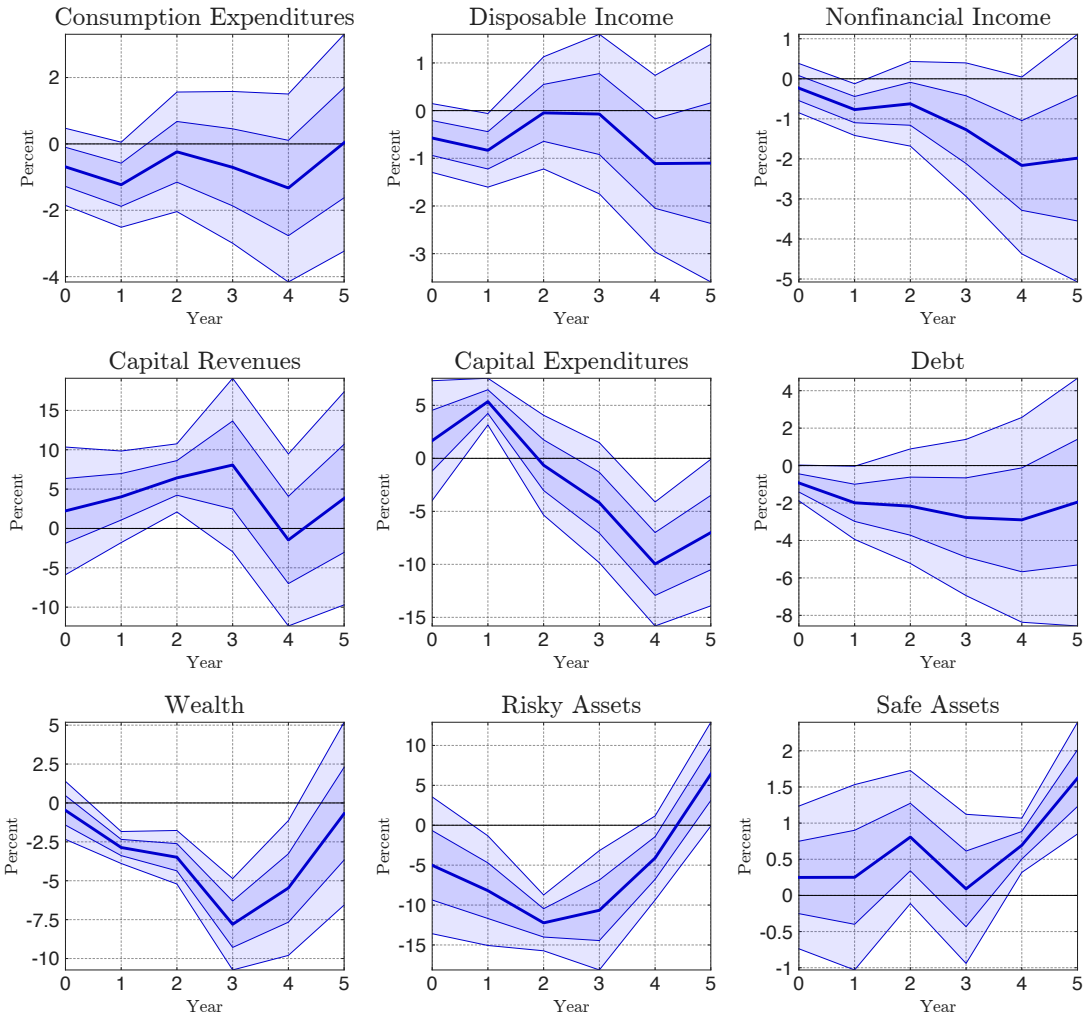


Figure 5: Impulse Responses of Household Variables

Notes: Impulse responses to a one percentage point contractionary monetary policy shock, estimated by panel local projections pooling all households. Each variable is normalized by the sample mean. Shaded areas denote 68 and 95 percent confidence bands based on Driscoll-Kraay standard errors.

for yield: a higher safe interest rate reduces households' risk appetite. On impact, the response is significant at the 95 percent level for all groups above the median, and its magnitude increases with liquid wealth, from essentially zero at p25 to roughly -1.0 point through the upper part of the distribution. The uniform sign shows that reaching for yield operates across the entire liquid wealth distribution, while its strength and timing, which we examine in turn, reveal substantial heterogeneity across households.

The reduction builds over the first two years rather than arriving all at once. The decline deepens from impact through horizon 2, where the risky share reaches its trough, ranging from -0.50 points at p25 to -1.50 points at the top. This trough leads the economy by one to two year, as aggregate output bottoms out at horizon 4 (Figure 4). As the contraction dissipates and the macroeconomy recovers, the risky share reverts

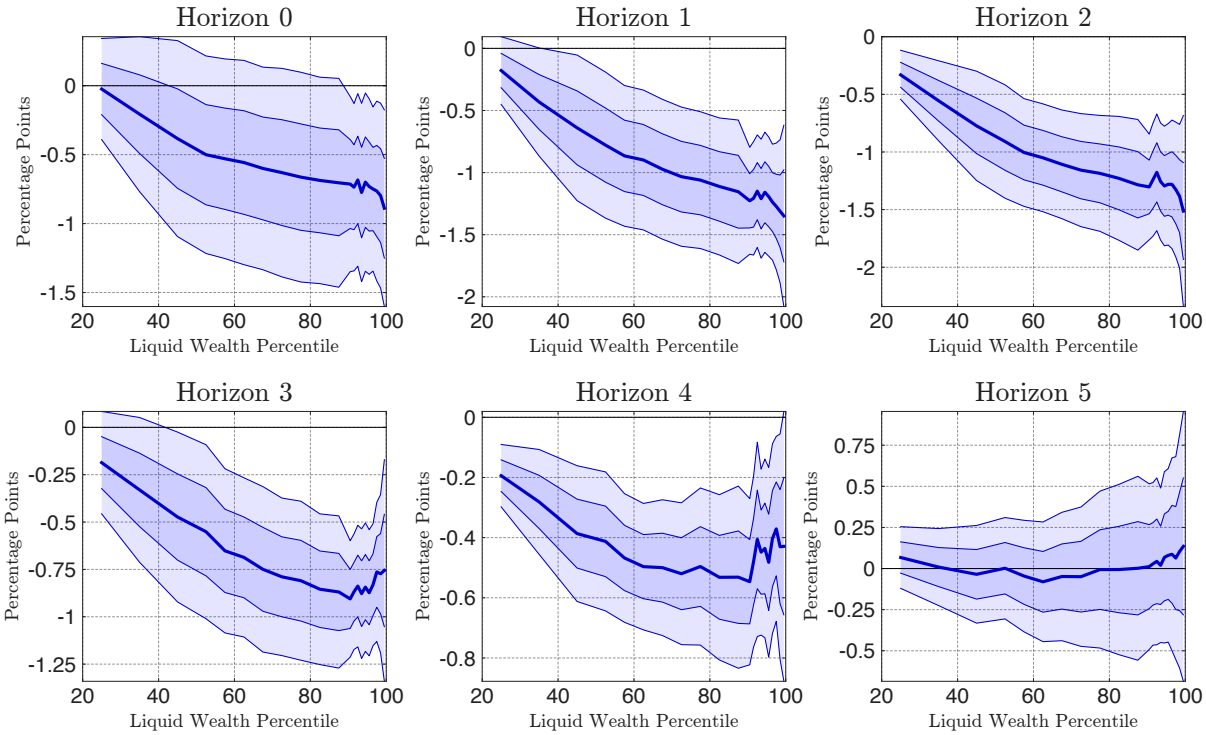


Figure 6: Impulse Response of the Risky Share Across the Liquid Wealth Distribution

Notes: Each panel plots the impulse response of the risky share ($\theta_{i,t+h} - \theta_{i,t-1}$) to a one percentage point contractionary monetary policy shock at horizon h, estimated separately for each of the 23 liquid wealth groups. Groups below the 20th percentile are excluded. Shaded areas denote 68 and 95 percent confidence bands based on Driscoll-Kraay standard errors.

toward zero, and by horizon 5 the response has largely unwound. The one-period lead points to forward-looking risk consideration rather than a mechanical response to past outcomes: households pull back from risk in anticipation of the downturn and begin rebuilding exposure before the recovery is realized.

Wealth gradient. The cross-sectional response is negative throughout but convex in wealth. On impact and through the trough, the reduction in the risky share deepens as one moves up the distribution but at a diminishing margin, steepening across the middle and then flattening over the upper part. During the recovery the convexity sharpens into an upward tilt: the top reverts first, and by horizon 5 the risky share of the wealthiest households has turned positive while the middle remains slightly below zero. Reaching for yield is thus universal in direction, but because the risky share is the household's chosen exposure to risk, the convex gradient says directly that the wealthy depress their risk-taking the least at the margin and rebuild it the soonest.

Participation. The unconditional response averages participants, who adjust their risky share, with non-participants, whose share is zero before and after the shock. Because the

participation rate rises with liquid wealth (Figure 3), this dilution alone could generate a downward slope in the unconditional response even if every participant adjusted by the same amount, and the convexity could be an artifact of a participation rate that flattens at the top. We therefore re-estimate equation (12) on stockholders only. Figure 7 reports the result. Among participants, the gradient remains convex, with a flat segment that now extends from roughly the 60th percentile to the top at horizons 0 to 1, and at longer horizons an upward tilt at the very top. Participation therefore shapes the level of the unconditional response but not its curvature; the flattening at the top is a property of the choices of households who hold stocks. This convex gradient with flat tail is the regularity our theory is built to explain, and the economic force it formalizes in Section 5.

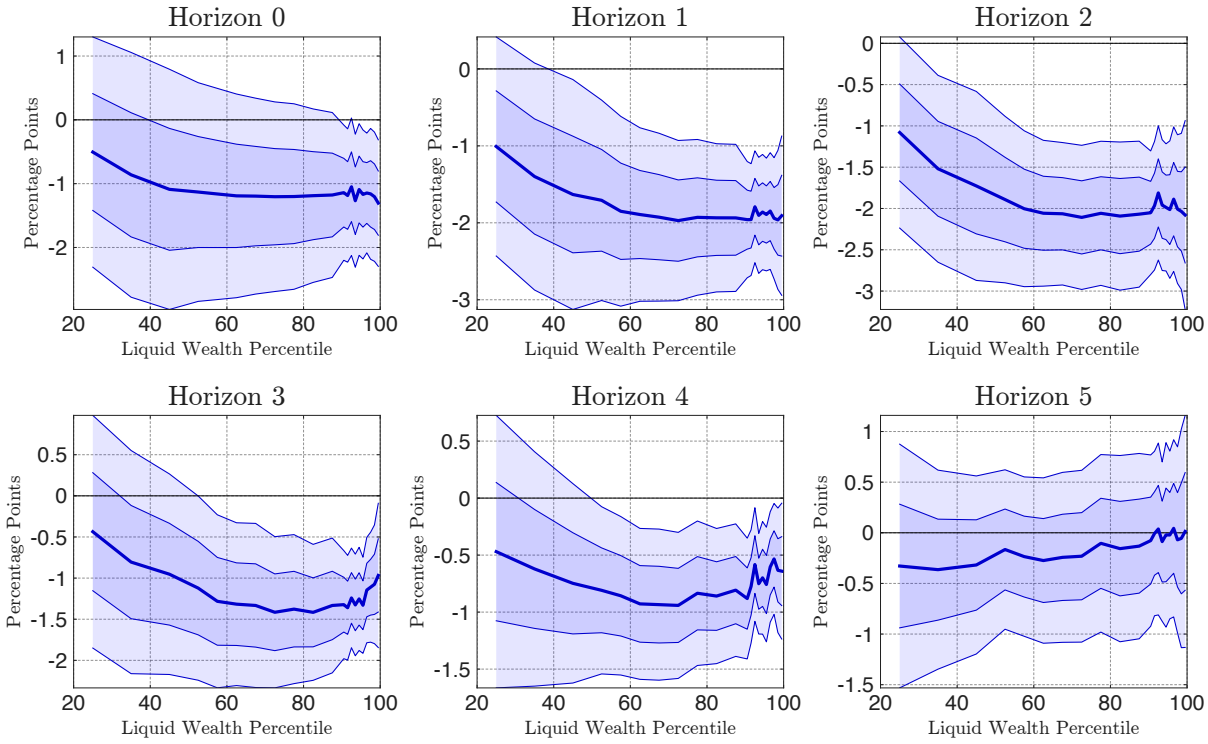


Figure 7: Responses of Participants' Risky Share Across the Liquid Wealth Distribution

Notes: Each panel plots the impulse response of the risky share from stock market participants, to a one percentage point contractionary monetary policy shock at horizon h, estimated separately for each of the 23 liquid wealth groups. Groups below the 20th percentile are excluded. Shaded areas denote 68 and 95 percent confidence bands based on Driscoll-Kraay standard errors.

Taken together, the timing and the gradient establish *reaching for yield* as a headline result: the risky share falls following a tighter monetary policy, and recovers when the monetary policy eases later. It is an expectations-driven retreat from risk that spans the whole distribution but weakens toward the top. The share, however, is only one of the forces that determine where money actually flows. A household that lowers its chosen portfolio risk exposure can still be a net buyer of risky assets once its saving and its capital gains are taken into account. What the reaching for yield documented here implies for

the purchases and sales from the compositional structure, and how that net flow responds and varies across the distribution is what the rest of the section takes up.

4.5 Liquid Wealth Across the Distribution

The risky share only tells us how households adjust their portfolio composition, but not the level of investment. Before analyzing net flows, we look at the two balance-sheet quantities behind the share: risky asset holdings and total liquid wealth.

Figure 8 shows the total liquid wealth responses. On impact its decline is statistically significant only at the top of the distribution, where it reaches about -1 percentage point of pre-shock liquidity above the 90th percentile. Below that, the point estimates are indistinguishable from zero. Combined with our findings of universal reduction in risky share, this suggests that lower level of liquid wealth after a monetary contraction is not a primary driver of the reaching for yield behavior. At longer horizons the top recovers to zero by horizon 2 without turning positive, while the groups below it turn positive by horizon 5.

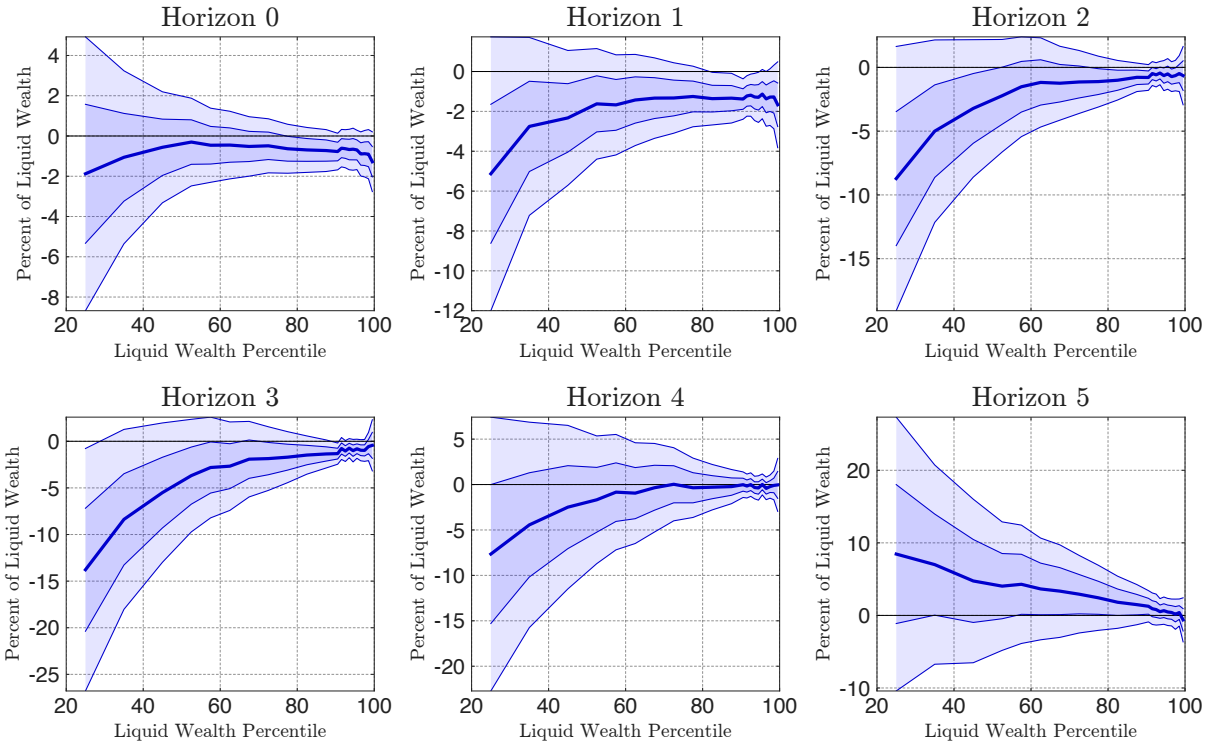


Figure 8: Impulse Response of Total Liquid Wealth Across the Liquid Wealth Distribution

Notes: Each panel plots the impulse response of total liquid wealth, normalized by lagged liquid wealth, to a one percentage point contractionary monetary policy shock at horizon h, estimated separately for each of the liquid wealth groups. Groups below the 20th percentile are excluded. Shaded areas denote 68 and 95 percent confidence bands based on Driscoll-Kraay standard errors.

A level change, however, cannot reveal the within-portfolio transactions beneath it.

Risky wealth falls whether a household sells its equities or even buys them as prices drop sharply, so it mixes active trading with passive valuation.[1] Total liquid wealth measures only the size of the portfolio, not its composition: a household that sells safe assets to buy risky ones leaves it unchanged. These responses are thus informative about the level changes, but silent on the flows, the active purchases and sales that reallocate the portfolio between safe and risky assets.

4.6 Portfolio Flow

Figure 9 plots the portfolio flow estimate across the liquid wealth distribution. The portfolio flow into risky assets is constructed as the difference between risky wealth and capital gains. Four features stand out. First, on impact the direct flow is statistically insignificant at every wealth percentile. Second, at longer horizons the estimates sharpen and the gradient resolves into a sign reversal. By horizon 2 when monetary contraction is at the trough, the flow is significantly negative for households below roughly the 90th percentile, who are net sellers of risky assets, and significantly positive at the very top, where households are net buyers, reaching about +0.25 percentage points at the top percentile. Horizon 1 is transitional, carrying the same upward gradient with the top slightly above zero. Third, the positive wealth gradient is present in horizons 1 to 3 when monetary policy is in the tight phase: the gap between the top and the middle widens, and the top crosses from negative into positive territory. Lastly, the flow response reverts toward zero and becomes insignificant at horizons 3 and 4, and by horizon 5 the estimates are positive and largely significant across the distribution, with a negative gradient.

To sum up, our estimates suggest that portfolio flow is sluggish on impact, potentially due to inertia or the friction in portfolio adjustment. As the monetary contraction deepens, the wealthy starts to net buy risky assets and the less affluent withdraw. When the macroeconomy recovers from the trough, the mid-wealth households become net buyers as well. In the following empirical analysis, we decompose the flow responses to identify the driving forces across the wealth distribution at each horizon.

Flow and the Channels

Our accounting identity decomposes the flow responses into three channels that each associates with a separate variable. The first, the risky share, has been documented above; we now present the remaining two, net saving and capital gains, and then combine all three through the accounting identity (4). Each is estimated separately across liquid wealth groups using equation (12), with the outcome normalized by pre-shock liquidity so that the channels share common units.

[1] Figure C.1 in the appendix plots the response of risky asset holdings.

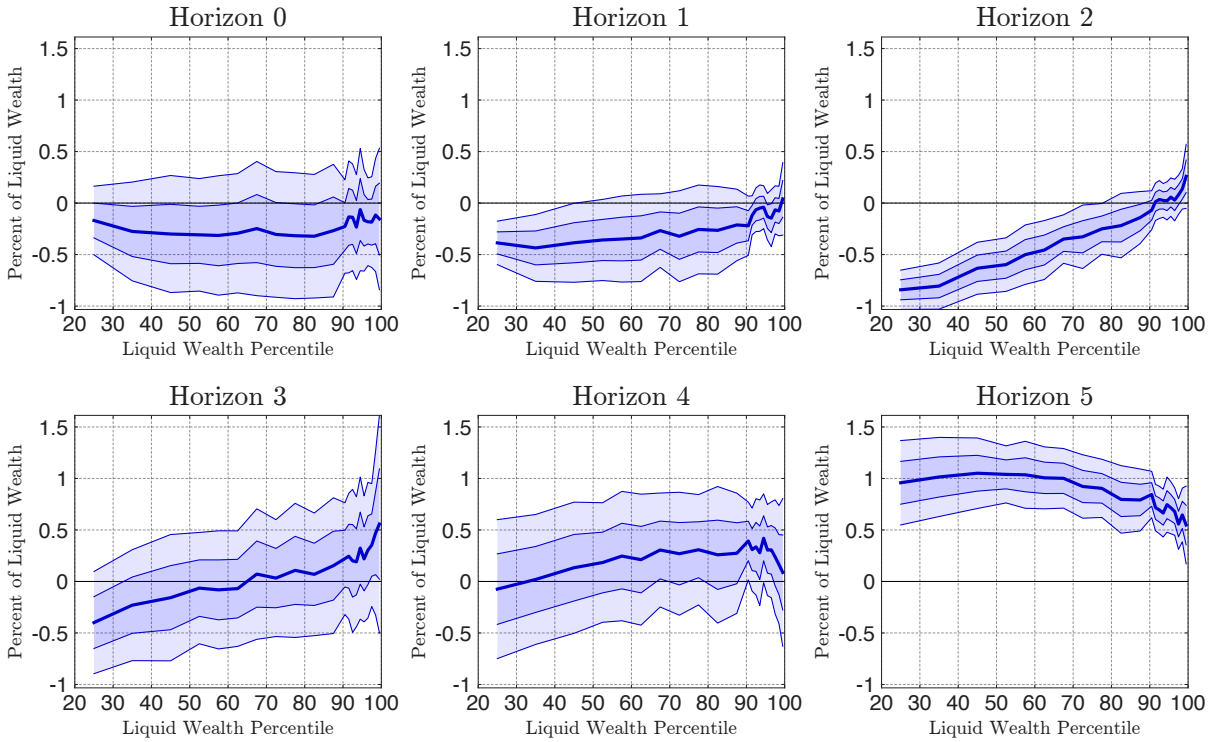


Figure 9: Impulse Response of Portfolio Flow Across the Liquid Wealth Distribution

Notes: Each panel plots the impulse response of direct portfolio flow with 1% Flow-Tail Trimming, normalized by lagged liquid wealth, to a one percentage point contractionary monetary policy shock at horizon h, estimated separately for each of the liquid wealth groups. Groups below the 20th percentile are excluded. Shaded areas denote 68 and 95 percent confidence bands based on Driscoll-Kraay standard errors.

Net Saving. Figure 10 presents the net saving channel. On impact, the response is insignificant across most of the distribution. Below the 90th percentile the point estimates trace a mild hump peaking near the median, but it lies entirely within the confidence bands. Above the 90th percentile the response turns significantly positive at the 68 percent level and rises toward the very top, from about 0.3 to 0.7 percent of pre-shock liquidity. The significant pattern is therefore flat at zero across the distribution and rising only at the top. This rise is consistent with higher net capital income for the wealthy, whose larger deposit holdings earn more as rates rise, partly offsetting the fall in non-financial income. At horizons 1 through 3, the responses are small and largely insignificant, except for households at the top who increase their net savings until the interest rate falls back to the pre-shock level at horizon 3. At horizons 4 and 5, net saving turns clearly positive with a downward slope and statistically significant across most of the distribution, reaching about 11 percent at p20–p30 and 1.5 percent at p99–p100 by horizon 5. This delayed response likely reflects households rebuilding their liquid positions as the contraction dissipates, and the much larger normalized response at the bottom reflects the small liquidity base of those households rather than larger absolute saving flows.

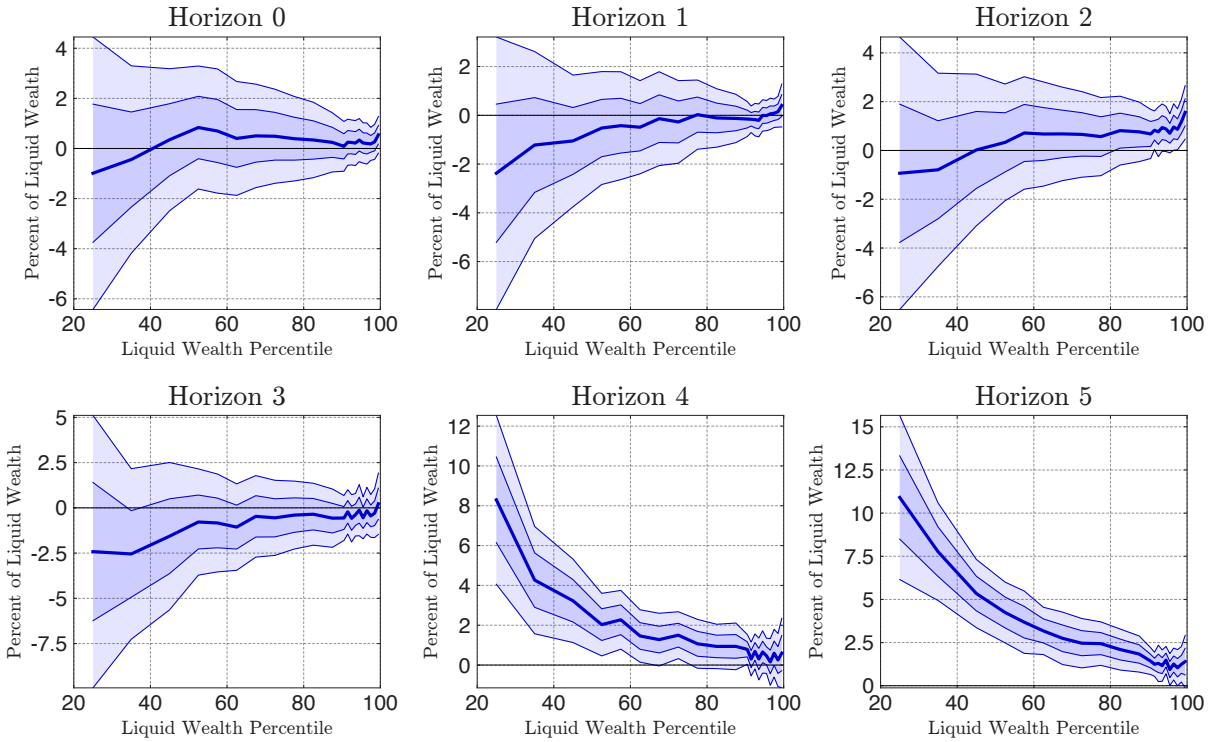


Figure 10: Impulse Response of Net Saving Across the Liquid Wealth Distribution

Notes: Each panel plots the impulse response of net saving, normalized by lagged liquid wealth, to a one percentage point contractionary monetary policy shock at horizon h, estimated separately for each of the liquid wealth groups. Groups below the 20th percentile are excluded. Shaded areas denote 68 and 95 percent confidence bands based on Driscoll-Kraay standard errors.

Capital Gains. Figure 11 presents the capital gains channel. On impact, capital gains decline across the entire distribution, and the response is statistically significant at the 68 percent level throughout and at the 95 percent level above roughly the 75th percentile. The point estimate increases in magnitude from about -0.85 percent of pre-shock liquidity at p20–p30 to about -1.2 percent by the median, is flat across the upper-middle of the distribution, and steepens at the very top, reaching about -1.9 percent at p99–p100. The profile tracks equity exposure: the risky share rises only mildly through the upper-middle, where capital losses are flat, and jumps at the top, where they steepen, so that the top percentile, with a risky share roughly twice that of the p20–p30 group, suffers losses roughly twice as large relative to liquid wealth. This is consistent with a common negative shock to equity prices filtered through heterogeneous portfolio compositions.

The dynamic pattern is also informative. The negative response persists at horizons 1 and 2, while the statistical significance weakens starting at horizon 2, consistent with the sustained decline in stock prices that follows a monetary tightening and the gradual build-up of real effects on corporate earnings. By horizon 3, the capital gains response has largely reverted to zero, and at horizons 4 and 5 it turns positive and statistically significant for many groups, reflecting a recovery in equity valuations as the monetary

contraction dissipates and the policy rate reverts toward its pre-shock level. The timing of this reversal corresponds closely to the reversion of the risky share toward zero observed in Figure 6.

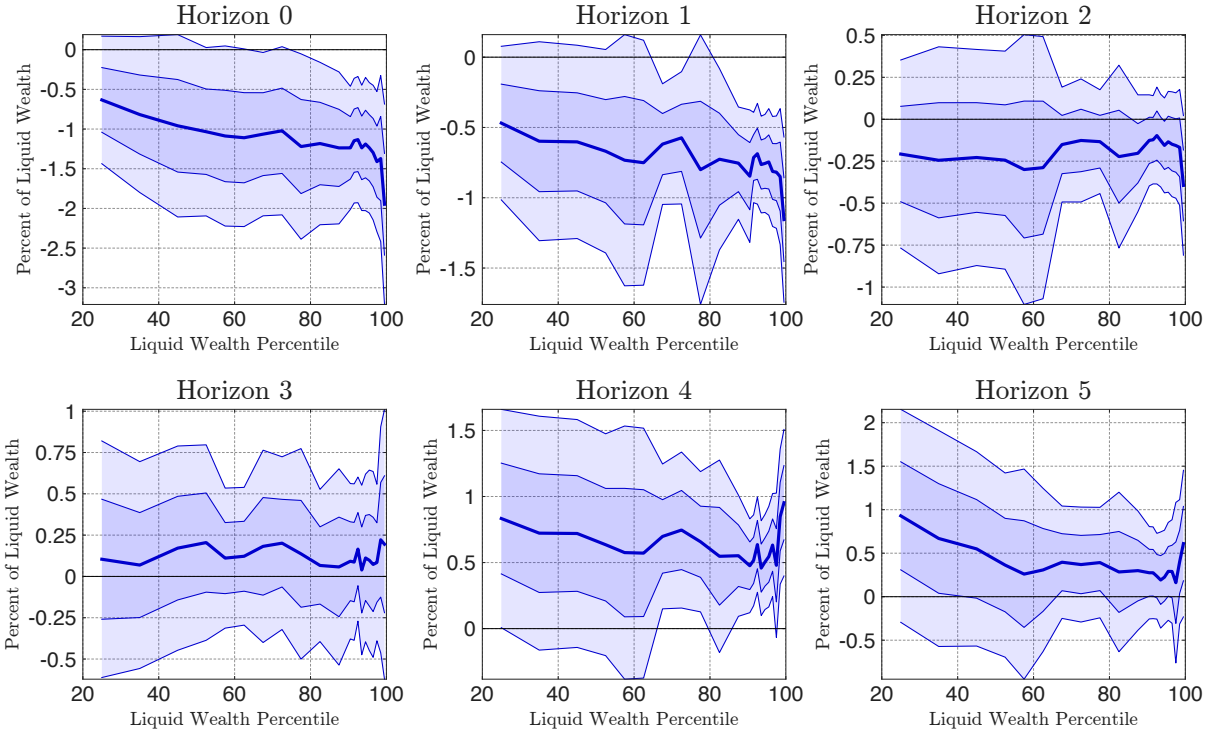


Figure 11: Impulse Response of Capital Gains Across the Liquid Wealth Distribution

Notes: Each panel plots the impulse response of capital gains on risky assets, normalized by lagged liquid wealth, to a one percentage point contractionary monetary policy shock at horizon h, estimated separately for each of the liquid wealth groups. Groups below the 20th percentile are excluded. Shaded areas denote 68 and 95 percent confidence bands based on Driscoll-Kraay standard errors.

While the confidence bands are wide for some groups and horizons, particularly for net saving at short horizons and for capital gains among lower-wealth households, the estimates are informative in several important respects. First, for each channel, the cross-sectional profile at a given horizon is mostly smooth and economically coherent: point estimates move monotonically across neighboring wealth groups rather than fluctuating erratically, lending credibility to the distributional patterns even where some individual group estimates might contain large noise. Second, the dynamic profiles are internally consistent across channels: the capital gains reversal at horizons 4–5 coincides with the risky share reversion and the net saving recovery. Third, the estimates that are statistically significant, capital gains for the upper quartile on impact, risky share for all groups above the median at horizons 0–4, and net saving at horizons 4–5, etc., are sufficient to discipline the decomposition at the horizons that matter most for the portfolio response. The imprecision in individual channel estimates at certain horizons reflects the

inherent limitations of annual data in capturing portfolio dynamics that may unfold at higher frequencies.

Decomposition

We decompose the flow responses in Figure 9 according to the accounting identity (4). Specifically, we estimate *portfolio rebalancing*, *cash flow*, and *asset valuation* channels and report their average contributions, together with the point estimate of flow responses across the liquidity distribution.[2]

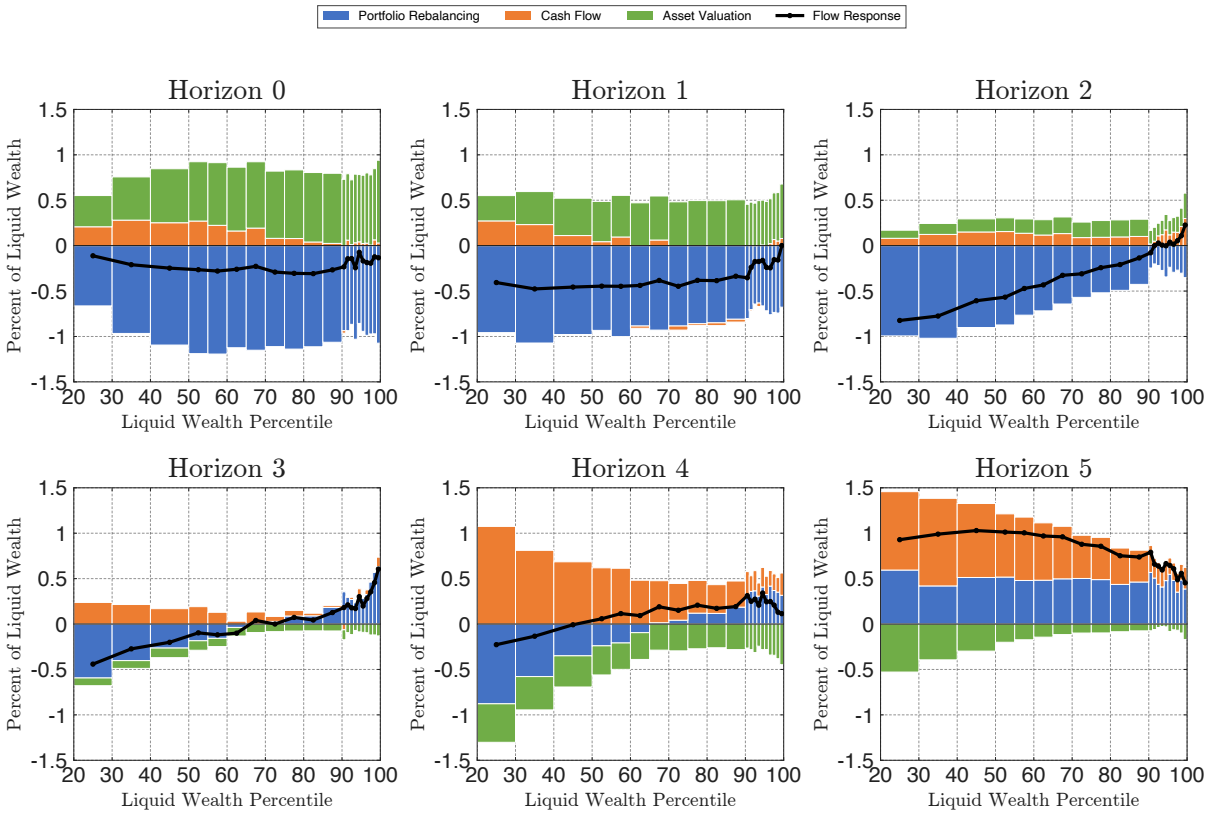


Figure 12: Flow Decomposition with Accounting Identity

Notes: Stacked bars decompose the portfolio flow responses into three terms from the accounting identity: portfolio rebalancing (blue), cash flow (orange), and asset valuation (green). The black line plots the point estimates of flow responses. All values are in percentage points of pre-shock liquid wealth. Groups below the 20th percentile are excluded.

Figure 12 presents the flow decomposition across the liquid wealth distribution. Several findings emerge. First, on impact and in the following year, the flow response in the top quintile is muted: the passive change in risky wealth through the asset valuation channel offsets the decline in risk exposure through the portfolio rebalancing channel,

[2]For the consistency of accounting identity, portfolio rebalancing channel is expressed as $a_{i,t}\frac{d\theta_{i,t+1}}{dR_t^b}$ in the decomposition model, while the other two channels remain the expression in equation (4). We show in appendix that the decomposition is robust to different expressions.

and cash flow is a negligible contributor. The other four quintiles look different. Their rebalancing outflow is stronger, but less affluent households also experience capital loss and they save more, keeping their liquid portfolio roughly balanced at horizon 0. However, their reduced risky exposure contributes significantly more at horizon 1, making them net equity sellers. Second, the convex gradient of the risky-share response drives the cross-sectional heterogeneity in the flow response, whose gradient follows that of the portfolio rebalancing channel closely from horizon 0 to horizon 3. The exception is the top 10 percent at horizon 2, when their net saving builds up and reverses the sign of their flow response. This is our third finding: before the trough of the monetary contraction, the wealthy are net buyers, because of strong cash flow and a weak rebalancing outflow, while the rest are net sellers because of the decline in their risky share. Fourth, from horizon 3 onward, when monetary policy enters its easing phase, the risky-share response of the wealthy turns positive first and their net purchases continue. The reversal in the lower- and middle-wealth groups follows, and by horizon 5 the rebalancing contribution to the flow is roughly equal across the distribution. Fifth, the cash flow contribution has a different profile. In the longer run it dominates portfolio rebalancing in the lower half of the distribution and becomes the decisive force behind the flow gradient. Lastly, the contribution from capital gains is strongest in the beginning (horizon 0 and 1) when the interest rate rises and at horizon 4 and 5 in the longer run where monetary policy loosens.

4.7 Summary

The empirical analysis documents four facts on the household portfolio choice responses to a monetary policy shock.

First, reaching for yield is universal and persistent. After a contractionary shock, households at every level of liquid wealth reduce their risky share on average, the reduction deepens to a trough at horizon 2, and it unwinds as the contraction passes, with the wealthiest reverting first and turning positive by horizon 5. Because risky share reaches its trough before aggregate output, the retreat from risk is forward-looking rather than a mechanical response to realized losses.

Second, the response is convex in wealth. Its magnitude grows with liquid wealth but with a diminishing margin, steepening through the lower half of the distribution and flattening across the upper half. The wealthy cut their risk-taking the least at the margin and rebuild it the soonest. This convexity sets the cross-sectional shape of the flow through the tightening phase, and it is what leaves room for the other channels to reverse the sign at the top.

Third, trading and risk-taking diverge. Estimated directly, the flow is insignificant on impact, because it is the small residual of two large and offsetting quantities, but a positive wealth gradient builds through the tightening phase and resolves by horizon 2

into a reversal: middle-wealth households are significant net sellers of risky assets while the top are net buyers. The flow reverts toward zero as policy eases and is positive across the distribution by horizon 5. The reversal matches the one Gomes et al. (2025) document in brokerage transactions, but the two differ in horizon. Their monthly data resolve the sign of the flow within weeks of a rate change, whereas our annual data cannot pin it down on impact and show the reversal only once it has built up over two years. Read together, the two suggest that the reversal appears quickly in high-frequency trading and persists through the tightening phase.

Fourth, the decomposition shows where that reversal comes from. The wealth gradient of the flow is inherited from the risky share: through the monetary tightening the flow tracks the rebalancing channel closely, so the convexity of the share response is what makes the flow flatten and then reverse toward the top. The reversal itself has two sources at two horizons. On impact, capital losses that scale with equity exposure offset the rebalancing outflow of the wealthy and hold their flow near zero while the middle begins to sell; by horizon 2, net saving at the top builds up and turns the wealthiest into net buyers. The channels also run on different clocks: valuation moves with equity prices and is strongest when the policy rate moves, portfolio rebalancing follows the risky share and unwinds with the contraction, and cash flow builds slowly and dominates the lower half of the distribution only in the longer run.

Taken together, these facts shed light on the development of portfolio choice theory. The flow reversal is a result of the choices on portfolio composition and size: given the accounting identity and the mechanical response of capital gains to equity exposure, it follows from the convex risky-share response and the saving behavior of the wealthy. The saving response to monetary policy is well understood from the heterogeneous-agent literature (Kaplan et al., 2018; Auclert, 2019), the open question concerns the risky share: why reaching for yield exists at all under a standard portfolio problem, and why its strength flattens at the top of the wealth distribution. Section 5 takes up both.

5 A Portfolio Choice Model

The empirical analysis documented several patterns in how household portfolios respond to monetary policy across the liquid wealth distribution. Following a monetary contraction, the risky share declines monotonically in wealth with a convex gradient, flattening toward the top; capital losses grow steeply with equity exposure; and the flow is negative for middle-wealth households yet positive for the wealthiest, who actively buy risky assets at the trough of contraction. No standard model delivers all three. This section develops a theory of portfolio choice to explain them. The model is deliberately stylized: it isolates the consumption-saving and portfolio-allocation decisions in the partial equilibrium framework, and delivers analytical results that link the wealth gradient in the risky share and flow

responses to primitive features of the household problem.

5.1 Environment

Consider a discrete-time real economy with a unit mass of households indexed by i. Each household i faces idiosyncratic risk to its labor efficiency $e_{i,t}$, which follows a discrete Markov chain with transition matrix Π_e. Labor income is $y_{i,t} = e_{i,t}w_t$, where w_t is the aggregate net-of-tax/transfer real wage, taken as exogenous by the household. Households have CRRA preferences over consumption with coefficient of relative risk aversion γ:

$$U(c) = \frac{c^{1-\gamma}}{1-\gamma} \tag{13}$$

On top of the textbook idiosyncratic income process, we introduce countercyclical idiosyncratic disaster risk in the labor efficiency $e_{i,t}$: the transition probabilities Π_e depend on the aggregate state of the economy, so that the probability of transitioning to a low-e state increases in recessions and decreases in booms. This captures the empirical finding that business cycle variation in individual income risk operates primarily through left-skewness: large downward income movements become more likely in recessions while upward movements become less likely (Guvenen et al., 2014b). This pattern has been confirmed in Norwegian data (Halvorsen et al., 2024) and shown to affect household portfolio decisions in Sweden (Catherine et al., 2024). As we will show later, this feature is crucial for generating the heterogeneity in the risky share response to monetary policy shocks that we empirically document.

Households can save in two assets: risk-free bonds $b_{i,t}$ whose price is normalized to one, which pay a gross return R^b_{t-1} pre-determined by the monetary authority, and stocks $s_{i,t}$, which have price Q^s_t and pay a dividend d_t each period, yielding a gross return $R^s_t = \frac{Q^s_t + d_t}{Q^s_{t-1}}$. Stock returns are subject to an aggregate productivity shock $\varepsilon_{z,t}$ with mean zero and variance σ^2_z. For tractability, we assume that this shock affects labor income only transitorily: $\frac{\partial \mathbb{E}\{y_{i,t+s}\}}{\partial \varepsilon_{z,t}} = 0$ for all $s > 0$, so that a date-t innovation moves current income but leaves expected income at all future dates unchanged. Households choose consumption $c_{i,t}$, bond holdings $b_{i,t+1}$, and stock holdings $s_{i,t+1}$ to maximize expected lifetime utility:

$$\max_{\{c_{i,t}, b_{i,t+1}, s_{i,t+1}\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t U(c_{i,t}) \tag{14}$$

subject to the budget constraint

$$c_{i,t} + Q^s_t s_{i,t+1} + b_{i,t+1} = (Q^s_t + d_t)s_{i,t} + R^b_{t-1} b_{i,t} + y_{i,t} \tag{15}$$

and the constraint that both bond and stock holdings cannot be negative:

$$a_{i,t+1} \equiv Q_t^s s_{i,t+1} + b_{i,t+1} \tag{16}$$

$$s_{i,t+1}, b_{i,t+1} \geq 0 \tag{17}$$

We impose no additional portfolio constraints beyond non-negativity of assets. The net flow into stock holdings is expressed as:

$$F_{i,t} = Q_t^s s_{i,t+1} - Q_t^s s_{i,t} \tag{18}$$

The household's problem yields two optimality conditions. The Euler equation for bonds:

$$U'(c_{i,t}) = \beta R_t^b \mathbb{E}_t\{U'(c_{i,t+1})\} \tag{19}$$

and the Euler equation for stocks:

$$U'(c_{i,t}) = \beta \mathbb{E}_t\{R_{t+1}^s U'(c_{i,t+1})\} \tag{20}$$

Combining these two conditions gives the no-arbitrage condition:

$$0 = \mathbb{E}_t\{(R_{t+1}^s - R_t^b) U'(c_{i,t+1})\} \tag{21}$$

Equation (19) governs the consumption-saving decision that marginal utilities balance inter-temporally in the optimum, while equation (21) governs the portfolio allocation where expected marginal utilities balance across assets intra-temporally. We define the portfolio risky share as $\theta_{i,t+1} \equiv Q_t^s s_{i,t+1}/a_{i,t+1}$. We study this problem in partial equilibrium: the bond rate R_t^b, the distribution of stock returns R_{t+1}^s, the wage w_t, and the dividend d_t are all taken as exogenous by households. This allows us to isolate the consumption and portfolio choice mechanisms without specifying the production side of the economy or imposing market clearing. The monetary policy shock enters as an exogenous change in R_t^b, which in turn affects stock returns R_{t+1}^s and labor income $y_{i,t}$ through their dependence on the aggregate economy.

The optimality conditions (19) and (21) must hold for all realizations of shocks and at all points in time. We start from them and study first-order responses by perturbation around the deterministic steady state in two independent dimensions. First, we expand to first order in the monetary policy shock dR^b to characterize the impulse responses of consumption and portfolio choice following the method in Zhang (2026b). These first-order responses depend only on the deterministic steady-state consumption policy and its derivatives. Second, we expand to second order in the standard deviation σ of the aggregate productivity shock and represent it by the first-order perturbation terms, to

determine the optimal risky share $\theta^*(a)$ following the latest portfolio choice literature (e.g., Bhandari et al., 2023; Auclert et al., 2024b). This pins down the portfolio composition without higher-order perturbation of the whole equilibrium.

5.2 Portfolio Composition in the Steady State

We first characterize the optimal portfolio risky share in the steady state following Auclert et al. (2024b). In the deterministic steady state (DSS), the no-arbitrage condition implies $R^s_t = R^b_{t-1}$: bonds and stocks must offer identical returns, and the portfolio composition is indeterminate. Only the consumption-saving decision is pinned down by the Euler equation. To determine the optimal portfolio risky shares, we must account for aggregate risk.

The absence of portfolio adjustment frictions, together with the transitory nature of the aggregate risk, is what makes this problem tractable. With no adjustment costs, households rebalance freely every period, so the portfolio chosen at date t looks only one period ahead: it depends on the joint distribution of returns and income between t and $t+1$ and on the value of the wealth carried forward. And because the aggregate shock is transitory, once the single period of aggregate risk has passed, the household has nothing further to forecast in its inputs: life from $t+2$ onward is the steady state. The infinite-horizon portfolio problem thus collapses to a static, one-period-ahead problem. We expand the Euler Equation (21) to second order in σ to pin down the optimal risky share. First, the steady state.

Proposition 2 (Optimal Steady State Portfolio Share)**.** *Assume the labor income response is independent of idiosyncratic productivity for simplicity. Up to the second order in aggregate risk, the optimal portfolio choice of a household i with portfolio size $a_{i,t+1}$ is given by:*

$$\begin{aligned}\theta(a_{i,t+1}) = &\underbrace{\frac{1}{a_{i,t+1}}\frac{\mathbb{E}\{U'(c_{i,t+1})\}}{\mathbb{E}\{-U''(c_{i,t+1})\cdot MPC_{i,t+1}\}}\cdot\lambda_z}_{\textit{Myopic Demand}} \\ &\underbrace{-\frac{1}{a_{i,t+1}}\frac{\mathbb{E}\left\{\frac{dy_{i,t+1}}{d\varepsilon_{z,t+1}}\right\}}{\frac{dR^s_{t+1}}{d\varepsilon_{z,t+1}}}}_{\textit{Income Hedging Demand}} \underbrace{+\frac{1}{a_{i,t+1}}\frac{\xi_i}{\mathbb{E}\{-U''(c_{i,t+1})\cdot MPC_{i,t+1}\}\cdot\frac{dR^s_{t+1}}{d\varepsilon_{z,t+1}}}}_{\textit{Disaster Hedging Demand}}\end{aligned} \tag{22}$$

where $\lambda_z \equiv \frac{1}{2}\mathbb{E}\left\{\frac{d^2R^s_{t+1}}{d\varepsilon^2_{z,t+1}}\right\}/\mathbb{E}^2\left\{\frac{dR^s_{t+1}}{d\varepsilon_{z,t+1}}\right\} > 0$ *is the unit reward (price) of bearing risk, which is common to all agents and represents the return-risk tradeoff of stocks.* $\xi_i \equiv \sum_{k\geq 1}\int \frac{\partial\,\mathbb{E}_t\{U'(c_{i,t+1})\}}{\partial P_{t+k}(e)}\frac{dP_{t+k}(e)}{d\varepsilon_{z,t+1}}de < 0$ *captures the countercyclical income risk, as the response of expected marginal utility to the aggregate shock through the shift in the future path of idiosyncratic productivity probability distribution* $P_{i,t+k}(e)$*. First-order sufficient*

statistic $MPC_{i,t+1} \equiv \frac{\partial c_{i,t+1}}{\partial y_{i,t+1}}$ *is the Marginal Propensity to Consume, and the expectations are taken over the idiosyncratic shock conditional on the deterministic steady-state path.*

Proof. See Appendix D.2. □

Equation (22) has a familiar shape: a Mertonian-style mean-variance demand for stocks, corrected by two hedging motives that relate to the labor income process. All three terms are expressed per unit of wealth, so the equation speaks to portfolio composition, the share of wealth in equity, rather than the absolute stock value.

Myopic demand. The first term is what the household would hold if it cared only about the one-period risk-return tradeoff. It multiplies the price of risk λ_z, the reward per unit of aggregate risk that is common to every household, by the inverse of household's effective risk aversion $\frac{\mathbb{E}\{U'(c_{i,t+1})\}}{\mathbb{E}\{-U''(c_{i,t+1})\cdot MPC_{i,t+1}\}}$: expected marginal utility over its first-order sensitivity to a marginal unit of expected capital income.[3] The ratio measures how hard expected marginal utility is to move by carrying an additional unit of aggregate risk: the harder it is, the larger the portfolio risky share a given price of risk λ_z can support.[4] The term is positive. Because the inverse of effective risk aversion scales with consumption and the inverse of MPC under CRRA without preference heterogeneity, the factor $1/a_{i,t+1}$ largely offsets it, and the myopic share is close to scale-invariant as in the familiar Merton benchmark in which portfolio composition is independent of the level of wealth. However, if wealthier households are effectively less risk averse, the myopic share could rise with wealth.

The other two terms are *hedging demands*: a household values assets that pay off in recessions when it expects its labor income to fall or the risk of a large income loss to rise, whereas stocks actually do the opposite, falling in bad aggregate states. A background risk that strikes in those same states thus makes equity a poor hedge, and the household holds less of it.

Income hedging demand. The second term is the standard labor-income channel of Viceira (2001). The expectation of labor income falls with the market, so equity and human capital load on the same shocks, and the household trims equity to avoid doubling up: an offset that fades with wealth as income risk becomes a smaller share of resources. Since labor income and capital income both enter marginal utility through household's budget, the income hedging depends directly on the relative strength of expected labor

[3] When there is no idiosyncratic income risk and we perturb around the deterministic steady state, the inverse of wealth-normalized effective risk aversion, $\frac{1}{a_{i,t+1}}\frac{U'(c_{i,t+1})}{-U''(c_{i,t+1})\cdot MPC_{i,t+1}}$, is approximately $\frac{1}{\gamma}$ (equality holds exactly in the continuous-time limit), the inverse of relative risk aversion.

[4] Multiply both the numerator and denominator of the price of risk by the aggregate risk variance σ_z^2, we have $\lambda_z = \frac{\mu_{t+1}}{\sigma^2_{s,t+1}}$, where $\mu_{t+1} = \frac{\sigma_z^2}{2} E\left\{\frac{d^2 R^s_{t+1}}{d\varepsilon^2_{z,t+1}}\right\}$ is the risk premium and $\sigma^2_{s,t+1} = \mathbb{E}^2\left\{\frac{dR^s_{t+1}}{d\varepsilon_{z,t+1}}\right\}\sigma_z^2$ is the variance of stock return. It thus shares the classic mean-variance evaluation of risk.

income response and stock return response to the aggregate shock. The wealth gradient of the income hedging term is negative: a fixed amount of labor-income risk is a shrinking fraction of a larger portfolio, so its drag on the risky share weakens as households grow richer, pushing θ up.

Disaster hedging demand. The last term, a feature absent from textbook theory, is the tail-risk counterpart of income hedging: because idiosyncratic disaster risk is countercyclical, a large income loss becomes more likely exactly when stocks fall and induces investors' precautionary motive, so equity again fails to insure. We capture this parsimoniously by allowing the transition probabilities $P(e_{i,t+1})$ over idiosyncratic labor efficiency states to depend on the aggregate output. A positive aggregate shock in the background creates positive output gap and reduces the probability of transitioning to a low-e' state, so $\frac{dP_{i,t+1}(e)}{d\varepsilon_z}$ shifts mass away from the left tail and lowers expected marginal utility. Formally $\xi_i < 0$, and the term is negative, just like income hedging.

Countercyclical idiosyncratic disaster risk makes the steady-state portfolio risky share realistic. With CRRA preferences and a plausible degree of risk aversion, the myopic and income-hedging demands alone imply risky shares far above those observed in the data, since safe labor income acts as an implicit bond (Cocco et al., 2005), the household-side counterpart of the equity premium puzzle. Left-skewed, countercyclical income risk closes much of this gap. In asset pricing, models in which households face countercyclical rare idiosyncratic disasters, disciplined by administrative earnings data, match the level and dynamics of the equity premium without implausible risk aversion or equity return variance (Constantinides and Ghosh, 2017; Schmidt, 2025). On the household side, the same risk lowers the optimal equity share in a life-cycle model (Catherine, 2022), and heterogeneous exposure to it explains much of the cross-sectional variation in stock market shares across Swedish households (Catherine et al., 2024). The disaster-hedging term thus matters for the level of the risky share as well as for its response to monetary policy shocks.

For the steady-state portfolio, we let the wealth gradient of income hedging demand dominate that implied by disaster-hedging. The two hedging demands differ in scale. Income hedging operates on the entire labor-income stream, in every state and every period, and its weight in the risky share is governed by the ratio of human to financial wealth, which falls steeply as financial wealth grows. Disaster hedging operates only through a rare event: its contribution is of the order of the disaster probability times its shift with aggregate conditions, both small. Whatever its variation across households, it is therefore second-order relative to the attenuation of income hedging. We discuss the wealth gradient of disaster hedging in more detail later, when we study the impulse response to a monetary policy shock.

Therefore, assuming the gradient of disaster hedging is weaker than the attenuation of the income hedging, the three demands combined suggest that the steady-state risky

share is increasing in wealth,

$$\frac{\partial \theta_{i,t+1}}{\partial a_{i,t+1}} \geq 0. \tag{23}$$

The disaster channel shapes the steady-state portfolio, and its central role in what follows is through the *impulse response* to the monetary shock, where the relevant object is the shift in expected marginal utility the contraction induces.

5.3 A Theory of Reaching for Yield

We now study how portfolio choice responds to an unexpected monetary policy shock. The monetary authority raises the bond rate R_t^b by dR^b as a one-time MIT shock. We characterize the first-order responses under a simplifying assumption on the exogenous environment: the aggregate shock $\varepsilon_{z,t+1}$ affects stock returns and labor income only at $t+1$. From $t+2$ onward, all exogenous inputs: returns, wages, and dividends, revert to their deterministic steady-state values. The same assumption applies to the monetary policy shock: the change in the bond rate dR_t^b affects exogenous inputs only at t and $t+1$, with the economy returning to the steady state thereafter. This reduces the household's problem to a two-period structure that is tractable analytically while preserving the essential risk-return tradeoff that determines portfolio composition. As a consequence, the monetary contraction begins and peaks in the same period and is followed by no easing phase: the bond rate returns to its steady-state value rather than falling below it. The model's responses therefore correspond to the impact and tightening phase of the empirical impulse responses, not to the recovery at longer horizons.

It is widely acknowledged that a fully rational investor with CRRA preferences has an optimal risky share that is not affected by the interest rate. In the textbook Merton solution:

$$\theta^* \approx \frac{\mu_{t+1}}{\gamma\, \sigma^2_{s,t+1}}, \tag{24}$$

where μ_{t+1} is the risk premium, γ is the relative risk aversion, and $\sigma^2_{s,t+1}$ is the variance of stock returns.[5] Our optimal share in Proposition 2 nests this benchmark: without labor income, idiosyncratic risk, and disaster risk, the income-hedging and disaster-hedging demands vanish, the expectations collapse, and the myopic demand reduces to the Merton share.

The risky share depends on the price of risk and risk aversion alone, so the level of interest rate is *irrelevant* and reaching for yield has no place in it. This is because a fully rational investor re-optimizes against the full non-linear effects of an interest rate change. A higher interest rate lowers her expected marginal-utility level, which on its own lowers

[5] Equation (24) is a consistent approximation to optimal portfolio risky share, as there is no closed form solution to the discrete-time model. It holds exactly only in the corresponding continuous-time limit.

the value she places on the equity premium. However, the same rate change also revises the sensitivity of her marginal utility to aggregate risk, and under CRRA the second adjustment offsets the first. Just as in first-order perturbation analysis in macroeconomics, the two effects of the shock differ in their stakes for the household. The change in her resources moves consumption and utility at first order. The revision of her sensitivity to aggregate risk moves only higher-order decisions such as the portfolio share, and utility is flat in the share up to terms of order σ_z^2: at the deterministic steady state the two assets pay the same return, so the welfare cost of a mis-set share is proportional to the variance of the excess return. We therefore assume that agents are "procedurally rational" (Simon, 1976) and rebalance their portfolios to the first-order effects of the two shocks on their marginal utility, evaluating their sensitivity to aggregate risk at the pre-shock interest rate.

Assumption 1 (Certainty-Equivalent Portfolio Adjustment)**.** *Households adjust their portfolios using the first-order (certainty-equivalent) responses of expected marginal utility to both shocks $d\varepsilon_{z,t+1}$ and dR_t^b, according to the portfolio condition of Proposition 2. Specifically, for any given portfolio $(\theta_{i,t+1}, a_{i,t+1})$, the sensitivity of expected marginal utility to the aggregate shock does not vary with the monetary shock:*

$$\left.\frac{d^2\,\mathbb{E}_t\{U'(c_{i,t+1})\}}{dR_t^b\,d\varepsilon_{z,t+1}}\right|_{\theta_{i,t+1},\,a_{i,t+1}} = 0 \tag{25}$$

evaluated at the deterministic steady state.

In Assumption 1, the sensitivity in the portfolio condition is kept, with its dependence on the portfolio, and evaluated at the pre-shock interest rate rather than revised. Under CRRA that revision is the second-order interaction of the two shocks, and its stakes for welfare are smaller by a factor of order σ_z^2 than those of the resource change it retains, because it bears only on the portfolio margin while the resource change also bears on consumption. This is the sense in which the portfolio adjustment is certainty-equivalent: its inputs are the first-order responses to each shock that certainty equivalence delivers in the perturbation of dynamic models (Boppart et al., 2018; Auclert et al., 2021), the same objects that characterize the household's consumption response in Proposition 4, so that the portfolio adjustment and the rest of the household's behavior are described by one representation.

It is worth noting that *Certainty Equivalent* in our assumption is used in the sense of the perturbation literature, where first-order responses are computed without re-evaluating the risk structure, rather than the sense of a certainty-equivalent return. Its precise content is the no-interaction restriction (25): the monetary shock is not allowed to revise the household's sensitivity to aggregate risk, the cross-effect of the two shocks. And it applies to the adjustment of the risky share, not to its steady-state level, which is pinned down

by aggregate risk at second order and is not certainty-equivalent. At the deterministic steady state the household is fully rational, and the simplification bears only on how she responds to a departure from it; this is also the structure of the sparse max, in which the agent is rational at her default and economizes on attention to deviations from it.

Conceptually, our assumption is in line with the behavioral economic literature on bounded rationality (Gabaix, 2014, 2023), making the portfolio response to the monetary shock a corner solution of a sparse-max problem: agents operate on a simplified, sparse model in which they pay attention to the effect of the shock on their resources (which they must track for consumption in any case), and leave their sensitivity to aggregate risk at its default.[6]

Proposition 3 (Portfolio Share Response to Monetary Policy Shock)**.** *Assume the labor income response is independent of idiosyncratic productivity for simplicity.*[7] *Under Assumption 1, to first order in the monetary policy shock dR_t^b, the portfolio risky share responds as*

$$\begin{aligned}\frac{d\theta_{i,t+1}}{dR_t^b} = &\underbrace{\frac{\partial\theta_{i,t+1}}{\partial a_{i,t+1}}\cdot\frac{da_{i,t+1}}{dR_t^b}}_{\text{wealth effect}} + \underbrace{(-1)\cdot\lambda_z}_{\text{reach for yield}} + \underbrace{\frac{\partial\theta_{i,t+1}}{\partial\Lambda_z}\cdot\frac{d\Lambda_z}{dR_t^b}}_{\text{belief on price of risk}} \\ &+ \underbrace{\frac{\lambda_z}{a_{i,t+1}}\mathbb{E}\left\{-\frac{dy_{i,t+1}}{dR_t^b}\right\}}_{\text{income effect}} + \underbrace{\frac{\lambda_z}{a_{i,t+1}}\cdot\frac{\psi_i}{E\{-U''(c_{i,t+1})\cdot MPC_{i,t+1}\}}}_{\text{precautionary motive}}\end{aligned} \tag{26}$$

where $\Lambda_z = \lambda_z \cdot E\{dR^s_{t+1}/d\varepsilon_{z,t+1}\} = \frac{1}{2}\frac{E\{d^2R^s_{t+1}/d\varepsilon^2_{z,t+1}\}}{E\{dR^s_{t+1}/d\varepsilon_{z,t+1}\}} > 0$ is the price of aggregate shock (return-risk tradeoff) expressed as per unit of exposure to the aggregate shock and is common to all assets and households, $\psi_i \equiv \left(\sum_{k\geq 1}\int \frac{\partial\,\mathbb{E}_t\{U'(c_{i,t+1})\}}{\partial P_{t+k}(e)}\frac{\partial P_{t+k}(e)}{\partial Y_{t+1}}de\right)\cdot\frac{dY_{t+1}}{dR_t^b} > 0$ captures the shift in expected marginal utility through the change in future path of idiosyncratic productivity probability distribution induced by the monetary shock, and all sensitivities are evaluated at the deterministic steady state.

Proof. See Appendix D.4. □

To read Proposition 3, recall that the optimal risky share, conditional on portfolio size $a_{i,t+1}$, tracks the inverse of household's effective risk aversion $\frac{E\{U'\}}{E\{-U''\cdot MPC\}}$, scaled by the price of risk. Under our certainty equivalent assumption, the sensitivity of expected marginal utility to income $E\{-U''\cdot MPC\}$ is fixed, so the risky share response is driven entirely by change in the belief on price of risk,[8] and the expected marginal utility $E\{U'\}$,

[6] We show in Appendix D.3 how our Assumption 1 can be derived as the corner solution of a sparse-max optimization problem in Gabaix (2014).

[7] We relax this assumption in the quantitative exercise later, it does not change the results.

[8] Because agents fix their sensitivity to the aggregate risk, the anticipated change in Λ_z is decided without fully rational expectations and it thus does not necessarily coincide with change in the risk premium priced by the stock market.

which measures how much the household values the reward for bearing risk. Apart from the wealth effect that works through portfolio size and belief on price of risk that works through the attractiveness of stocks, each channel in (26) moves the risky share by moving $E\{U'\}$. These channels share one logic: a household made safely better off, either richer in secure resources or further from an income disaster, values the reward for bearing risk less, and gives up risk in exchange, so it holds a smaller risky share; when it is made worse off, it values that reward more, takes on risk, and the risky share rises. They differ in how they move $E\{U'\}$, and in how they vary across the wealth distribution.

Wealth effect. Two of the five terms fall outside this logic. The wealth effect, $\frac{\partial \theta_{i,t+1}}{\partial a_{i,t+1}} \frac{da_{i,t+1}}{dR_t^b}$, works through portfolio size rather than risk appetite: the shock changes the household's wealth, and its risky share moves along the steady-state policy mechanically, which is increasing in wealth ($\partial \theta_{i,t+1}/\partial a_{i,t+1} \geq 0$). Its sign therefore turns on how wealth responds, $\frac{da_{i,t+1}}{dR_t^b}$, which combines net saving and capital gains and is characterized in Section 5.6. However, the wealth effect is not the main driver of the decline in risky share we documented, which is observed in the middle of liquidity distribution where the wealth effect is insignificant.

Belief on price of risk. When the price of risk is expected to fall, agents anticipate a lower reward for bearing risk and reduce their risky share. Survey evidence suggests that such expectations move with the business cycle: households and retail investors expect lower stock returns after prices fall and when they perceive the economy as weak (Greenwood and Shleifer, 2014; Amromin and Sharpe, 2014), and turned more pessimistic about short-run returns during the COVID-19 crash (Giglio et al., 2021b). After a monetary contraction, this channel would therefore reinforce the decline in the risky share. Moreover, portfolio allocations respond only weakly to changes in households' own return expectations (Giglio et al., 2021a), so even a sizable shift in perceived premia would pass through little to the risky share. On the other hand, Lian et al. (2019) find that reaching for yield exists in experiments that hold the price of risk fixed, with a constant equity premium and variance. We therefore shut this channel down and assume $d\Lambda_z/dR_t^b = 0$, which makes our account of reaching for yield conservative and consistent with the literature. In a companion paper, Zhang (2026a) shows in a heterogeneous-agent model that a common belief about the price of risk, determined by stock market clearing, contributes to both the magnitude and the wealth gradient of the risky share responses.

Setting both aside, we turn to the three channels that move $E\{U'\}$.

Reach for yield. After a contractionary monetary policy shock dR_t^b, the household is made safely richer by the return on its own portfolio. A higher interest rate raises the portfolio's gross return one-for-one by no-arbitrage in the first order, so conditional on

portfolio size, next period's cash-on-hand rises by exactly $a_{i,t+1}$, a safe gain that carries no risk at all and lowers the expected marginal utility. Recall that the expected marginal utility enters optimal portfolio risky share through the inverse of household's effective risk aversion, so a lower $E\{U'\}$ means the household becomes more effectively risk averse and values the reward for bearing risk less, reducing its risky share. As both the higher interest income and lower equity income enter the same budget constraint in the same period, they have the same impact on the risky share through expected marginal utility. The household thus gives up an amount of equity premium equal to the increase in interest income, with everything else in place. As a fraction of the portfolio this is $-\frac{a_{i,t+1}\, dR_t^b}{dR_t^b} \cdot \frac{1}{a_{i,t+1}} = -1$, and priced by the risk-return ratio λ_z, the risky share falls by λ_z, common to every household. This is reaching for yield: risk appetite is lower when the safe rate is high, because a higher interest rate lets the household reach the same wealth while bearing less risk.

Existing literature has turned to behavioral factors for the explanations of this behavior ruled out by the rational benchmark. Lian et al. (2019), in experiments that hold the risk premium and risk fixed, attribute it to reference dependence and salience. Under reference dependence, investors judge returns against a reference point formed from past experience; when the safe rate falls below that anchor, its yield feels like a shortfall, and they move into risky assets to make up the difference. Salience operates alongside it: with the safe rate low, the relatively high return on risky assets stands out and pulls demand toward them. Using brokerage data, Gomes et al. (2025) find reaching for yield stronger when investors trade at a loss, which they read through prospect theory: loss aversion makes investors risk-seeking in the domain of losses, so an investor sitting on a loss gambles on risky assets in the hope of breaking even. In each case, reaching for yield is a departure from standard optimization.

Our framework departs from full rationality as well, but in a different and more limited way, with three advantages. First, it predicts a response that is symmetric in the sign of the shock and independent of the level of the interest rate: a rate increase lowers the risky share by λ_z whether rates are high or low, whether the household is at a gain or a loss, and by the same amount that a rate cut raises it. These are consistent with our estimates in Section 4, and they are where the reference-dependent and prospect-theoretic accounts make different predictions. Second, while other theories explain reaching for yield by adding a new mechanism to the standard model, a reference return, a kink in the value function, or a salience distortion, ours explains it by removing one: the household adjusts her portfolio to the first-order effects of the shocks and omits a second-order interaction whose welfare stakes are an order σ_z^2 below those of the effects she keeps. This is the behavior implied by the literature on bounded rationality and rational inattention (Gabaix, 2014; Sims, 2003), in which agents rationally leave aside consequences of shocks that are not worth the cost of incorporating. Third, our assumption preserves the first-order structure that dynamic macroeconomic models already rely on. The portfolio adjustment

uses the same first-order responses and sufficient statistics that characterize consumption and saving in heterogeneous-agent models (Auclert et al., 2024b; Zhang, 2026a), so the portfolio channel of monetary transmission can be measured with the same objects as the consumption channel and embedded in a heterogeneous-agent economy efficiently. Together, these make reaching for yield a feature of standard monetary transmission rather than a phenomenon requiring a theory of its own.

Income effect. The income effect is reaching for yield in reverse. A monetary contraction lowers expected labor income, so next period's safe cash-on-hand falls: the household is safely poorer. Valuing the reward for bearing risk more, it takes on risk to make up the shortfall, raising its risky share. This prediction is in line with Campbell and Sigalov (2022), who show that an investor who must fund their spending only from portfolio income takes on more risk when the safe component of that income falls short. Although the source of the shortfall differs, the underlying economic intuition is the same: once safe resources no longer cover the household's needs, the compensation for bearing risk is worth more at the margin. The one difference from reaching for yield lies in the safe change itself. There it was the return on the entire portfolio, of size $a_{i,t+1}$, so dividing by portfolio size cancelled it and the response was common to all. Here it is the income shortfall $\mathbb{E}\{-dy_{i,t+1}/dR_t^b\}$, which does not scale with wealth; divided by $a_{i,t+1}$ and priced by λ_z, the offset $\frac{\lambda_z}{a_{i,t+1}}\mathbb{E}\{-dy_{i,t+1}/dR_t^b\}$ therefore varies across households.

Precautionary motive. The precautionary channel points the same way as the income effect but works through a different route. A monetary contraction raises the probability of a rare idiosyncratic income disaster, shifting probability toward low-consumption states and lifting expected marginal utility directly, $\psi_i > 0$, with no change in cash-on-hand. Valuing the reward for bearing risk more, the household again becomes less effectively risk averse and takes on risk, raising its risky share. This positive response is a different object from the steady-state disaster-hedging demand of Proposition 2, where the same countercyclical risk *lowered* the share. In the steady state, disaster risk enters as a return covariance: equity is a poor hedge, so the household holds less of it. In the response, the contraction's rise in disaster risk lifts expected marginal utility, and it is that level shift in $E\{U'\}$, not the hedging covariance, that moves the share.

In the last two channels, which disturbed expected marginal utility only indirectly through resources, the marginal-utility sensitivity cancelled out within the budget constraint. The precautionary channel disturbs it directly: ψ_i is a change in $E\{U'\}$ with no resource change behind it, so the household must translate it into resource terms by dividing by the sensitivity $E\{-U'' \cdot MPC\}$. The offset is $\frac{\lambda_z}{a_{i,t+1}} \frac{\psi_i}{E\{-U'' \cdot MPC\}}$, and this surviving sensitivity is what makes the precautionary channel behave so differently across wealth, as we show next.

5.4 Heterogeneity in Portfolio Share Responses

The previous section discussed the five channels while holding aside how each varies across households at different levels of wealth. That variation, the wealth gradient in portfolio share responses to monetary policy shock, is the focus in this section, and it is where we confront the theory with the data. Empirically, the impact response of the portfolio share to a contractionary shock is negative at every wealth level, and its magnitude grows with wealth but unevenly: it deepens through the middle of the distribution and then flattens, staying nearly flat across the richer half. Plotted against wealth, the response is convex. We now show that the five channels reproduce this shape, taking the wealth gradient of each in turn. Two do not bear on the gradient: reaching for yield is flat across wealth, and the belief on price of risk is assumed to be fixed. The wealth effect and the income effect both slope downward, deepening the response as wealth rises. The precautionary motive slopes upward, and it is this channel, strengthening with wealth, that halts the decline at the top and gives the profile its convex shape.

Wealth effect. This channel slopes down for a straightforward reason: wealthier households hold more equity, so a contraction inflicts larger capital losses on them and their liquid wealth declines more significantly (Figure 8). Since the optimal risky share rises with wealth ($\partial\theta_{i,t+1}/\partial a_{i,t+1} > 0$), this larger wealth loss produces a larger fall in the risky share, deepening the response toward the top. We derive and discuss separately the net-saving and capital-gains responses, which sum up to the wealth response $da_{i,t+1}/dR_t^b$, in Section 5.6.

Income effect. Expected labor income loss raises the risky share as well, but its magnitude fades with wealth and so also slopes down. It runs through the income response relative to wealth: because wealth is far more unequal than income responses, the shortfall $\mathbb{E}\{-dy_{i,t+1}/dR_t^b\}$ is a shrinking share of liquidity up the distribution, so the offset $\frac{\lambda_z}{a_{i,t+1}}\mathbb{E}\{-dy_{i,t+1}/dR_t^b\}$ is large for the poor, whose income response is big relative to their liquid wealth, and negligible for the rich.

Precautionary motive. The last channel runs the other way, and it is the decisive one: precautionary motive slopes *up*, and it is why the response flattens at the top rather than deepening without end. The precautionary offset, $\frac{\lambda_z}{a_{i,t+1}}\frac{\psi_i}{E\{-U''\cdot MPC\}}$, grows with wealth and depends on both the fear of disaster and the size of the portfolio move that fear requires.

The disturbance itself, ψ_i, changes only modestly across wealth because two forces offset: the wealthy sit on the flatter area of their marginal-utility curve, so a rise in disaster risk moves their expected marginal utility less, but rare disasters are more cyclical and more persistent for wealthier households. Greater cyclicality means a given aggregate

shock moves their disaster probability by more, raising $|\psi_i|$; greater persistence means a disaster, once it strikes, removes a larger share of lifetime resources, raising the stakes of that exposure. Both features are documented in administrative earnings data. The incomes of high-income households are markedly more exposed to aggregate fluctuations (Parker and Vissing-Jorgensen, 2009, 2010), earnings losses in recessions are largest at the top of the distribution (Guvenen et al., 2014a), and exposure to aggregate risk peaks among top earners (Guvenen et al., 2017). Persistence follows the same gradient: in US administrative data, only 20 to 25 percent of a large earnings decline is reversed within ten years for workers in the top percentile of recent earnings, against about 75 percent at the bottom (Guvenen et al., 2021), and Norwegian register data show the same asymmetry (Halvorsen et al., 2024). The intuition is a job ladder: a high-income earner who loses a job falls from a high rung that is hard to climb back to, much as a dismissed chief executive rarely finds a comparable position soon, whereas a low-income earner's loss is typically a short spell out of work followed by a return to a similar job. The wealthy are therefore more exposed to aggregate shocks, and their disaster risk is more persistent, which compensates for the insensitivity of their expected marginal utility to a given monetary contraction.[9]

The main driver of the gradient of precautionary motive lies in the denominator, the sensitivity $E\{-U'' \cdot MPC\}$, which differs sharply across the wealth distribution and it can be interpreted as how much a marginal unit of reward moves expected marginal utility. It is large for the poor whose marginal utility is steep, and small for the wealthy whose marginal utility is flat. To offset the same disturbance, then, the poor need only a small trade, while the wealthy need a large one. The formula makes this precise. Because the price of risk λ_z is common to all households, the wealth gradient of the offset is governed by $\frac{\psi_i}{a_{i,t+1}\, E\{-U'' \cdot MPC\}}$: the disturbance ψ_i over the sensitivity $E\{-U'' \cdot MPC\}$, scaled by portfolio size. The disturbance changes only modestly across wealth, for the reason just given. The denominator falls far more steeply: under CRRA, marginal utility flattens while MPC falls as consumption rises, so a marginal unit of reward moves it less and less, and even after scaling by portfolio size the denominator keeps falling. A numerator that changes modestly over a denominator that falls sharply leaves an offset that rises with wealth.

Taken together, the convex shape and flat slope at the top of distribution that we documented in portfolio risky share responses are the balance of two opposing tendencies. Through the middle of the distribution, the wealth effect deepens the portfolio risky share reduction directly, as capital losses grow with wealth, while the income effect lifts it but by less and less, its positive offset fading. Toward the top, the precautionary motive has

[9]Rather than deriving the resulting pattern, we impose it directly here for simplicity. In a companion paper, Zhang (2026a) show that the same pattern emerges endogenously in a heterogeneous-agent model with idiosyncratic disaster risk calibrated to labor-market data.

an opposite wealth gradient. Its force grows with wealth for the reason at the heart of this section: the wealthy fear disaster no more than others, but they are so rich and their marginal utility is so insensitive, that answering the same increase in the tail risk demands much more reward from bearing risk, and so a far larger increase in their risky share. That rising force bends the risky-share response into the convex, flattening profile the data show, and, through the accounting identity, it is the source of flow reversal at the top.

5.5 Simulation of Risky Share Responses

We simulate our stylized model and other counterfactual models to visualize the theory. Figure 13 plots the results. For comparability across models, we assume aggregate risk does not affect the probability distribution of idiosyncratic rare disaster, $\xi_i = 0$, so that all models in our simulation share the same steady state. For the purpose of demonstrating the theoretical mechanism, we do not strictly pursue the specific magnitude of the results.

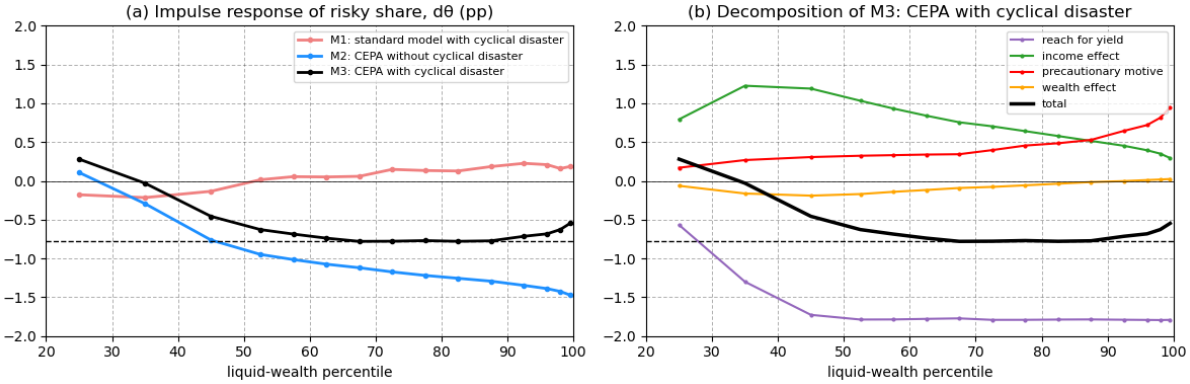


Figure 13: Participants' Risky Share Response across the Liquid Wealth Distribution

Notes: This figure reports simulated risky share responses $d\theta$ to a 1pp contractionary monetary policy shock, of stock market participants across the wealth distribution. Panel (a) compares the results of three models: M1 is the canonical model in which agents are fully rational and the rare disaster is counter-cyclical. Agents in M2 model are subject to our "Certainty Equivalent portfolio Adjustment" (CEPA) Assumption 1 but the rare disaster is acyclical. M3 model also features CEPA agents and the rare disaster is counter-cyclical as well. Panel (b) decomposes the transmission channels of model M3 according to Proposition 3. We show details of the model and calibration strategy in Appendix D.5.

Panel (a) shows risky share responses $\frac{d\theta_{i,t+1}}{dR_t^b}$ across the distribution in three models that differ in two aspects. First, the comparison between standard model (M1) in which agents are fully rational and consider how their sensitivity to risk is affected by the monetary policy shock, and our benchmark model (M3) in which agents do not update how their sensitivity to risk changes, highlights the effect of our "Certainty Equivalent Portfolio Adjustment" (CEPA) assumption: the canonical portfolio choice model does not predict

reaching for yield, while models with CEPA assumption do. Second, the counterfactual model (M2) differs from our model (M3) only in the cyclicality of the probability of rare disaster. After a monetary tightening, our model (M3) generates a general reduction in risky share, with a convex gradient that flattens from p65-p85 and slopes up at the top. This is consistent with our evidence, especially the one conditional on stock market participation. However, the counterfactual model where individual rare disasters are independent of aggregate states, although it delivers reaching for yield with the same CEPA assumption, predicts instead a strictly negative gradient of risky share responses that is even slightly concave at the top, which directly contradicts the data.

Panel (b) of Figure 13 decomposes the risky share responses in the benchmark model (M3) as a quantitative counterpart of Proposition 3, to further investigate channels through which the gradient is decided. There are four channels (the belief channel being shut down) and counterfactual model (M2) shares three of them: reaching for yield channel from higher interest income is flat across the distribution because it is proportional to portfolio size; income effect is decreasing in wealth because labor income accounts for a small fraction of wealth for the rich agents at the top; and the wealth effect does feature a positive gradient, but the magnitude makes it negligible because net saving offset capital loss.[10] Therefore, the main driver of a flat slope must bring a strong and positive gradient. It turns out to be the last channel, precautionary motive from a larger likelihood of falling into rare disaster. More importantly, in our stylized partial equilibrium model, this is the only channel that is able to activate a positive gradient of risky share responses at the top of distribution in a monetary contraction when portfolio size does not change significantly, which we documented at horizon 3 and 4 in Figure 6. As we explained earlier, the effect of cyclical rare disaster grows with wealth for two reasons, both supported by evidence: first, the tail risk of high-income earners is more exposed to aggregate fluctuations, and their tail risk is more persistent; second, the wealthy are far less sensitive to marginal resource changes and the exchange rate of marginal utility to portfolio risky share is much larger for them. The precautionary motive channel thus reverses the gradient of risky share responses, as we move from the acyclical counterfactual model (M2) to the benchmark model (M3) with countercyclical tail income risk.

5.6 Responses of Consumption, Saving, and Capital Gains

The previous section derived the portfolio share response, how households actively rebalance after the shock. Rebalancing is only one term in the accounting identity behind the flow: a household's risky holdings also change as it saves out of income and as the assets it already holds gain or lose value. This section supplies those other terms. We begin with

[10] In the two-period model net saving response is stronger than the impact responses we document. As the contraction deepens, the empirical evidence on wealth effect at horizon 1 and 2 exhibits the similar positive gradient.

the consumption response, from which net saving follows, and then combine net saving with capital gains into the response of total liquid wealth. That last object does double duty: it is the input $da_{i,t+1}/dR_t^b$ that the wealth effect of the previous section left to be derived here. Two results carry over to the flow. Capital losses grow steeply with wealth, as richer households hold more equity and absorb larger valuation losses; net saving is, to a large extent, muted at the top, because the higher return on bonds carried into the next period is not yet received, so the implied flow in our simple exercise is shaped mainly by capital gains and risky share.

Proposition 4 (Consumption Response)**.** *Assume non-financial income responses* $\frac{dy_{i,t+1}}{dR_t^b}$ *are independent of idiosyncratic productivity. Up to the first order, the perturbation of the Euler equation* (19) *with respect to* dR_t^b*, given the portfolio* $\theta_{i,t}$*,* $a_{i,t}$ *and* $a_{i,t+1}$ *in the steady state, yields the consumption response:*

$$\begin{aligned}\frac{dc_{i,t}}{dR_t^b} =& MPC_{i,t}\cdot\left[\theta_{i,t}a_{i,t}\frac{dR_t^s}{dR_t^b}+\frac{dy_{i,t}}{dR_t^b}+\frac{1}{R_t^b}\left(a_{i,t+1}+E\left\{\frac{dy_{i,t+1}}{dR_t^b}\right\}\right)\right]\\ &-MPS_{i,t}\cdot\frac{c_{i,t}}{\gamma R_t^b}-MPS_{i,t}\cdot\frac{\beta R_t^b}{\gamma}c_{i,t}^{\gamma+1}\psi_i\end{aligned} \tag{27}$$

where $MPS_{i,t}\equiv 1-MPC_{i,t}$ *denotes the marginal propensity to save, and* $\psi_i \equiv \left(\sum_{k\geq 1}\int \frac{\partial \mathbb{E}_t\{U'(c_{i,t+1})\}}{\partial P_{t+k}(e)}\frac{\partial P_{t+k}(e)}{\partial Y_{t+1}}de\right)\cdot\frac{dY_{t+1}}{dR_t^b} > 0$ *captures the response of expected marginal utility to the monetary policy shock through the shift in idiosyncratic disaster probability.*

Proof. We perturb the Euler equation for bonds with respect to dR_t^b using the method in Zhang (2026b), which introduces marginal propensities as sufficient statistics for the impulse response. The complete proof is in Appendix D.6. □

The effect of the monetary policy shock on consumption in equation (27) can be categorized into four groups. First, consumption reacts to changes in current budget $(\theta_{i,t}a_{i,t}\frac{dR_t^s}{dR_t^b}+\frac{dy_{i,t}}{dR_t^b})$ due to financial and non-financial income changes, as well as the wealth effect as stock price fluctuates. Second, it also responds to discounted future budget $\frac{1}{R_t^b}(a_{i,t+1}+E\{\frac{dy_{i,t+1}}{dR_t^b}\})$, the former of which the literature refers to as is always referred to as the Unhedged Interest Rate Exposure from Auclert (2019). Third, there is the intertemporal substitution effect: a higher bond rate induces households to postpone consumption, with the strength of this effect proportional to the EIS γ^{-1} and the level of current consumption. Fourth, the disaster term $-\frac{\beta R_t^b}{\gamma}c_{i,t}^{\gamma+1}\psi_i$ captures the precautionary response to countercyclical tail risk. Since $\psi_i > 0$, this term is negative: the increase in disaster probability following a contractionary shock raises expected marginal utility tomorrow, inducing households to cut current consumption and build a precautionary buffer. The $c_{i,t}^{\gamma+1}$ scaling implies that this precautionary effect is larger for wealthier households with higher consumption, as the curvature of marginal utility is lower and a given shift in disaster probability translates into a larger consumption adjustment.

Net saving The net saving response follows directly from the consumption response. Since $NS_{i,t} = y_{i,t} - c_{i,t}$, normalizing by pre-shock liquidity $a_{i,t}$ and substituting equation (27):

$$\begin{aligned}\frac{1}{a_{i,t}}\frac{dNS_{i,t}}{dR_t^b} =& MPC_{i,t} \cdot \left[-\frac{1}{a_{i,t}R_t^b}\left(a_{i,t+1} + E\left\{\frac{dy_{i,t+1}}{dR_t^b}\right\}\right) - \theta_{i,t}\frac{dR_t^s}{dR_t^b}\right] \\ &+ MPS_{i,t} \cdot \frac{1}{a_{i,t}}\left(\frac{c_{i,t}}{\gamma R_t^b} + \frac{dy_{i,t}}{dR_t^b}\right) + MPS_{i,t} \cdot \frac{\beta R_t^b}{\gamma a_{i,t}} c_{i,t}^{\gamma+1}\psi_i\end{aligned} \tag{28}$$

The normalized net saving response in equation (28) is governed by three groups of terms with distinct wealth gradients. The first group, scaled by $MPC_{i,t}$, contains the relative discounted (financial and non-financial) income term in the next period, $-\frac{1}{a_{i,t}R_t^b}\left(a_{i,t+1} + E\{\frac{dy_{i,t+1}}{dR_t^b}\}\right)$, and the capital loss term $-\theta_{i,t} \cdot dR_t^s/dR_t^b$ that increase in wealth. The second group, scaled by $MPS_{i,t}/a_{i,t}$, contains the intertemporal substitution effect and the current income response, and shrinks with wealth. The third group is the disaster term $\frac{\beta R_t^b}{\gamma a_{i,t}} c_{i,t}^{\gamma+1}\psi_i$, which is positive and increases net saving as households build a precautionary buffer against rising tail risk. The overall wealth gradient depends on the interaction of these competing forces, and the model does not deliver a sharp prediction for the shape of the normalized net saving profile without precise calibration of the exogenous inputs. However, the disaster term provides an additional positive contribution to net saving that is increasing in wealth through the $c_{i,t}^{\gamma+1}$ scaling. This is consistent with the positive wealth gradient in the net saving response documented in Section 4 and in Holm et al. (2021). In their empirical framework, the positive impact response of net saving for the wealthy is attributed primarily to higher interest income on deposits. In our theoretical model, however, the higher bond return R_t^b is only earned on savings carried into the next period and does not raise income on impact. The disaster channel provides a complementary mechanism that can generate a positive net saving response on impact through precautionary motives, even before higher interest income materializes.

Total liquid wealth The relative response of total liquid wealth as the sum of changes in net savings and capital gains follows directly:

$$\begin{aligned}\frac{d(a_{i,t+1}/a_{i,t})}{dR_t^b} =& - MPC_{i,t} \cdot \frac{1}{a_{i,t}R_t^b} \cdot \left(a_{i,t+1} + E\left\{\frac{dy_{i,t+1}}{dR_t^b}\right\}\right) \\ &+ MPS_{i,t} \cdot \left(\frac{1}{a_{i,t}}\frac{c_{i,t}}{\gamma R_t^b} + \frac{1}{a_{i,t}}\frac{dy_{i,t}}{dR_t^b} + \theta_{i,t}\frac{dR_t^s}{dR_t^b}\right) + MPS_{i,t} \cdot \frac{\beta R_t^b}{\gamma a_{i,t}} c_{i,t}^{\gamma+1}\psi_i\end{aligned} \tag{29}$$

Capital gains are exogenous in partial equilibrium and mechanically increasing in the risky share: $d(CG_{i,t}/a_{i,t})/dR_t^b = \theta_{i,t} \cdot dR_t^s/dR_t^b < 0$, with wealthier households suffering larger normalized capital losses due to their higher equity exposure. The disaster term $\frac{\beta R_t^b}{\gamma a_{i,t}} c_{i,t}^{\gamma+1}\psi_i$ is positive, partially offsetting these capital losses through precautionary saving.

However, since the capital gains term is negative, increasing in magnitude with wealth, and substantially larger than both the diluted intertemporal substitution and income effects and the disaster precautionary term, the change in total liquid wealth is negative and increasingly so for wealthier households. This is consistent with the empirical evidence in Figure 8, which shows that total liquid wealth declines significantly only at the top, with the response below it indistinguishable from zero.

The wealth gradient in $da_{i,t+1}$ is the key input into the portfolio response through the first-order effect $(\partial\theta/\partial a) \cdot da$: since $\partial\theta_{i,t+1}/\partial a_{i,t+1} > 0$ and $da_{i,t+1}$ is more negative for the wealthy, this term pushes the risky share decline to be steeper at the top of the distribution.

5.7 Flow and the Wealth Gradient

We can now map the model responses back through the accounting identity to understand the wealth gradient of flow responses. Recall from accounting identity (4) that portfolio flow is decomposed into three channels: portfolio rebalancing, cash flow, and asset valuation. The model provides predictions for each.

Portfolio rebalancing The risky share response $\frac{d\theta_{i,t+1}}{dR_t^b}$ is convex in wealth by the analysis above. When scaled by the liquidity growth factor $\frac{a_{i,t+1}}{a_{i,t}}$ which on average has a negative wealth gradient and close to one for groups above the median, this term is negative and roughly constant at about -0.7 percent across the upper half of the distribution, deepening to about -0.95 percent only in the top percentile. The convexity ensures that this term does not continue to grow with wealth, leaving room for the other two channels to offset.

Net saving The net saving response $dNS_{i,t} = dy_{i,t} - dc_{i,t}$ depends on how consumption adjusts across the distribution. The model predicts that the MPC is decreasing in wealth, so consumption falls less for wealthier households in response to the income decline. Combined with the positive inter-temporal substitution effect (which scales with consumption level), the net saving response is modestly increasing in wealth. This is consistent with the empirical pattern in Figure 10.

Capital gains The capital gains response is exogenous in partial equilibrium and increases monotonically with equity exposure $\theta_{i,t}$, portfolio composition determined prior to the shock. Since the risky share is increasing in wealth, capital losses grow steeply with liquid wealth. In the accounting identity, the contribution of asset valuation is positive and increasing in wealth.

The wealth gradient. Flow is the sum of these three channels. The portfolio rebalancing channel is negative and roughly flat; The net saving channel is small in magnitude. The capital gains channel is positive and increasing. At the bottom of the upper half

of the distribution, the portfolio rebalancing term dominates and the flow is negative: these households are net sellers facing a contractionary monetary policy shock. As wealth increases, the net saving term grows while the portfolio rebalancing term remains flat and even tilts upward, gradually offsetting the negative rebalancing effect. At the very top, the cash flow channel is strong enough and the flow turns positive, making the wealthiest households net buyers. The model therefore qualitatively predicts the empirical wealth gradient through a single mechanism: the precautionary motive that rises with wealth and flattens the risky-share response at the top. Because wealthier households' precautionary motive has a stronger demand for capital income, their conditional portfolio response to the rate change is less negative, producing the convexity in $\frac{d\theta}{dR^b}$ that allows the net saving to dominate at the top.

6 Conclusion

This paper studies how household risk-taking responds to monetary policy across the liquid wealth distribution. We document two responses, linked by an accounting identity: the portfolio risky share, the household's chosen exposure to risk, and the net purchases of risky assets, its actual trading. In Norwegian administrative data, reaching for yield is universal and persistent: households at every level of liquid wealth cut their risky share on average after a monetary contraction, and hold it down until the contraction passes. The response is convex in wealth, flattening at the top. Trading diverges from risk exposure. Through the tightening phase, middle-wealth households become net sellers of risky assets while the wealthiest become net buyers. The accounting identity shows where the reversal comes from. The convexity of the risky-share response sets the shape of the flow: because portfolio rebalancing flattens at the top while capital losses keep growing, the asset valuation channel holds the flow of the wealthiest near zero while households below them sell. By the trough of the contraction, the net saving of the wealthiest builds up and turns them into net buyers. A partial-equilibrium model in which households adjust their portfolios to the first-order effects of monetary policy on their marginal utility gives reaching for yield a foundation: higher interest income makes households safely richer and raises their effective risk aversion, which reduces their risky share. With countercyclical idiosyncratic disaster risk, the same model generates the convex wealth gradient: because the marginal utility of the wealthy is less sensitive to resources, a rise in disaster risk takes a larger offsetting increase in their risky share, dampening their reaching for yield at the top.

These findings point in several directions. The accounting identity is a portable tool: it applies wherever portfolio composition, income, and returns are observed, and it decomposes the flow, which a direct estimate cannot separate and which it measures only noisily when capital gains and changes in risky wealth nearly offset. For structural

work, our results point to a feature standard portfolio models do not deliver: a risky-share response to monetary policy that is convex in wealth, flattening at the top. It is this flattening, together with capital losses that scale with equity exposure, that keeps the wealthiest from selling as the rest of the distribution does and leaves their saving to turn them into net buyers; a model without it will mischaracterize portfolio dynamics at the top of the distribution. The heterogeneity also has aggregate consequences that a partial-equilibrium analysis cannot reach. These are open questions, and we are pursuing two of them in a companion paper that places this framework in general equilibrium (Zhang, 2026a). The first, and more important, is what countercyclical disaster risk, which standard models omit, implies for aggregate transmission and the wealth distribution. The second is whether the endogenous stock trades between households amplify aggregate demand fluctuations even when net flows sum to zero.

Our analysis has limitations. The partial equilibrium framework does not endogenize asset prices. The annual frequency may attenuate short-lived responses, and the imputation of capital gains introduces measurement error. The Norwegian institutional setting may limit the external validity of the magnitudes. Even so, the flows reverse sign at the top of the wealth distribution over the tightening phase just as they do in independent brokerage evidence from a very different economy (Gomes et al., 2025), which suggests that the underlying forces are general.

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Online Appendix to

"Who Reaches for Yield?"

Marek Jasinski Hanwen Zhang

October 2026

A List of registry data sources

Source	Variables
Income and wealth from tax returns Annual, 1993–	Labor income Business income Capital income Transfers received Taxes paid Asset holdings (e.g., deposits, mutual funds, bonds, real estate) Debt (total debt)
https://www.ssb.no/en/omssb/tjenester-og-verktoy/data-til-forskning/inntekt	
Housing wealth database Annual, 1993–	Value of housing (including cabins and secondary homes) (as in Fagereng et al. (2020b))
Stockholder registry Annual, 2004–	ISIN / firm ID Owner ID Quantity owned of stock
https://www.ssb.no/383782/utlan-av-data-om-aksjonaerer-aksjeselskaper-og-allmennaksjeselskaper	
The central population register Annual, 1964–	Region of residence at the end of the year Date (month) of birth Gender, indicator variable for gender Marital status indicator variable Spousal id (unique identifier of spouse)
https://www.ssb.no/en/omssb/tjenester-og-verktoy/data-til-forskning/befolkning	
Norwegian mapping authority Event data, 1993–	Buyer/seller ID Price
https://www.ssb.no/bygg-bolig-og-eiendom/eiendom/statistikk/eiendomsomsetning	
Other public data sources:	
Consumer price index, Statistics Norway https://www.ssb.no/en/priser-og-prisindekser/statistikker/kpi	
Flow of funds, Statistics Norway https://www.ssb.no/en/nasjonalregnskap-og-konjunkturer/statistikker/finsekv	
Stock price index and general stock prices, Oslo Børs https://www.oslobors.no/ob_eng/markedsaktivitet/#/details/OBX.OSE	
House price indices, Eitrheim and Erlandsen (2004) https://www.norges-bank.no/en/topics/Statistics/Historical-monetary-statistics/	
Exchange rate data, Norges Bank https://www.norges-bank.no/en/topics/Statistics/exchange_rates/	

B Monetary Policy Shocks and Interest Rates

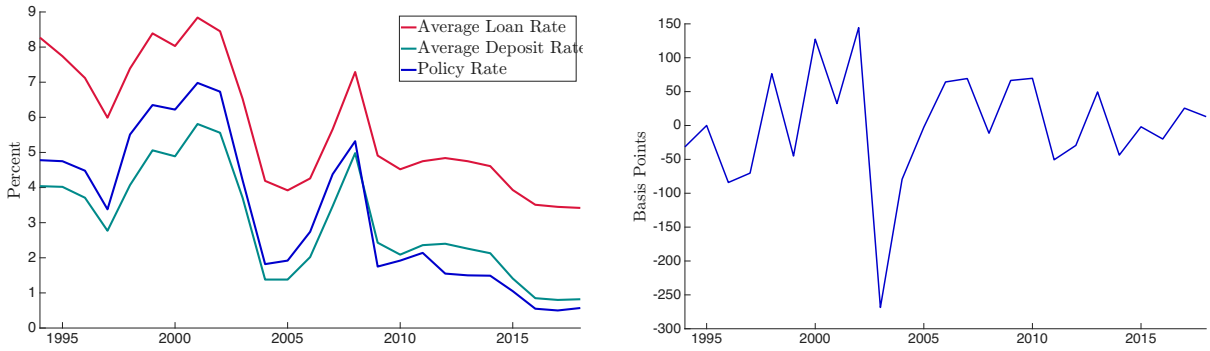


Figure B.1: Norwegian Interest Rates and Monetary Policy Shocks

Notes: Left panel: annual average loan rate, deposit rate, and policy rate in Norway, 1994–2018. Right panel: annual monetary policy shock series, constructed by aggregating monthly narrative shocks from Holm et al. (2021) within each calendar year. Units are basis points.

C Additional Estimations and Robustness

Unless noted otherwise, figures are estimated on the 1 percent flow-trimmed panel, as in the main text. Appendix C.3 to Appendix C.4 use the untrimmed panel and compare with the untrimmed baseline; the trim is defined on the flow and, as shown in Appendix C.2, moves the flow estimates little and the other outcomes not at all.

C.1 Responses of Risky Wealth across the Distribution

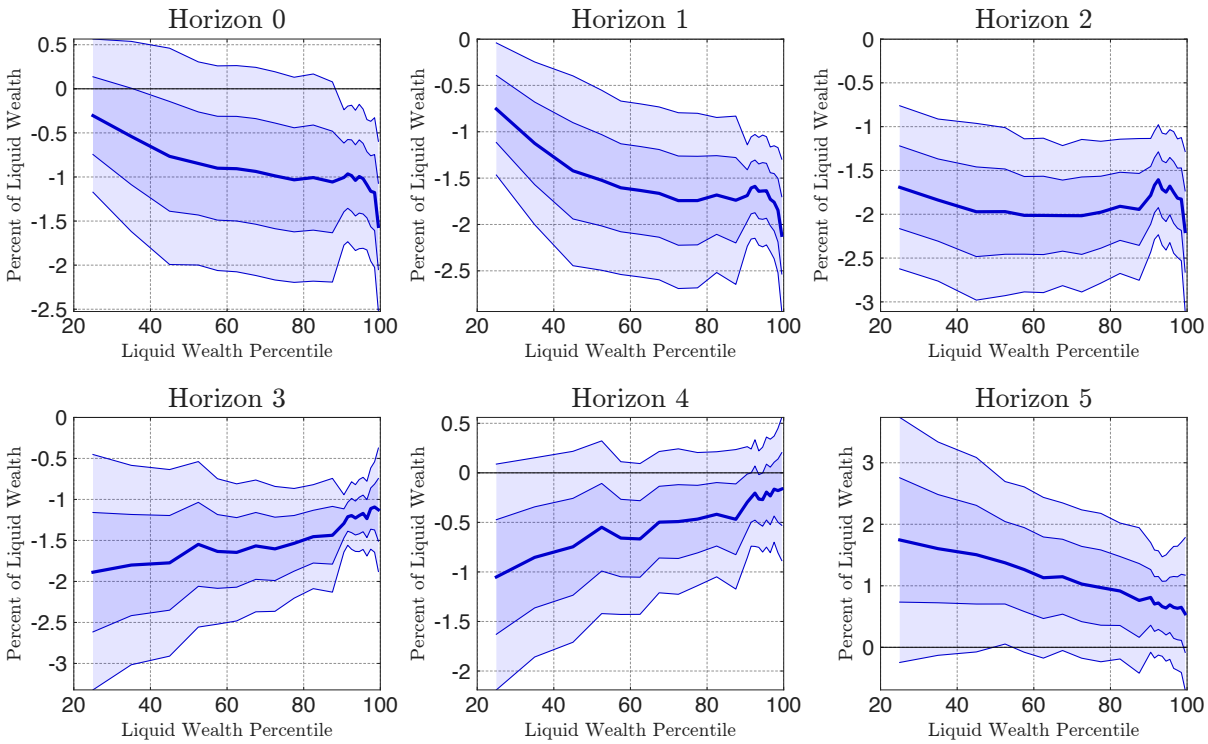


Figure C.1: Response of Risky Asset Holdings Across the Liquid Wealth Distribution

Notes: Each panel plots the impulse response of risky asset holdings, normalized by lagged liquid wealth, to a one percentage point contractionary monetary policy shock at horizon h, estimated separately for each of the liquid wealth groups. Groups below the 20th percentile are excluded. Shaded areas denote 68 and 95 percent confidence bands based on Driscoll-Kraay standard errors.

Figure C.1 plots the response of risky asset holdings, normalized by pre-shock liquidity. On impact, risky wealth falls across the entire distribution, with the magnitude increasing smoothly in liquid wealth, from -0.25% at p25 to -1.90% at the top, and deepening further by horizon 2 before recovering.

C.2 Measurement of the flow

The portfolio flow is a residual: the change in risky wealth net of imputed capital gains. This appendix reports how its estimates depend on the measurement choices behind Figures 9 and 12.

Trimming. Figure C.2 plots the direct flow estimate on the untrimmed panel and with the top and bottom 1 and 2 percent of the flow distribution removed within year and liquid wealth group. The 1 percent trim is the baseline of the main text. The point estimates are close under the three choices at every horizon, and the reversal at the top at horizon 2 is present in all. Trimming changes the estimates in two places. In the lower half of the distribution, where liquid wealth is a few thousand dollars, the untrimmed profile has jumps between neighboring groups around the 30th to 40th percentiles: a handful of household-years with large trades relative to a small liquidity base dominate a group-year and move the group coefficient. Removing the tails smooths the profile across groups and narrows the confidence bands, most in those groups. The 1 and 2 percent trims give nearly the same point estimates, so the result does not depend on how far into the tails the trim reaches; we use the lighter of the two.

Imputation of capital gains. Capital gains on directly held stocks are computed stock by stock from the Shareholder Registry from 2005 onward and imputed from the Oslo Stock Exchange index before 2005; gains on mutual funds are imputed from the aggregate fund price index throughout. To check that the stock-level gains do not drive the result, Figure C.3 re-estimates the decomposition with the index imputation applied in every year, on the same 1 percent trimmed panel as Figure 12, and Figure C.4 compares the capital gains channel under the two imputations. The gradient of capital losses across the distribution and the decomposition at the top are close under the two; the index imputation flattens the profile at the very top at horizon 0, where the wealthiest hold concentrated stock positions whose prices fell by more than the index.

C.3 Labor Income Across the Distribution

Figure C.5 plots the response of labor income across the liquid wealth distribution, normalized by pre-shock liquid wealth as the other channel responses are. Two features of the figure are the liquidity base rather than the income response. The scale of the response falls steeply with wealth because liquid wealth is a few thousand dollars for households below the median and several times annual income at the top (Figures 1 and 2): converted to income units, the responses lie within one to two percent of pre-shock labor income across the distribution, in line with the pooled income response in Figure 5. The timing is that of the real economy rather than of the policy rate. Labor income is close to zero

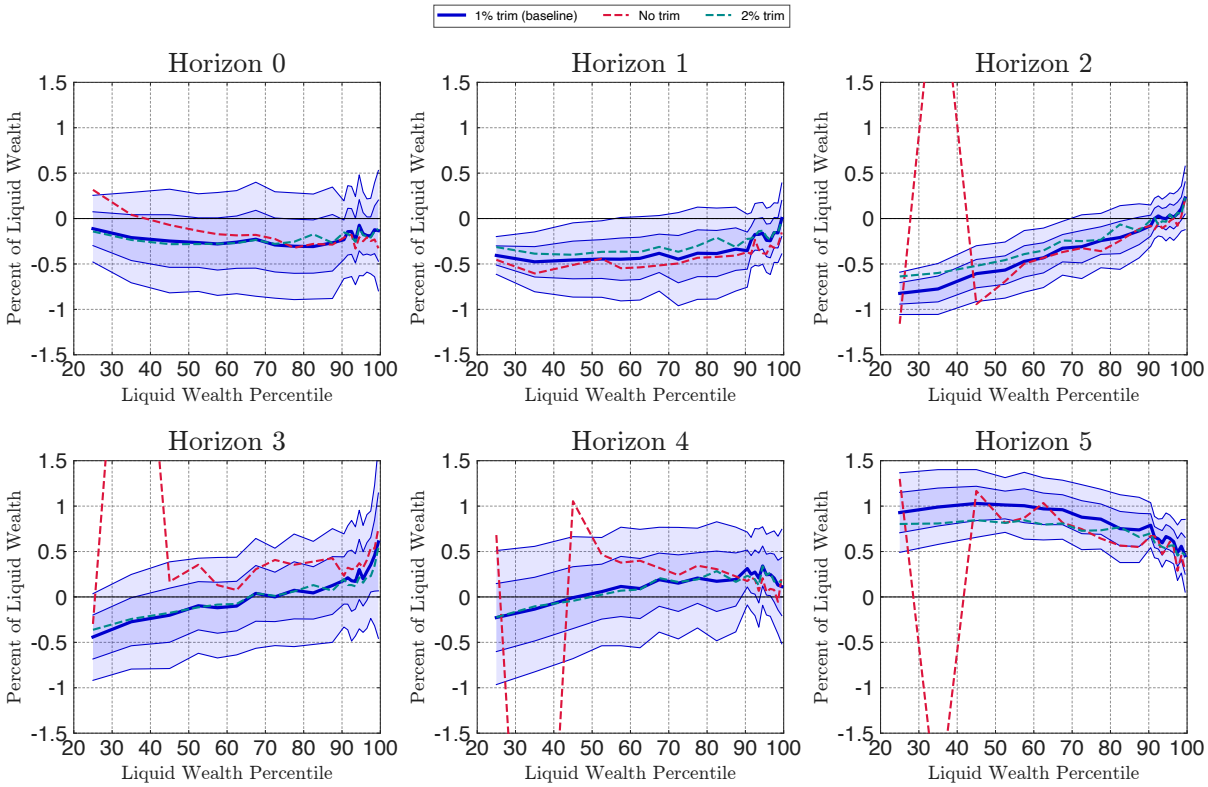


Figure C.2: Direct Flow Response Under Alternative Trimming

Notes: Each panel plots the impulse response of the direct portfolio flow, normalized by pre-shock liquid wealth, to a one percentage point contractionary monetary policy shock at horizon h, estimated separately for each liquid wealth group on the untrimmed panel (dashed), the 1 percent flow-trimmed panel (solid, baseline) and the 2 percent flow-trimmed panel (dashed). Shaded areas denote 68 and 95 percent confidence bands for the baseline, based on Driscoll-Kraay standard errors. Groups below the 20th percentile are excluded.

on impact below the top, significantly negative only at the top at horizons 1 and 2, and falls across the distribution at horizons 4 and 5, when unemployment peaks and output reaches its trough (Figure 4). The earlier decline at the top is consistent with the greater cyclicality of top earnings documented by Guvenen et al. (2014b) and Guvenen et al. (2017).

The figure is the input to the income effect of Proposition 3, $\frac{\lambda_z}{a_{i,t+1}}\mathbb{E}\{-dy_{i,t+1}/dR_t^b\}$, which is the income loss relative to the portfolio. It is close to zero on impact, which is why the impact gradient of the risky share is governed by reaching for yield and the precautionary motive rather than by the income effect, as argued in Section 5.

The income effect in the model runs through income expected one period ahead, so the responses at horizons 3 to 5 are the relevant ones for the tightening phase. There the loss relative to liquidity is largest for the least liquid households and shrinks as wealth rises, which is the wealth gradient of the income effect in Proposition 3: it offsets reaching for yield from below, and is negligible at the top.

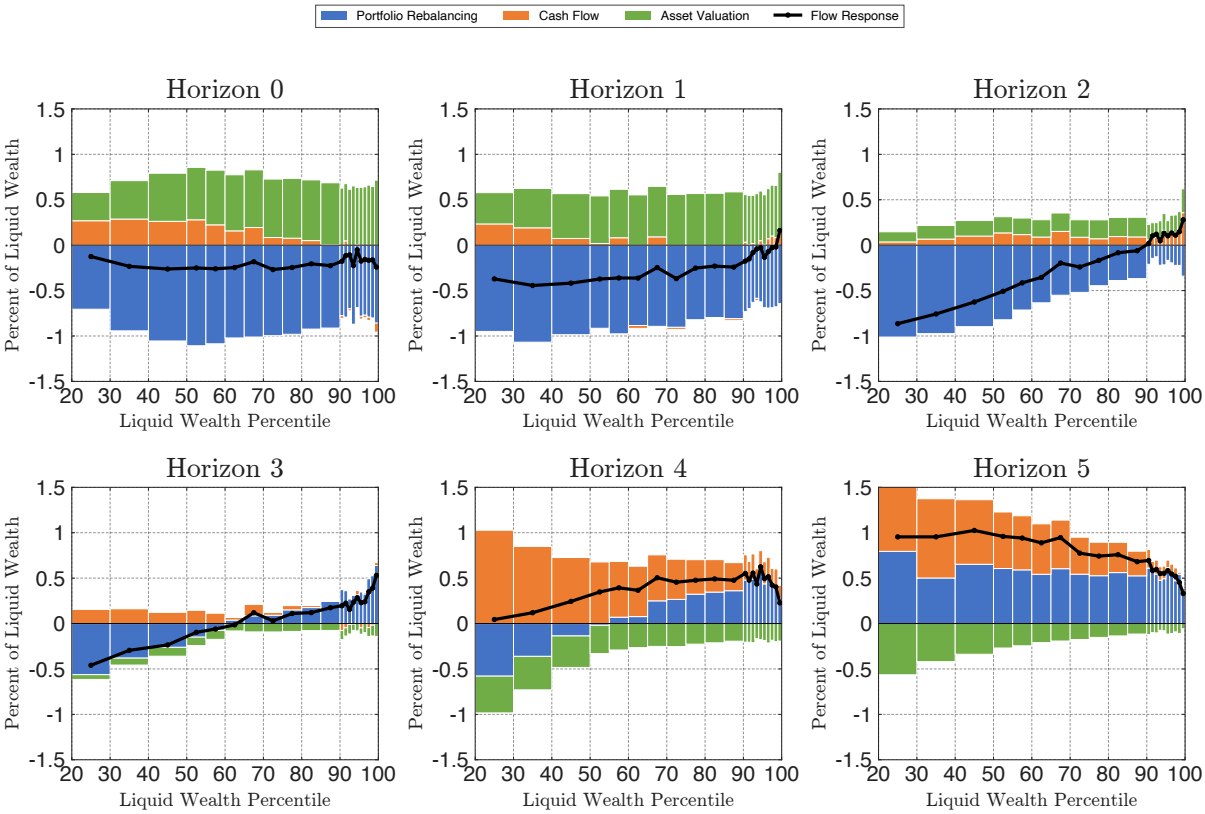


Figure C.3: Flow Decomposition with Index-Imputed Capital Gains

Notes: As Figure 12, with capital gains on directly held stocks imputed from the Oslo Stock Exchange index in every year rather than computed stock by stock from 2005. Stacked bars decompose the portfolio flow response into portfolio rebalancing (blue), cash flow (orange) and asset valuation (green); the black line plots the point estimate of the flow response. All values are in percentage points of pre-shock liquid wealth. Groups below the 20th percentile are excluded.

C.4 Participation

Section 4 studies the risky share of all households, participants and non-participants alike, and Figure 7 the share of pre-shock participants. This appendix reports the participation margin itself. Figures C.6 to C.8 plot the responses of the participation rate, of entry among pre-shock non-participants and of staying in among pre-shock participants, by liquid wealth group. Against the risky-share responses of Figure 6, of 1 to 1.5 percentage points on a base of 8 to 20 percent, the participation response is significant but small, so the unconditional risky-share response is mostly the intensive margin, which is what Figure 7 shows directly. Table C.1 reports participation, the unconditional and conditional risky share, the median debt ratio and the number of person-years by group.

Liquid wealth percentile	Participation (%)	Risky share (%)	Risky share, participants (%)	Median debt ratio	Person-years
20–30	22.7	6.6	28.9	32.26	7,412,739
30–40	29.8	8.5	28.5	17.17	7,412,825
40–50	36.2	10.1	27.8	9.54	7,412,856
50–55	40.4	10.9	27.1	6.28	3,706,431
55–60	42.9	11.4	26.6	4.65	3,706,413
60–65	45.2	11.7	25.9	3.41	3,706,433
65–70	47.5	12.1	25.5	2.43	3,706,421
70–75	49.8	12.4	24.8	1.63	3,706,429
75–80	52.0	12.6	24.2	1.00	3,706,437
80–85	54.3	12.8	23.5	0.53	3,706,438
85–90	57.4	13.1	22.8	0.21	3,706,424
90–91	59.5	13.4	22.5	0.10	741,290
91–92	60.4	13.5	22.4	0.07	741,282
92–93	61.4	13.6	22.2	0.05	741,295
93–94	62.4	13.9	22.3	0.03	741,281
94–95	63.6	14.2	22.3	0.02	741,284
95–96	64.8	14.6	22.6	0.01	741,290
96–97	66.3	15.2	23.0	0.01	741,294
97–98	67.8	16.0	23.6	0.01	741,277
98–99	70.4	17.4	24.7	0.00	741,289
99–100	77.6	23.2	29.9	0.00	741,268

Table C.1: Portfolio Descriptives by Liquid Wealth Group

Notes: Person-year weighted averages over 1993–2019 on the untrimmed panel. Participation is the share of households with positive risky asset holdings; the risky share is the ratio of risky assets to liquid wealth, averaged over all households and over participants; the debt ratio is pre-shock debt over pre-shock liquid wealth among participants. Groups below the 20th percentile are omitted.

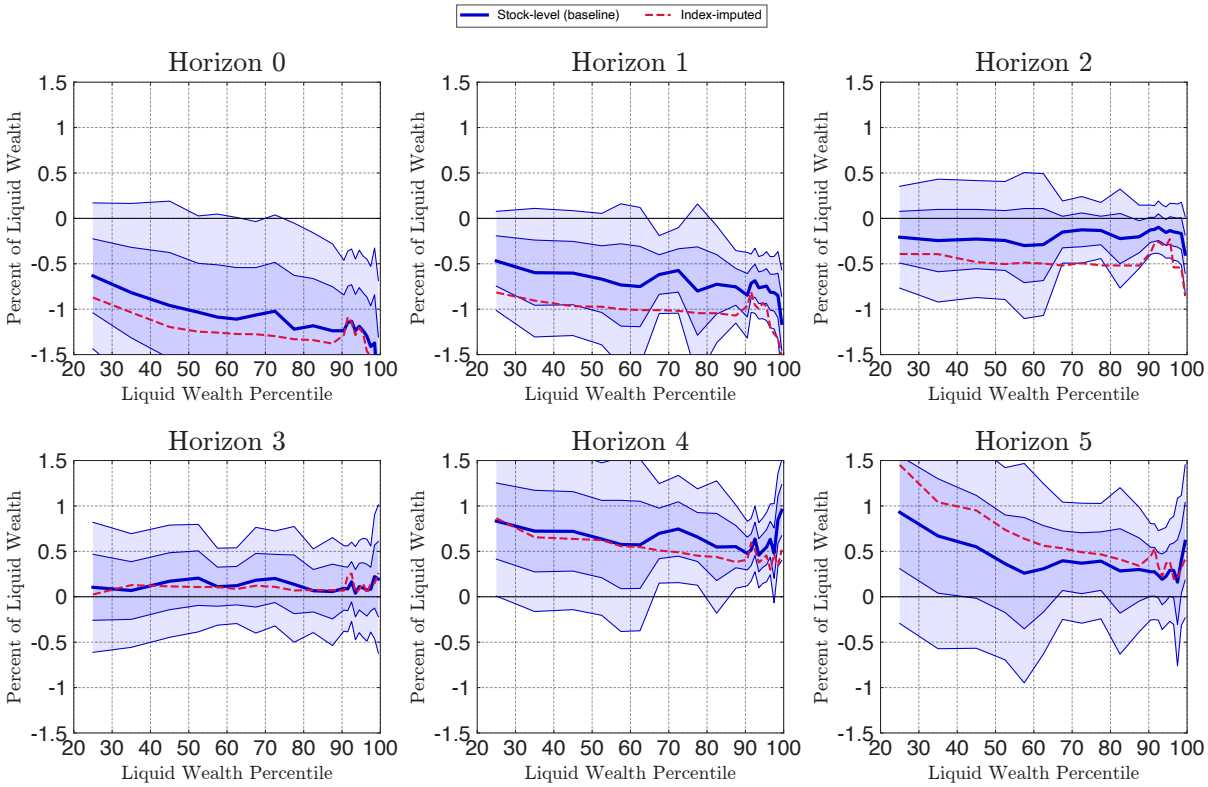


Figure C.4: Capital Gains Response Under Alternative Imputations

Notes: Each panel plots the impulse response of capital gains on risky assets, normalized by pre-shock liquid wealth, to a one percentage point contractionary monetary policy shock at horizon h, with gains on directly held stocks computed stock by stock from 2005 (solid, with 68 and 95 percent Driscoll-Kraay confidence bands) and imputed from the Oslo Stock Exchange index in every year (dashed). Both are estimated on the untrimmed panel. Groups below the 20th percentile are excluded.

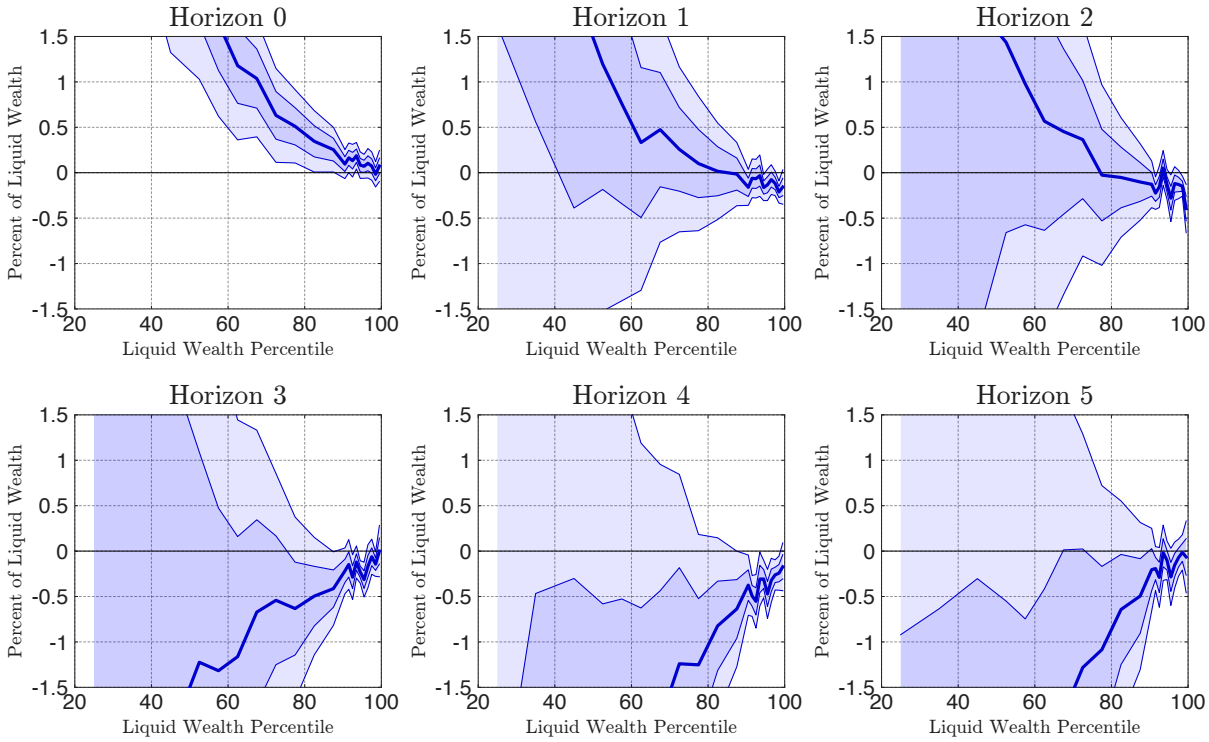


Figure C.5: Impulse Response of Labor Income Across the Liquid Wealth Distribution

Notes: Each panel plots the impulse response of labor income, normalized by pre-shock liquid wealth, to a one percentage point contractionary monetary policy shock at horizon h, estimated separately for each of the liquid wealth groups with equation (12) on the untrimmed panel; the flow-tail trim of Section 3 is defined on the flow and does not bear on income. Groups below the 20th percentile are excluded. Shaded areas denote 68 and 95 percent confidence bands based on Driscoll-Kraay standard errors.

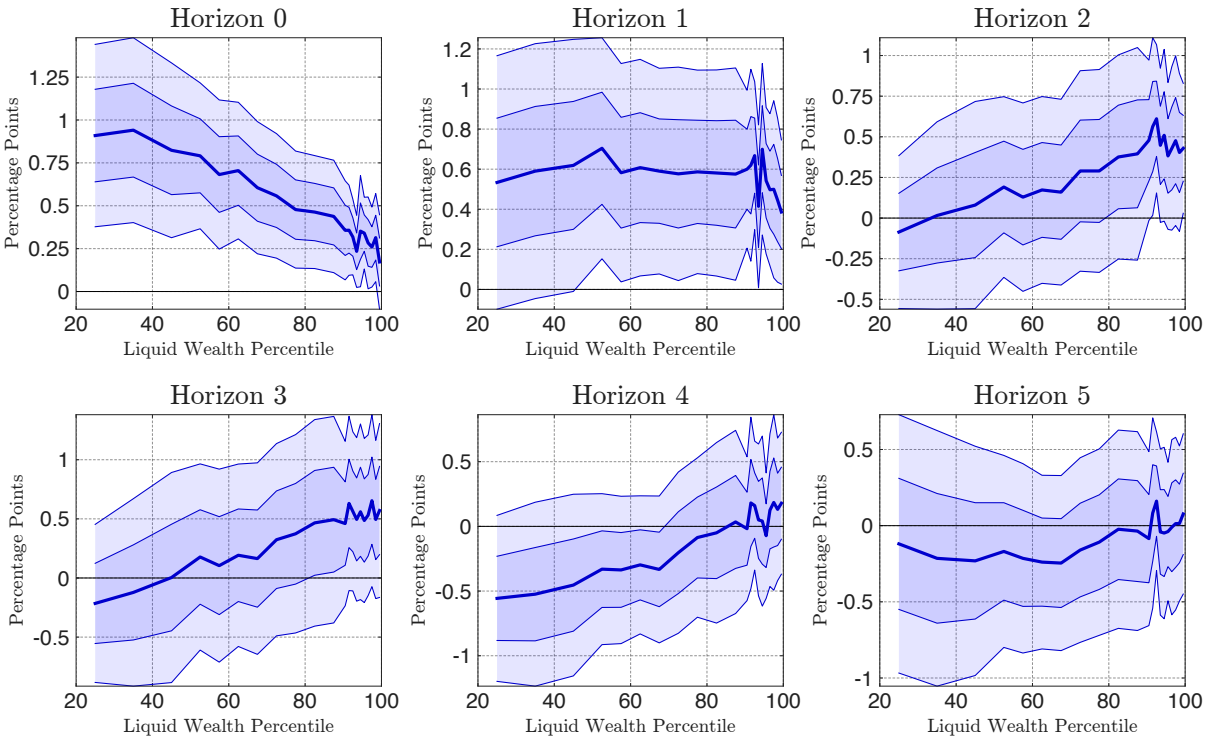


Figure C.6: Response of the Participation Rate Across the Liquid Wealth Distribution

Notes: Each panel plots the impulse response of the stock market participation rate, in percentage points, to a one percentage point contractionary monetary policy shock at horizon h, estimated separately for each liquid wealth group on the untrimmed panel (Appendix **??** shows the trim does not affect non-flow outcomes). Participation is an indicator for positive risky asset holdings. Groups below the 20th percentile are excluded. Shaded areas denote 68 and 95 percent confidence bands based on Driscoll-Kraay standard errors.

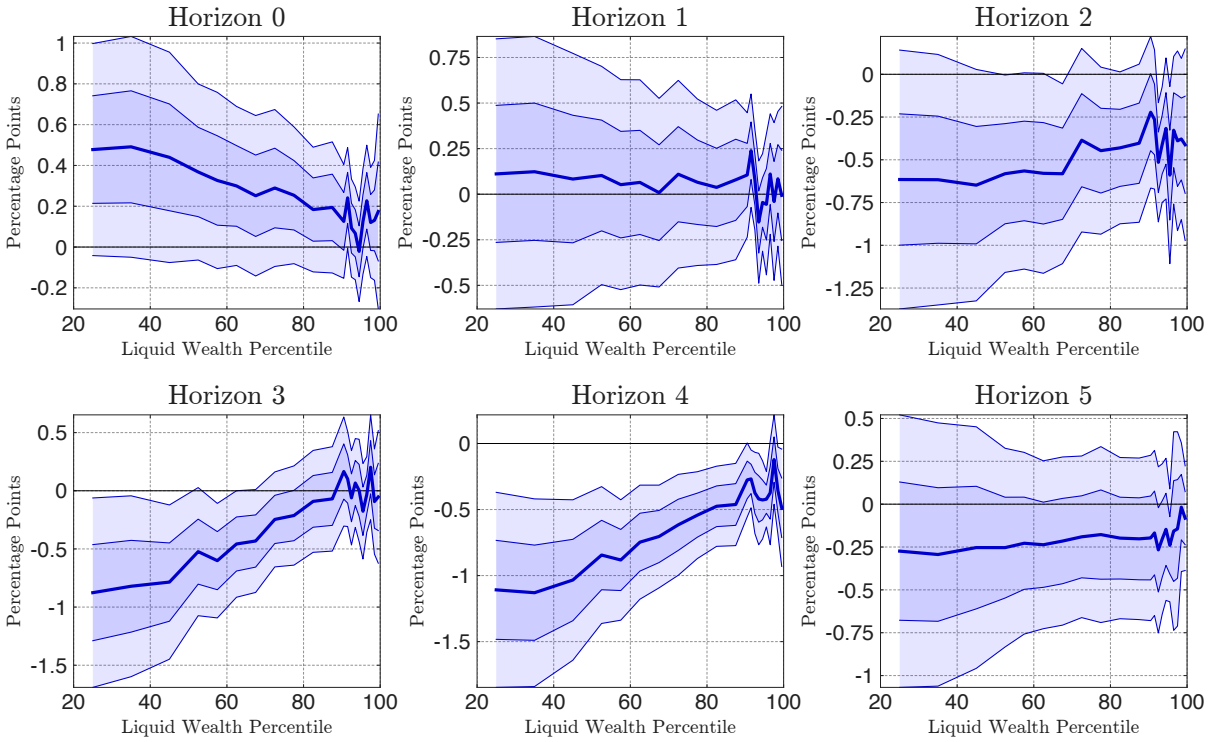


Figure C.7: Entry Among Pre-Shock Non-Participants Across the Liquid Wealth Distribution

Notes: Each panel plots the impulse response of participation at horizon h, in percentage points, among households with no risky assets in the pre-shock year, to a one percentage point contractionary monetary policy shock, estimated separately for each liquid wealth group on the untrimmed panel. Groups below the 20th percentile are excluded. Shaded areas denote 68 and 95 percent confidence bands based on Driscoll-Kraay standard errors.

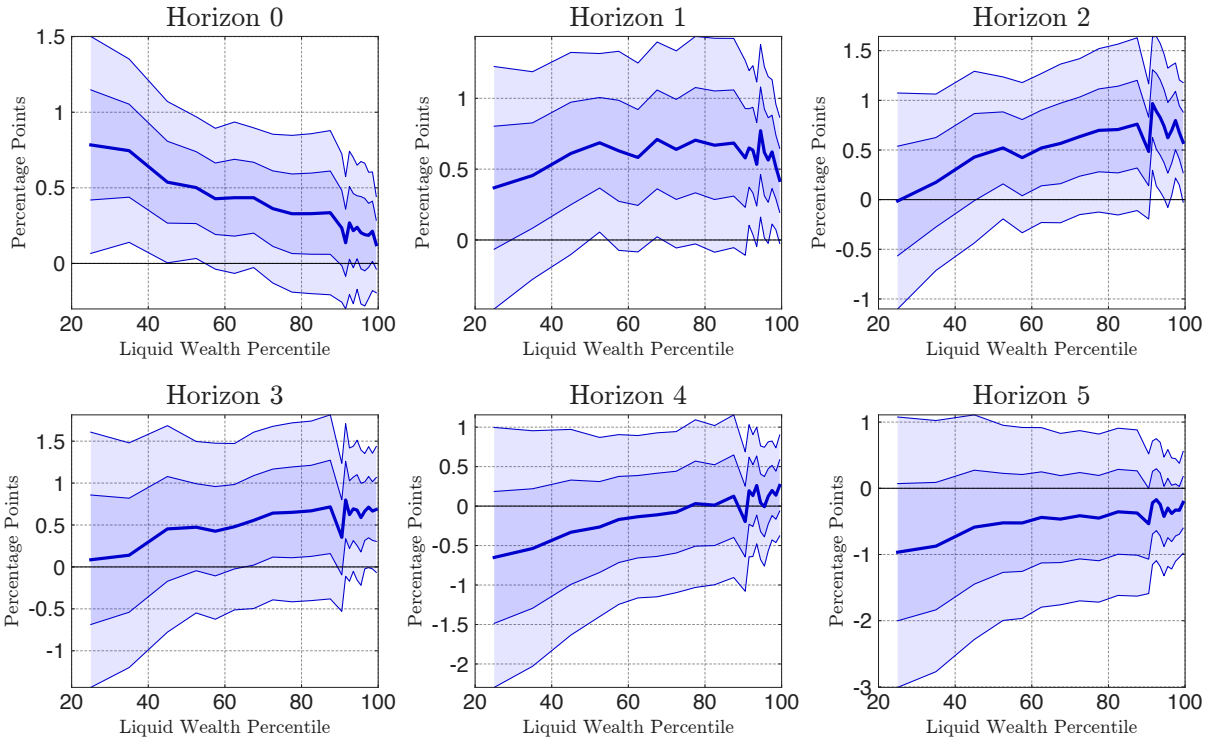


Figure C.8: Staying In Among Pre-Shock Participants Across the Liquid Wealth Distribution

Notes: Each panel plots the impulse response of participation at horizon h, in percentage points, among households with positive risky assets in the pre-shock year (100 minus the exit rate), to a one percentage point contractionary monetary policy shock, estimated separately for each liquid wealth group on the untrimmed panel. Groups below the 20th percentile are excluded. Shaded areas denote 68 and 95 percent confidence bands based on Driscoll-Kraay standard errors.

D Theoretical Appendix

D.1 Proof of Proposition 1

The proof follows from differentiating the definitions of net flows, net saving, and capital gains with respect to the monetary policy shock.

Step 1: Definitions. Total liquid wealth at the end of period t is:

$$a_{i,t+1} = a_{i,t+1}^{risky} + a_{i,t+1}^{safe} \tag{D.1}$$

The portfolio risky share is $\theta_{i,t+1} = a_{i,t+1}^{risky}/a_{i,t+1}$, so risky wealth is:

$$a_{i,t+1}^{risky} = \theta_{i,t+1} \cdot a_{i,t+1} \tag{D.2}$$

Net flows into risky assets are:

$$F_{i,t} = a_{i,t+1}^{risky} - a_{i,t}^{risky} - CG_{i,t} = \theta_{i,t+1} \cdot a_{i,t+1} - a_{i,t}^{risky} - CG_{i,t} \tag{D.3}$$

Capital gains on risky assets are the change in risky wealth due to price movements:

$$CG_{i,t} = a_{i,t}^{risky} - a_{i,t-1}^{risky} - F_{i,t-1} \tag{D.4}$$

or equivalently, the revaluation of existing holdings. For the current period:

$$a_{i,t+1}^{risky} = F_{i,t} + a_{i,t}^{risky} + CG_{i,t} \tag{D.5}$$

where $a_{i,t}^{risky} + CG_{i,t}$ is the value of risky assets after price movements but before active trading.

Net saving is the change in total liquid wealth net of capital gains:

$$NS_{i,t} = a_{i,t+1} - a_{i,t} - CG_{i,t} \tag{D.6}$$

Step 2: Express flows in terms of observables. From the definition of risky wealth:

$$F_{i,t} = \theta_{i,t+1} \cdot a_{i,t+1} - (a_{i,t}^{risky} + CG_{i,t}) \tag{D.7}$$

Since $a_{i,t+1} = a_{i,t} + NS_{i,t} + CG_{i,t}$ (from the definition of net saving):

$$F_{i,t} = \theta_{i,t+1} \cdot (a_{i,t} + NS_{i,t} + CG_{i,t}) - a_{i,t}^{risky} - CG_{i,t} \tag{D.8}$$

Expanding:

$$F_{i,t} = \theta_{i,t+1} \cdot a_{i,t} + \theta_{i,t+1} \cdot NS_{i,t} + \theta_{i,t+1} \cdot CG_{i,t} - \theta_{i,t} \cdot a_{i,t} - CG_{i,t} \tag{D.9}$$

where we used $a_{i,t}^{risky} = \theta_{i,t} \cdot a_{i,t}$. Rearranging:

$$F_{i,t} = (\theta_{i,t+1} - \theta_{i,t}) \cdot a_{i,t} + \theta_{i,t+1} \cdot NS_{i,t} - (1 - \theta_{i,t+1}) \cdot CG_{i,t} \tag{D.10}$$

Step 3: Differentiate with respect to R_t^b. Applying the product rule to equation (D.10):

$$\frac{dF_{i,t}}{dR_t^b} = a_{i,t}\frac{d(\theta_{i,t+1} - \theta_{i,t})}{dR_t^b} + (\theta_{i,t+1} - \theta_{i,t})\frac{da_{i,t}}{dR_t^b} \tag{D.11}$$

$$+ \theta_{i,t+1}\frac{dNS_{i,t}}{dR_t^b} + NS_{i,t}\frac{d\theta_{i,t+1}}{dR_t^b} \tag{D.12}$$

$$- (1 - \theta_{i,t+1})\frac{dCG_{i,t}}{dR_t^b} + CG_{i,t}\frac{d\theta_{i,t+1}}{dR_t^b} \tag{D.13}$$

Step 4: Evaluate at the pre-shock steady state. At the pre-shock values, $\theta_{i,t+1} = \theta_{i,t}$ (no shock has occurred yet, so the risky share is at its steady-state value), $da_{i,t}/dR_t^b = 0$ (pre-shock wealth is predetermined), and $\theta_{i,t}$ is predetermined. Therefore:

$$\frac{dF_{i,t}}{dR_t^b} = a_{i,t}\frac{d\theta_{i,t+1}}{dR_t^b} + \theta_{i,t+1}\frac{dNS_{i,t}}{dR_t^b} + NS_{i,t}\frac{d\theta_{i,t+1}}{dR_t^b} - (1 - \theta_{i,t+1})\frac{dCG_{i,t}}{dR_t^b} + CG_{i,t}\frac{d\theta_{i,t+1}}{dR_t^b} \tag{D.14}$$

Collecting the $d\theta_{i,t+1}/dR_t^b$ terms:

$$\frac{dF_{i,t}}{dR_t^b} = (a_{i,t} + NS_{i,t} + CG_{i,t})\frac{d\theta_{i,t+1}}{dR_t^b} + \theta_{i,t+1}\frac{dNS_{i,t}}{dR_t^b} - (1 - \theta_{i,t+1})\frac{dCG_{i,t}}{dR_t^b} \tag{D.15}$$

Since $a_{i,t+1} = a_{i,t} + NS_{i,t} + CG_{i,t}$:

$$\frac{dF_{i,t}}{dR_t^b} = a_{i,t+1}\frac{d\theta_{i,t+1}}{dR_t^b} + \theta_{i,t+1}\frac{dNS_{i,t}}{dR_t^b} - (1 - \theta_{i,t+1})\frac{dCG_{i,t}}{dR_t^b} \tag{D.16}$$

which is equation (4).

Step 5: Decompose $d\theta_{i,t+1}/dR_t^b$ using MPR. The total change in the risky share can be decomposed into the conditional response (holding portfolio size fixed) and the wealth effect:

$$\frac{d\theta_{i,t+1}}{dR_t^b} = \frac{\partial\theta_{i,t+1}}{\partial R_t^b} + \frac{\partial\theta_{i,t+1}}{\partial a_{i,t+1}} \cdot \frac{da_{i,t+1}}{dR_t^b} \tag{D.17}$$

The marginal propensity to take risk is defined as:

$$MPR_{i,t} \equiv \frac{d(\theta_i a_i)}{da_i} = \theta_{i,t+1} + a_{i,t+1}\frac{\partial \theta_{i,t+1}}{\partial a_{i,t+1}} \tag{D.18}$$

Substituting $d\theta_{i,t+1}/dR_t^b$ into equation (D.16) and using $da_{i,t+1}/dR_t^b = dNS_{i,t}/dR_t^b + dCG_{i,t}/dR_t^b$:

$$\frac{dF_{i,t}}{dR_t^b} = a_{i,t+1}\frac{\partial \theta_{i,t+1}}{\partial R_t^b} \tag{D.19}$$

$$+ a_{i,t+1}\frac{\partial \theta_{i,t+1}}{\partial a_{i,t+1}}\left(\frac{dNS_{i,t}}{dR_t^b} + \frac{dCG_{i,t}}{dR_t^b}\right) \tag{D.20}$$

$$+ \theta_{i,t+1}\frac{dNS_{i,t}}{dR_t^b} - (1-\theta_{i,t+1})\frac{dCG_{i,t}}{dR_t^b} \tag{D.21}$$

Collecting terms in $dNS_{i,t}/dR_t^b$ and $dCG_{i,t}/dR_t^b$:

$$\frac{dF_{i,t}}{dR_t^b} = a_{i,t+1}\frac{\partial \theta_{i,t+1}}{\partial R_t^b} \tag{D.22}$$

$$+ \underbrace{\left(\theta_{i,t+1} + a_{i,t+1}\frac{\partial \theta_{i,t+1}}{\partial a_{i,t+1}}\right)}_{=MPR_{i,t}}\frac{dNS_{i,t}}{dR_t^b} \tag{D.23}$$

$$- \underbrace{\left(1 - \theta_{i,t+1} - a_{i,t+1}\frac{\partial \theta_{i,t+1}}{\partial a_{i,t+1}}\right)}_{=1-MPR_{i,t}}\frac{dCG_{i,t}}{dR_t^b} \tag{D.24}$$

which is equation (3). □

D.2 Proof of Proposition 2

We derive the optimal steady-state portfolio share by perturbing the no-arbitrage condition to second order in the scale of aggregate risk.

Step 1: No-arbitrage condition. The optimality conditions for bonds and stocks yield the no-arbitrage condition:

$$0 = E_t\left\{\left(R_{t+1}^s - R_t^b\right)U'(c_{i,t+1})\right\} \tag{D.25}$$

where the expectation is taken over both aggregate and idiosyncratic uncertainty. At the deterministic steady state (DSS), $R_{t+1}^s = R_t^b$ for all realizations, so the condition holds trivially and the portfolio composition is indeterminate. To pin down the optimal portfolio, we expand to second order in the aggregate productivity shock ε_z.

Step 2: First-order expansion. Write $R^s_{t+1} - R^b_t = \frac{dR^s_{t+1}}{d\varepsilon_z}\varepsilon_z + \frac{1}{2}\frac{d^2R^s_{t+1}}{d\varepsilon_z^2}\varepsilon_z^2 + O(\varepsilon_z^3)$ and $U'(c_{i,t+1}) = U'(c_{i,t+1}) + \frac{dU'(c_{i,t+1})}{d\varepsilon_z}\varepsilon_z + O(\varepsilon_z^2)$, where $c_{i,t+1}$ denotes consumption evaluated at the DSS. Since $R^s - R^b = 0$ at the DSS, the first-order term of the product is:

$$E\left\{\frac{dR^s_{t+1}}{d\varepsilon_z} \cdot U'(c_{i,t+1})\right\} \cdot E[\varepsilon_z] = 0 \tag{D.26}$$

which vanishes because $E[\varepsilon_z] = 0$. This yields no information about the portfolio.

Step 3: Second-order expansion. Collecting terms of order $E[\varepsilon_z^2]$ from the product $(R^s_{t+1} - R^b_t) \cdot U'(c_{i,t+1})$:

$$E\left\{\frac{d^2R^s_{t+1}}{d\varepsilon_z^2}\right\} \cdot E\{U'(c_{i,t+1})\} + 2 \cdot \frac{dR^s_{t+1}}{d\varepsilon_z} \cdot \frac{dE\{U'(c_{i,t+1})\}}{d\varepsilon_z} = 0 \tag{D.27}$$

where we have used the fact that $\frac{dR^s_{t+1}}{d\varepsilon_z}$ is an aggregate variable (common across idiosyncratic states) and can therefore be separated from the expectation over idiosyncratic risk $E\{\cdot\} = \int(\cdot)\, P(e')\, de'$.

Step 4: Expanding $dE\{U'(c)\}/d\varepsilon_z$. The expected marginal utility is $E\{U'(c)\} = \int U'(c_i(e'))\, P(e')\, de'$. Differentiating with respect to ε_z:

$$\frac{dE\{U'(c)\}}{d\varepsilon_z} = \underbrace{\int \frac{dU'(c_i(e'))}{d\varepsilon_z}\, P(e')\, de'}_{\text{consumption response}} + \underbrace{\sum_{k\geq 1}\int \frac{\partial\, \mathbb{E}_t\{U'(c_{i,t+1})\}}{\partial P_{t+k}(e)}\, \frac{dP_{t+k}(e)}{d\varepsilon_{z,t+1}}\, de}_{\equiv \xi_i} \tag{D.28}$$

The first integral captures how marginal utility at each idiosyncratic state responds to the aggregate shock through the consumption channel. The second integral, ξ_i, captures the shift in the probability distribution over idiosyncratic states induced by the aggregate shock (the countercyclical disaster risk channel).

Step 5: Expanding the consumption response. The aggregate shock ε_z affects consumption at each idiosyncratic state e' through two channels: the portfolio return on risky assets and the response of non-financial income. At the DSS:

$$\frac{dU'(c_i(e'))}{d\varepsilon_z} = U''(c_i(e')) \cdot \frac{dc_i(e')}{d\varepsilon_z} = U''(c_i(e')) \cdot MPC_i(e') \cdot \left(\theta_i \cdot a_i \cdot \frac{dR^s_{t+1}}{d\varepsilon_z} + \frac{dy_i(e')}{d\varepsilon_z}\right) \tag{D.29}$$

where $MPC_i(e') = dc_i/dm_i$ is the marginal propensity to consume out of beginning-of-period resources at state e', $\theta_i \cdot a_i$ is the dollar amount invested in risky assets that is determined in the last period, and $dy_i(e')/d\varepsilon_z$ is the non-financial income response to the aggregate shock.

Integrating over idiosyncratic states:

$$\int \frac{dU'(c_i(e'))}{d\varepsilon_z} P(e')\, de' = E\{U''(c)\cdot MPC\}\cdot\left(\theta_i\cdot a_i\cdot\frac{dR^s}{d\varepsilon_z}+E\left\{\frac{dy'}{d\varepsilon_z}\right\}\right) \tag{D.30}$$

where we have used the fact that $\theta_i\cdot a_i$ and $dR^s/d\varepsilon_z$ are determined before the realization of idiosyncratic uncertainty and can be taken outside the expectation.

Step 6: Substitution. Combining equations (D.27), (D.28), and (D.30):

$$0 = E\left\{\frac{d^2R^s}{d\varepsilon_z^2}\right\}\cdot E\{U'\} \tag{D.31}$$

$$+\,2\cdot\frac{dR^s}{d\varepsilon_z}\left[E\{U''\cdot MPC\}\left(\theta_i\cdot a_i\cdot\frac{dR^s}{d\varepsilon_z}+E\left\{\frac{dy'}{d\varepsilon_z}\right\}\right)+\xi_i\right]. \tag{D.32}$$

Step 7: Solving for θ_i. Re-arrange and we yield the optimal portfolio risky share:

$$\theta_i = \frac{1}{a_i}\left(-\frac{E\{U'\}}{E\{U''\cdot MPC\}}\cdot\lambda_z-\frac{E\left\{\frac{dy'}{d\varepsilon_z}\right\}}{\frac{dR^s}{d\varepsilon_z}}-\frac{\xi_i}{E\{U''\cdot MPC\}\cdot\frac{dR^s}{d\varepsilon_z}}\right) \tag{D.33}$$

which is equation (22). □

D.3 Assumption 1 as a Corner Solution of the Sparse Max

This appendix derives Assumption 1 from the sparse-max framework of Gabaix (2014). In that framework an agent takes *actions* in response to *features* of her environment, and chooses how much attention to devote to each feature before acting. In our setting the actions are consumption $c_{i,t}$ and the risky share $\theta_{i,t+1}$. The features are identified below as the objects in the household's portfolio condition that react to the monetary policy shock.

The portfolio condition and its comparative static

Dividing the second-order expansion of the no-arbitrage condition (21) in aggregate risk

$$E\left\{\frac{d^2R^s_{t+1}}{d\varepsilon_z^2}\right\}\cdot E\{U'(c_{i,t+1})\}+2\,\frac{dR^s_{t+1}}{d\varepsilon_z}\cdot\frac{dE\{U'(c_{i,t+1})\}}{d\varepsilon_z}=0 \tag{D.34}$$

by $2\,E\{dR^s_{t+1}/d\varepsilon_z\}$ and using the definition of $\Lambda_z=\lambda_z\cdot\frac{dR^s_{t+1}}{d\varepsilon_z}$, the alternative expression of the price of aggregate risk per unit of exposure to the aggregate shock, with the price of risk λ_z in Proposition 2, the portfolio condition can be written as

$$F \;\equiv\; \Lambda_z\cdot E\{U'(c_{i,t+1})\} \;+\; \frac{dE\{U'(c_{i,t+1})\}}{d\varepsilon_z} \;=\; 0. \tag{D.35}$$

The risky share $\theta_{i,t+1}$ solves $F = 0$, and its response to the monetary policy shock follows from the implicit function theorem,

$$\frac{d\theta_{i,t+1}}{dR_t^b} = -\frac{\partial F/\partial R_t^b}{\partial F/\partial \theta_{i,t+1}}. \tag{D.36}$$

Condition (D.35) contains three objects. The price of risk Λ_z is a property of the return distribution that the household observes in the market; it is independent of her preferences and of her information processing, and is not an object over which attention is allocated. The other two are moved by the monetary shock, and they are the features. Write $\bar{E}$ for the level of expected marginal utility and $D(\theta_{i,t+1}, a_{i,t+1})$ for its sensitivity to aggregate risk, both evaluated at the pre-shock interest rate, so that $F = \Lambda_z \bar{E} + D(\theta_{i,t+1}, a_{i,t+1})$ at the default. The shock shifts each:

$$x_1 \equiv \frac{d\,\mathbb{E}\{U'(c_{i,t+1})\}}{dR_t^b}\bigg|_{a_{i,t+1}} dR_t^b, \qquad x_2 \equiv \frac{\partial}{\partial R_t^b}\frac{d\,\mathbb{E}\{U'(c_{i,t+1})\}}{d\varepsilon_z}\bigg|_{\theta_{i,t+1},\, a_{i,t+1}} dR_t^b, \tag{D.37}$$

with defaults $x_1^d = x_2^d = 0$. Feature 1 is the change in the household's resources; Feature 2 is the revision of her risk sensitivity at a given portfolio. In the framework of Gabaix (2014) the household perceives $x_j^s = m_j x_j + (1 - m_j) x_j^d$ and solves the perceived portfolio condition

$$\Lambda_z\left(\bar{E} + m_1 x_1\right) + D(\theta_{i,t+1}, a_{i,t+1}) + m_2 x_2 = 0. \tag{D.38}$$

Ignoring a feature does not remove it from the condition: with $m_2 = 0$ the sensitivity D remains, with its full dependence on the portfolio, and is evaluated at the pre-shock rate. Assumption 1 is the statement $(m_1, m_2) = (1, 0)$.

The two features differ in where they matter. Feature 1 is an input to the consumption–saving decision as well as to the portfolio: a household who does not perceive the change in her resources mis-sets her consumption, and in the sparse max attention attaches to the feature, not to a particular action, so a household who perceives her resources for consumption perceives them for the portfolio too. Feature 2 enters only the portfolio condition. This asymmetry is what makes the attention problem select $(1, 0)$ rather than $(0, 0)$, as we now show.

The sparse max with two features

Gabaix (2014, Section II) has the agent proceed in two steps. In the first, she chooses attention $m_1, m_2 \in \{0, 1\}$ to the two features, trading the utility loss from imperfect perception against a cost $\kappa > 0$ per feature attended. Because she optimizes given her perception, the loss from misperceiving a feature is second order in the resulting action error, by the envelope theorem: if ignoring feature j displaces action $\alpha \in \{c, \theta\}$ from its optimum by $\Delta\alpha_j$, the utility cost is $\frac{1}{2}\Omega_\alpha(\Delta\alpha_j)^2$, with $\Omega_\alpha \equiv -\partial^2 u/\partial\alpha^2 > 0$ the curvature

of utility in that action at the default. Writing Λ_j for the loss from ignoring feature j, summed across the actions it displaces, and Λ_{12} for the interaction of the two errors, the attention problem is[11]

$$\min_{m_1,m_2\in\{0,1\}} (1-m_1)(1-m_2)\,\Lambda_{12} \;+\; \tfrac{1}{2}(1-m_1)^2\Lambda_1 \;+\; \tfrac{1}{2}(1-m_2)^2\Lambda_2 \;+\; \kappa\,(m_1+m_2). \tag{D.39}$$

In the second step she maximizes utility, taking attended features at their true values and unattended ones at their defaults.[12]

Action errors and curvatures

By (D.36), the displacement of the risky share caused by ignoring a feature is that feature's contribution to $\partial F/\partial R_t^b$, divided by $\partial F/\partial\theta_{i,t+1}$:

$$\Delta\theta_1 = -\,\frac{\Lambda_z\, x_1}{\partial F/\partial\theta_{i,t+1}}, \qquad \Delta\theta_2 = -\,\frac{x_2}{\partial F/\partial\theta_{i,t+1}}. \tag{D.40}$$

Both are first order in dR_t^b, and under CRRA both are nonzero and of opposite sign: a higher safe rate lowers the level of expected marginal utility, which lowers the share, and revises its sensitivity in the direction that raises it.

Feature 1 changes consumption as well as the share: a household who does not consider the change in her resources mis-sets her consumption by the response of Proposition 4,

$$\Delta c_1 = \frac{dc_{i,t}}{dR_t^b}\,dR_t^b. \tag{D.41}$$

On the other hand, Feature 2 affects only the share. The sensitivity enters the consumption Euler equation (19) only through the precautionary term, which is of second order in aggregate risk, so $\Delta c_2 = O(\sigma_z^2)\,dR_t^b$ and vanishes at the order of our solution.

The curvature of utility in consumption, $\Omega_c \equiv -\partial^2 u/\partial c^2$, is that of the consumption–saving problem and is of order one. The curvature in the risky share is the second-order condition of the portfolio problem:

$$\Omega_\theta \;\equiv\; -\frac{\partial^2 u}{\partial\theta^2} \;=\; -\frac{\partial F}{\partial\theta_{i,t+1}}\cdot a_{i,t+1}\cdot\frac{dR_{t+1}^s}{d\varepsilon_z}\cdot\sigma_z^2 \;=\; O(\sigma_z^2) \tag{D.42}$$

the variance of the household's risky position priced by the slope of the portfolio condition. The cross-curvature is $O(\sigma_z^2)$ for the same reason, so it enters Λ_1 at the order of $\Omega_\theta(\Delta\theta_1)^2$

[11] Gabaix (2014) chooses attention against the ex-ante variance of each feature; here both features are proportional to the same shock, so using the realized dR_t^b or its variance gives the same ratio Λ_2/Λ_1 and the same window up to a common scale of κ.

[12] The default is the deterministic steady state, at which the household is fully rational; attention concerns only the response to a departure from it. This is why the simplification applies to the adjustment of the share and not to its level, which remains the second-order object of Proposition 2.

and does not affect what follows.

Collecting terms:

$$\Lambda_1 = \Omega_c \, (\Delta c_1)^2 + \Omega_\theta \, (\Delta \theta_1)^2 \tag{D.43}$$

$$\Lambda_2 = \Omega_\theta \, (\Delta \theta_2)^2 \tag{D.44}$$

$$\Lambda_{12} = \Omega_\theta \, \Delta\theta_1 \Delta\theta_2 \tag{D.45}$$

so that

$$\frac{\Lambda_2}{\Lambda_1} \;\leq\; \frac{\Omega_\theta \, (\Delta\theta_2)^2}{\Omega_c \, (\Delta c_1)^2} \;=\; O(\sigma_z^2), \tag{D.46}$$

and likewise $\Lambda_{12}/\Lambda_1 = O(\sigma_z^2)$. Since $\Delta\theta_1$ and $\Delta\theta_2$ have opposite signs, $\Lambda_{12} < 0$. The ratio is small for two reasons. Ignoring Feature 2 leaves the consumption plan intact and misallocates only the household's wealth between two assets whose returns differ by the excess return, an error priced by the variance of that return; ignoring Feature 1 misallocates resources between consumption and saving, an error priced by the curvature of utility in consumption. The portfolio margin is therefore cheap in welfare by a factor of order σ_z^2, and the displacement itself is bounded, since the risky share lies in $[0, 1]$ and moves by a few percentage points at most, whereas the consumption error is a first-order response of the household's entire flow of resources.[13]

The corner solution

Proposition 5 (Certainty-equivalent portfolio adjustment as a sparse-max corner)**.** *Suppose $\Lambda_2 < \Lambda_1$ and that attention is binary, $m_j \in \{0, 1\}$. For any attention cost*

$$\tfrac{1}{2}\Lambda_2 \;<\; \kappa \;<\; \tfrac{1}{2}\Lambda_1 + \Lambda_{12}, \tag{D.47}$$

the solution to (D.39) is $(m_1^*, m_2^*) = (1, 0)$. Her second-step problem is then (D.38) with $m_1 = 1$ and $m_2 = 0$, which is Assumption 1, and her share response is (D.36) with the Feature-2 term omitted, as in Proposition 3. By (D.46) the interval (D.47) is nonempty, with lower endpoint of order σ_z^2 and upper endpoint of order one.

Proof. The objective (D.39) takes four values,

$$L(1,1) = 2\kappa, \qquad L(1,0) = \tfrac{1}{2}\Lambda_2 + \kappa, \qquad L(0,1) = \tfrac{1}{2}\Lambda_1 + \kappa, \qquad L(0,0) = \tfrac{1}{2}\Lambda_1 + \Lambda_{12} + \tfrac{1}{2}\Lambda_2,$$

the interaction term appearing only when both features are ignored. Then $L(1,0) < L(1,1)$

[13] The ratio is formally $O(\sigma_z^2)$, but because $\Delta\theta_2$ scales with $\lambda_z = \mu/\sigma_s^2$ the small factor is partly undone by the equity premium, and it is worth seeing the size. Using $\Omega_\theta(\Delta\theta_2)^2/\Omega_c(\Delta c_1)^2 \approx \sigma_s^2 \, MPC \, (a_{i,t+1}\Delta\theta_2)^2/(\Delta c_1)^2$, with liquid wealth of 100,000, consumption of 50,000, $MPC = 0.1$, $\gamma = 2$, $\sigma_s = 0.2$ and $\lambda_z = 1.5$, a one percentage point shock gives $a_{i,t+1}\Delta\theta_2$ of about 1,500 against Δc_1 of about 350, and a ratio near 0.07. The portfolio margin is cheaper in welfare by roughly an order of magnitude.

iff $\frac{1}{2}\Lambda_2 < \kappa$; $L(1,0) < L(0,1)$ iff $\Lambda_2 < \Lambda_1$; and $L(1,0) < L(0,0)$ iff $\kappa < \frac{1}{2}\Lambda_1 + \Lambda_{12}$. All three hold under the stated conditions. In the second step the household solves (D.38) with $m_1 = 1$ and $m_2 = 0$: the level of expected marginal utility at its post-shock value, the sensitivity at the pre-shock rate.

$$\left.\frac{\partial^2 \mathbb{E}_t\{U'(c_{i,t+1})\}}{\partial R_t^b \, \partial \varepsilon_{z,t+1}}\right|_{\theta_{i,t+1},\, a_{i,t+1}} = 0,$$

which is Assumption 1. □

Remark 1 (Continuous attention)**.** *Under the linear attention cost of Gabaix (2014), $g(m) = m$ with $m \in [0,1]^2$, the first-order conditions give $m_2^* = 0$ exactly whenever $\kappa \geq \Lambda_2 + (1 - m_1^*)\Lambda_{12}$, which holds throughout the window (D.47) since $\Lambda_2, \Lambda_{12} = O(\sigma_z^2)$, and $m_1^* = 1 - \kappa/\Lambda_1 + O(\sigma_z^2)$. Feature 2 is therefore ignored exactly, and Feature 1 attended up to a correction of order κ/Λ_1, which would attenuate the consumption and reaching-for-yield responses by the same factor. If attention to the household's own resources is free, as it is natural to assume since she must track them for the consumption–saving decision whatever she does with her portfolio, then $m_1^* = 1$ and Assumption 1 holds without approximation under any convex attention cost.*

D.4 Proof of Proposition 3

The most intuitive proof would take the direct derivative of optimal portfolio share in Proposition 2 with respect to monetary policy shock dR_t^b, then show it collapses to Equation (26) from the decomposition of Assumption 1. In this appendix, we take another path that starts from the evaluation of optimality condition for generality. Following the same logic in Appendix D.3, we start from the second-order expansion of the no-arbitrage condition (21) and rewrite it as

$$F \;\equiv\; \Lambda_z \cdot E\{U'(c_{i,t+1})\} \;+\; \frac{dE\{U'(c_{i,t+1})\}}{d\varepsilon_z} \;=\; 0. \tag{D.48}$$

where $\Lambda_z = \lambda_z \cdot \frac{dR^s_{t+1}}{d\varepsilon_z}$ is an alternative expression of price of aggregate risk with per unit of portfolio exposure.

The risky share $\theta_{i,t+1}$ solves $F = 0$ in both the steady state and transitional dynamics. Since wealth level is a steady-state idiosyncratic state and agents have a portfolio share policy function with respect to it, we separate the wealth effect and conditional portfolio share response that follows from the implicit function theorem:

$$\frac{d\theta_{i,t+1}}{dR_t^b} \;=\; \frac{\partial \theta_{i,t+1}}{\partial a_{i,t+1}} \frac{da_{i,t+1}}{dR_t^b} \;-\; \left.\frac{\partial F / \partial R_t^b}{\partial F / \partial \theta_{i,t+1}}\right|_{a_{i,t+1}} \tag{D.49}$$

Now we expand each term in conditional response, starting with the numerator.

$$\begin{aligned}\left.\frac{\partial F}{\partial R_t^b}\right|_{a_{i,t+1}} &= \Lambda_z \cdot \frac{dE\{U'(c_{i,t+1})|_{a_{i,t+1}}\}}{dR_t^b} \;+\; E\{U'(c_{i,t+1})\} \cdot \frac{d\Lambda_z}{dR_t^b} \\ &+ \underbrace{\frac{d^2\,\mathbb{E}_t\{U'(c_{i,t+1})|_{\theta_{i,t+1},\,a_{i,t+1}}\}}{dR_t^b\;d\varepsilon_{z,t+1}}}_{\text{zero by Assumption 1}} \end{aligned} \tag{D.50}$$

$$\begin{aligned}&= \Lambda_z \cdot E\left\{U''(c_{i,t+1}) \cdot MPC_{i,t+1} \cdot \frac{d(R_t^b a_{i,t+1} + y_{i,t+1})|_{a_{i,t+1}}}{dR_t^b}\right\} \\ &+\; \Lambda_z \cdot \underbrace{\left(\sum_{k\geq 1} \int \frac{\partial\,\mathbb{E}_t\{U'(c_{i,t+1})\}}{\partial P_{t+k}(e)} \frac{\partial P_{t+k}(e)}{\partial Y_{t+1}}\,de\right)}_{\equiv \psi_i} \cdot \frac{dY_{t+1}}{dR_t^b} \;+ E\{U'(c_{i,t+1})\} \cdot \frac{d\Lambda_z}{dR_t^b}\end{aligned} \tag{D.51}$$

$$\begin{aligned}&= \Lambda_z \cdot E\{U''(c_{i,t+1}) \cdot MPC_{i,t+1}\} \cdot (a_{i,t+1} + E\{\frac{dy_{i,t+1}}{dR_t^b}\}) \;+\; \Lambda_z \cdot \psi_i \\ &+\; E\{U'(c_{i,t+1})\} \cdot \frac{d\Lambda_z}{dR_t^b}\end{aligned} \tag{D.52}$$

Where the covariance between $U''(c_{i,t+1}) \cdot MPC_{i,t+1}$ and $\frac{dy_{i,t+1}}{dR_t^b}$ is assumed away for simplicity. Similarly, the denominator can be expanded as:

$$\begin{aligned}\left.\frac{\partial F}{\partial \theta_{i,t+1}}\right|_{a_{i,t+1}} &= \Lambda_z \cdot \underbrace{\frac{\partial E\{U'(c_{i,t+1})\}}{\partial \theta_{i,t+1}}}_{\text{zero in Deterministic S.S.}} + \frac{\partial}{\partial \theta_{i,t+1}}\left(\frac{dE\{U'(c_{i,t+1})\}}{d\varepsilon_z}\right) \\ &= E\{U''(c_{i,t+1}) \cdot MPC_{i,t+1}\} \cdot a_{i,t+1} \cdot \frac{dR_{t+1}^s}{d\varepsilon_z}\end{aligned} \tag{D.53}$$

Collect terms and put together:

$$\begin{aligned}\frac{d\theta_{i,t+1}}{dR_t^b} &= \frac{\partial \theta_{i,t+1}}{\partial a_{i,t+1}} \frac{da_{i,t+1}}{dR_t^b} \;-\; \frac{\Lambda_z \cdot E\{U''(c_{i,t+1}) \cdot MPC_{i,t+1}\} \cdot (a_{i,t+1} + E\{\frac{dy_{i,t+1}}{dR_t^b}\})}{E\{U''(c_{i,t+1}) \cdot MPC_{i,t+1}\} \cdot a_{i,t+1} \cdot \frac{dR_{t+1}^s}{d\varepsilon_z}} \\ &- \frac{\Lambda_z \cdot \psi_i \;+\; E\{U'(c_{i,t+1})\} \cdot \frac{d\Lambda_z}{dR_t^b}}{E\{U''(c_{i,t+1}) \cdot MPC_{i,t+1}\} \cdot a_{i,t+1} \cdot \frac{dR_{t+1}^s}{d\varepsilon_z}} \\ &= \frac{\partial \theta_{i,t+1}}{\partial a_{i,t+1}} \frac{da_{i,t+1}}{dR_t^b} \;-\; \lambda_z \;-\; \frac{\lambda_z}{a_{i,t+1}} \cdot E\{\frac{dy_{i,t+1}}{dR_t^b}\} \\ &- \frac{\lambda_z}{a_{i,t+1}} \cdot \frac{\psi_i}{E\{U''(c_{i,t+1}) \cdot MPC_{i,t+1}\}} \\ &- \underbrace{\frac{E\{U'(c_{i,t+1})\}}{E\{U''(c_{i,t+1}) \cdot MPC_{i,t+1}\} \cdot a_{i,t+1} \cdot \frac{dR_{t+1}^s}{d\varepsilon_z}}}_{\equiv \frac{\partial \theta_{i,t+1}}{\partial \Lambda_z}} \cdot \frac{d\Lambda_z}{dR_t^b}\end{aligned} \tag{D.54}$$

This is Equation (26) in Proposition 3. □

D.5 Model Setup and Calibration of the Simulation

This appendix describes the computational version of the model of Section 5 that produces Figure 13, its calibration, and the experiment. The simulation implements the environment, Proposition 2, Assumption 1 and Proposition 3 as written; what it adds is a specification of the objects the propositions leave open, the discretization of the idiosyncratic process, the shape of the cyclical loading, and numbers. The code is a self-contained Python notebook built on the standard incomplete-markets routines of Auclert et al. (2021), and is available with the replication files.

Environment

Time is quarterly. A household is described by its liquid wealth a at the start of the period, an earnings (productivity) state e and a binary disaster indicator $d \in \{0, 1\}$. Log earnings follow a Markov chain on $n_e = 11$ points obtained by the Rouwenhorst method, normalized so that $\sum_e \pi_e e = 1$, and labor income is $y_{i,t} = \bar{y}\, e_{i,t}\, Y_t\, (1 - \ell d_{i,t})$, with $\bar{y}$ the labor share and Y_t aggregate output ($Y_{ss} = 1$); this is the paper's $y_{i,t} = e_{i,t} w_t$ with $w_t = \bar{y} Y_t$. The countercyclical disaster risk of Section 5 is written as a separate state: $d = 1$ means the household carries a large income loss of size ℓ, assumed deterministic for simplicity. The disaster transition is drawn after next period's earnings state e' and indexed by it,

$$\Pr(d' = 1 \mid d = 0, e') = P_t(e'), \qquad \Pr(d' = 1 \mid d = 1, e') = \rho_d(e'), \tag{D.55}$$

so that a household enters a disaster with probability $P_t(e')$ and, once in it, remains with probability $\rho_d(e')$. Writing the loss as a persistent state with its own arrival and exit probabilities is an equivalent way of writing the state-dependent transition matrix Π_e of Section 5. The arrival probability is cyclical and loads on high-income earners,

$$P_t(e) = \lambda_0 + w(e)\, \lambda_1 \Big(- \log Y_t / Y_{ss} \Big), \qquad w(e) = \frac{\min(e^\eta, \bar{w})}{\sum_{e'} \pi_{e'} \min(e'^\eta, \bar{w})}, \tag{D.56}$$

where λ_0 is the steady-state arrival probability, λ_1 the average cyclicality (since $\sum_e \pi_e w(e) = 1$) and η tilts the cyclicality toward high-income earners. The persistence $\rho_d(e)$ rises linearly in $\log e$ from ρ_d^{lo} at the lowest earnings state to ρ_d^{hi} at the highest. In the deterministic steady state $Y = Y_{ss}$ and $P = \lambda_0$ for every earner, so neither λ_1 nor η affects the steady state; they act only off steady state, through ψ_i in Proposition 3. Households save in a safe asset with gross return R^b and a stock with return $R^s = R^b + \mu + \sigma\varepsilon$, $\varepsilon \sim (0, 1)$ i.i.d., in share θ, and face the non-negativity constraints of Section 5. Preferences are CRRA with coefficient γ and discount factor β. There is a single household type; all heterogeneity is in (a, e, d).

Solution method

The model is solved by the two perturbations of Section 5. To zeroth order in aggregate risk, both assets earn R^b, and consumption solves the textbook Euler equation (19). We solve it by the endogenous grid method on a grid of 500 asset points between 0 and 250 quarters of average income, dense near the borrowing constraint, and obtain the stationary distribution by the histogram method; both routines are those of Auclert et al. (2021). To second order in aggregate risk, the portfolio share on the savings grid follows Proposition 2: $\mathbb{E}\{U'(c')\}$, $\mathbb{E}\{-U''(c')\,MPC'\}$ and the income-hedging term are conditional expectations of functions of next period's consumption under the steady-state policy and transition, with the marginal propensity to consume obtained by finite differences of the consumption policy in income. The share is clipped to $[0, 1]$ according to the no-short-sale constraints.

Two of the paper's assumptions are imposed in the steady-state rule. First, the labor-income response to the aggregate shock is the same in levels for every earner, $\mathbb{E}\{dy_{i,t+1}/d\varepsilon_{z,t+1}\} = b_Y \bar{y}$, as in Proposition 2. This is the term that makes the steady-state risky share rise with wealth: the hedging demand is the same amount for everyone and therefore a larger share of a small portfolio. Second, $\xi_i = 0$: we assume in partial equilibrium the aggregate productivity shock does not move tail risk, so the three models below share the same steady state and the steady-state share rises through income hedging alone, as in Section 5. The steady-state group-average share rises from 0.37 at p40–50 to 0.62 at p90–97 and is flat above, with participation complete above the median; we do not target the registry's profile, since a one-type model without a participation cost obtains the whole rise from the stockholders' share where the data obtain much of it from participation, and the object of the exercise is the response.

The experiment

The monetary experiment is the two-period experiment of Section 5 taken literally: the shock arrives at date 0 and moves the household's inputs at $t = 0$ and $t = 1$ only; from $t = 2$ on every input is back at its steady-state value, and the household chooses its date-0 portfolio looking one period ahead. Four inputs move.

- The safe rate on date-0 savings rises by $dR^b = 1$ percentage point for one period, and the expected stock return moves one for one (the belief channel is shut, $d\Lambda_z/dR^b = 0$).
- Output falls by 1 percent at date 0, the impact response per percentage point in US estimates, and by 5 percent at date 1, a one-period stand-in for the recession's income loss: if the impact loss decayed geometrically at 0.8 per quarter the cumulative loss would be five times the impact, which is the conservative end of our estimates (Figure 4), and a one-period loss is valued at a higher marginal propensity than a persistent one.

- Labor income moves with output with incidence $g(e) = \alpha_e + (1 - \alpha_e)/e$, so that $dy = \bar{y} \cdot (\alpha_e e + 1 - \alpha_e) \cdot dY$: the loss is a larger share of income for low earners, and aggregate labor income still moves one for one with Y. The disaster probability at date 1 is evaluated at the same cumulative gap, $P_1(e) = \lambda_0 - w(e) \cdot \lambda_1 \cdot dY_1$.

- The equity price Q_0^s falls by 1 percent. This is the no-arbitrage price of the claim when the one-quarter discount rate rises by 1 percentage point and dividends and the premium do not move.[14]

Solving the response. Given the inputs, the date-1 and date-0 consumption policies are solved backward from the steady state, with the capital loss $\theta_{ss}\, a'\, d \log Q_0^s$ entering date-0 cash on hand. For each pre-shock state the new share is then formed from the new level of $\mathbb{E}_0\{U'(c_1)\}$: under Assumption 1, with the denominator and the hedging term frozen at their steady-state values (M2 and M3); or with all three objects recomputed from the date-1 policy (M1). The first-order change in $\mathbb{E}_0\{U'\}$ recovers the channels of Proposition 3 exactly.

Proposition 3 assumes, for simplicity, that the labor-income response does not vary with idiosyncratic productivity, so that the income effect is $\frac{\lambda_z}{a_{i,t+1}} \mathbb{E}\{-dy_{i,t+1}/dR_t^b\}$. The simulation relaxes this: the income response varies with the earnings state through the incidence $g(e)$, and the income effect is the exact first-order change in expected marginal utility,

$$\frac{\lambda_z}{a_{i,t+1}} \frac{\mathbb{E}\{-U''(c_{i,t+1})\, MPC_{i,t+1} \cdot (-dy_{i,t+1}/dR_t^b)\}}{\mathbb{E}\{-U''(c_{i,t+1})\, MPC_{i,t+1}\}}, \tag{D.57}$$

which keeps the covariance between the marginal-utility weight and the income loss across next period's earnings states. The covariance is negative, since the loss is larger in levels for high-income earners while the weight is larger for low-income earners, and it lowers the income effect by about ten percent at every wealth level without changing its gradient.

The wealth effect is $\theta(a_0') - \theta(a_{ss}')$ along the steady-state distribution of wealth, with a_0' the household's date-0 savings after the capital loss, the income change and its own consumption choice. With a 1 percent capital loss the wealth effect is below 0.1 percentage point at every wealth level, so Figure 13 is, to a close approximation, the three marginal-utility channels of Proposition 3.

Reporting. Responses are group averages over all households in each bin of pre-shock liquid wealth, the registry's bins, with non-participants responding through the clipped rule. Above the median participation is complete, and the group averages coincide with the stockholders' responses.

[14] Event studies find larger responses, 4–5 percent per percentage point (Bernanke and Kuttner, 2005), because an identified policy surprise raises the expected rate path for several quarters rather than one.

The three models of Figure 13 share the steady state and differ in two switches: M1 is the fully rational household facing countercyclical disaster risk, which recomputes the numerator, the denominator and the hedging term after the shock; M2 imposes Assumption 1 with acyclical disaster risk ($\lambda_1 = 0$ along the transition); M3 imposes Assumption 1 with countercyclical disaster risk. The difference between M2 and M3 is therefore the precautionary motive alone, and the difference between M1 and M3 is Assumption 1 alone.

Calibration

The calibration is disciplined by evidence on US data: the model is about the mechanism, and the registry's profile is what the mechanism is meant to explain. Every parameter is either taken from a source or internally calibrated to a stated target; Table D.1 lists them. The price of risk uses the long US sample of Dimson et al. (2006): an arithmetic equity premium over bills of 7.4 percent a year and a standard deviation of real equity returns of 20.2 percent, which give $\lambda_z = \mu/\sigma^2 = 1.81$ and a Merton share $\lambda_z/\gamma = 0.36$. The level of every marginal-utility channel is proportional to λ_z, so the Norwegian moments (premium 5.7 percent, volatility 27 percent, $\lambda_z = 0.78$) or the textbook 4 percent and 18 percent ($\lambda_z = 1.23$) stretch Figure 13 without changing its shapes. Risk aversion is $\gamma = 5$, the baseline of Gomes and Michaelides (2005), whose elasticity of intertemporal substitution of 0.2 is also the one implied by CRRA utility with $\gamma = 5$; it is inside the range considered by Cocco et al. (2005). The discount factor is internal to liquid wealth of two years of GDP. In the Financial Accounts of the United States (Table B.101.e, end of 2024) households' deposits, directly held debt securities, directly held equities and mutual-fund shares sum to 76 trillion dollars against a GDP of 29 trillion, about 2.6 times GDP, and about 2 times once closely held business equity, which is part of the directly held equity line, is left out. The discount factor is low because $\gamma = 5$ with the earnings risk of Auclert et al. (2024a) generates strong precautionary saving; a higher β flattens every cross-sectional gradient, since the income channel is proportional to y/a'. The earnings process is the AR(1) of Auclert et al. (2024a), with the annual persistence of 0.91 and an innovation variance set to a standard deviation of log earnings of 0.92, discretized as an 11-point Rouwenhorst chain. The real rate is 2 percent a year.

The disaster block is calibrated on US displacement evidence and is the one used in the companion paper (Zhang, 2026a). The arrival rate of 3.5 percent a year is inside the 3–5 percent range of the Displaced Worker Survey (Farber, 2017); the loss of 53 percent is the uninsured share of a job loss under US unemployment insurance; the persistence rises from a present value of 0.2 years of income for the lowest earner to 2.2 years for the highest, the latter inside the 1.4–2.8 years of Davis and Von Wachter (2011) for high-tenure workers in mass layoffs, the gradient following tenure and prior earnings. The average cyclicality

Parameter		Value	Source or target
Preferences and returns (quarterly)			
γ	risk aversion	5	Gomes and Michaelides (2005)
β	discount factor	0.90	liquid wealth $= 2\times$ GDP
R^b	gross safe return	1.005	2 percent real a year
μ, σ	premium, stock volatility	0.0185, 0.101	Dimson et al. (2006); $\lambda_z = 1.81$
b_Y	income exposure to the market	0.37	internal: rising steady-state share; frozen under Assumption 1
Earnings			
ρ_e, sd$(\log e)$, n_e	Rouwenhorst chain	0.91 (annual), 0.92, 11	Auclert et al. (2024a)
$\bar{y}$	labor share	0.7	round up of the two-thirds convention
α_e	incidence of aggregate income change	0.69	Patterson (2023), Guvenen et al. (2017)
Disaster block			
λ_0	arrival probability	0.0087	3.5 percent a year, Displaced Worker Survey (Farber, 2017)
ℓ	income loss in the disaster	0.53	uninsured share of a job loss under US UI
$\rho_d^{lo} \to \rho_d^{hi}$	persistence by earnings	$0.30 \to 0.94$	PV of loss $0.2 \to 2.2$ years; Davis and Von Wachter (2011)
λ_1	average cyclicality	0.20	arrival doubles at a 5 percent gap; upper end of Farber (2017)
η, $\bar{w}$	loading on earnings, cap	1, 5	Guvenen et al. (2014b), Guvenen et al. (2014a), Guvenen et al. (2017)
The two-period experiment (per 1 pp shock)			
dR^b	safe rate, one period	1 pp	unit of inputs
dY_0, dY_1	output at $t = 0$, $t = 1$	-1%, -5%	first-year loss; cumulative loss (Coibion, 2012)
gap for P_1	output gap in (D.56)	5%	same path
dp_0	equity price	-1%	no-arbitrage response to a one-period 1 pp rate rise

Table D.1: Calibration of the Simulated Model

Notes: Quarterly. Liquid wealth, participation and the risky share are not targeted. The asset grid has 500 points between 0 and 250 quarters of average income.

$\lambda_1 = 0.20$ doubles the arrival probability of the average worker at a 5 percent output gap, the gap at the trough of 2009. Farber (2017) reports that one in six workers lost a job in 2007–09 against one in ten in a normal three-year window, a ratio of 1.6 over a period in which the gap averaged less than its trough value; a cumulative mapping of that ratio gives $\lambda_1 \approx 0.15$, so 0.20 is at the upper end of the evidence. The loading is proportional to earnings and capped at five times the average, which places the top 1 percent of earners at five times the median earner's cyclicality and the 95th–99th percentiles at 3.8 times, consistent with the finding of Guvenen et al. (2014b) that the top 1 percent is dramatically more exposed to recessions than those just below, with the top-versus-median exposure of two to three times in Guvenen et al. (2014a), and with the GDP beta of earnings of 3.7 at the 99.9th percentile against about 1 for the average worker in Guvenen et al. (2017). The proportional loading overstates the exposure of the 95th–99th percentiles, whose betas in Guvenen et al. (2017) are closer to 1.5–2 than to the 3.2 times the median earner's that $\eta = 1$ gives them. The incidence $\alpha_e = 0.69$ makes the own-percent income response 1.5–2.2 for the bottom quintile of earners, 0.95 at the median and 0.72–0.77 for the top decile, in line with earnings elasticities to GDP that decline with income over most of the distribution in US administrative data (Patterson, 2023); the rise of the elasticity in the top percentiles found by Guvenen et al. (2017) is carried by the disaster loading $w(e)$ rather than by the incidence. The one internal parameter of the portfolio block, b_Y, is set so that the steady-state share rises with wealth, the profile the paper assumes.

D.6 Proof of Proposition 4

We derive the consumption response to the monetary policy shock by perturbing the Euler equation for bonds. The perturbation method and the use of marginal propensities as sufficient statistics follow Zhang (2026b).

Step 1: Euler equation. The optimality condition for bonds is:

$$U'(c_{i,t}) = \beta R_t^b E\{U'(c_{i,t+1})\} \tag{D.58}$$

Step 2: Differentiate with respect to R_t^b. Taking the total derivative and using the Euler equation to substitute $\beta E\{U'(c_{i,t+1})\} = U'(c_{i,t})/R_t^b$:

$$U''(c_{i,t})\frac{dc_{i,t}}{dR_t^b} = \frac{U'(c_{i,t})}{R_t^b} + \beta R_t^b \frac{dE\{U'(c_{i,t+1})\}}{dR_t^b} \tag{D.59}$$

Step 3: Expand $dE\{U'\}/dR_t^b$. The expected marginal utility responds through the consumption channel and the disaster probability shift:

$$\frac{dE\{U'\}}{dR_t^b} = \underbrace{\int \frac{dU'(c_{i,t+1}(e))}{dR_t^b} P_{t+1}(e)\, de}_{\text{consumption channel}} + \underbrace{\left(\sum_{k\geq 1} \int \frac{\partial\, \mathbb{E}_t\{U'(c_{i,t+1})\}}{\partial P_{t+k}(e)} \frac{\partial P_{t+k}(e)}{\partial Y_{t+1}}\, de\right) \cdot \frac{dY_{t+1}}{dR_t^b}}_{\equiv \psi_i\text{: persistent idio. rare productivity disaster}} \tag{D.60}$$

For the consumption channel, the monetary shock affects next-period consumption through cash-on-hand $m_{i,t+1} = R_t a_{i,t+1} + y_{i,t+1}$:

$$\int \frac{dU'(c_i(e'))}{dR_t^b} P(e')\, de' = E\{U'' \cdot MPC\} \cdot E\left\{\frac{dm_{i,t+1}}{dR_t^b}\right\} \tag{D.61}$$

where $MPC_{i,t+1}(e') = dc_{i,t+1}/dm_{i,t+1}$. Expanding the cash-on-hand response using the budget constraint $a_{i,t+1} = y_{i,t} + R_{t-1}a_{i,t} - c_{i,t}$ and $R_t = R_t^b$ at the DSS:

$$E\left\{\frac{dm_{i,t+1}}{dR_t^b}\right\} = R_t^b \left(\frac{dy_{i,t}}{dR_t^b} + \theta_{i,t} a_{i,t} \frac{dR_t^s}{dR_t^b} - \frac{dc_{i,t}}{dR_t^b}\right) + a_{i,t+1} + E\left\{\frac{dy_{i,t+1}}{dR_t^b}\right\}. \tag{D.62}$$

Step 4: Sufficient statistics. Following Zhang (2026b), we first establish that $dm_{i,t+1}/da_{i,t+1} = R_t$ at the DSS. The relationship between the Euler equation perturbation with respect to savings and the marginal propensities gives:

$$\frac{\beta R_t^b}{U''(c_{i,t})} \cdot R_t^b \cdot E\{U'' \cdot MPC\} = \frac{MPC_{i,t}}{MPS_{i,t}} \tag{D.63}$$

Step 5: Substitute and collect $dc_{i,t}/dR_t^b$ terms. Dividing equation (D.59) by $U''(c_{i,t})$ and using CRRA ($U'/U'' = -c/\gamma$):

$$\frac{dc_{i,t}}{dR_t^b} = -\frac{c_{i,t}}{\gamma R_t^b} + \frac{\beta R_t^b}{U''(c_{i,t})} \left[E\{U'' \cdot MPC\} \cdot E\left\{\frac{dm_{i,t+1}}{dR_t^b}\right\} + \psi_i\right] \tag{D.64}$$

Substituting the cash-on-hand expansion from Step 3 and using the sufficient statistic from Step 4:

$$\frac{dc_{i,t}}{dR_t^b} = -\frac{c_{i,t}}{\gamma R_t^b} - \frac{MPC_{i,t}}{MPS_{i,t}} \frac{dc_{i,t}}{dR_t^b} + \frac{MPC_{i,t}}{MPS_{i,t}} \left(\frac{dy_{i,t}}{dR_t^b} + \theta_{i,t} a_{i,t} \frac{dR_t^s}{dR_t^b}\right) \tag{D.65}$$

$$+ \frac{MPC_{i,t}}{MPS_{i,t}} \frac{1}{R_t^b} \left(a_{i,t+1} + E\left\{\frac{dy_{i,t+1}}{dR_t^b}\right\}\right) + \frac{\beta R_t^b}{U''(c_{i,t})} \psi_i \tag{D.66}$$

Step 6: Solve for $dc_{i,t}/dR_t^b$. Collecting the $dc_{i,t}/dR_t^b$ terms on the left-hand side and using $1 + MPC/MPS = 1/MPS$:

$$\frac{dc_{i,t}}{dR_t^b} = MPC_{i,t}\left[\theta_{i,t}a_{i,t}\frac{dR_t^s}{dR_t^b} + \frac{dy_{i,t}}{dR_t^b} + \frac{1}{R_t^b}\left(a_{i,t+1} + E\left\{\frac{dy_{i,t+1}}{dR_t^b}\right\}\right)\right] \tag{D.67}$$

$$- MPS_{i,t}\frac{c_{i,t}}{\gamma R_t^b} + MPS_{i,t}\frac{\beta R_t^b}{U''(c_{i,t})}\psi_i. \tag{D.68}$$

Under CRRA, $U''(c_{i,t}) = -\gamma c_{i,t}^{-\gamma-1}$, so $\frac{\beta R_t^b}{U''(c_{i,t})} = -\frac{\beta R_t^b}{\gamma}c_{i,t}^{\gamma+1}$. Therefore:

$$\frac{dc_{i,t}}{dR_t^b} = MPC_{i,t}\left[\theta_{i,t}a_{i,t}\frac{dR_t^s}{dR_t^b} + \frac{dy_{i,t}}{dR_t^b} + \frac{1}{R_t^b}\left(a_{i,t+1} + E\left\{\frac{dy_{i,t+1}}{dR_t^b}\right\}\right)\right] \tag{D.69}$$

$$- MPS_{i,t}\frac{c_{i,t}}{\gamma R_t^b} - MPS_{i,t}\frac{\beta R_t^b}{\gamma}c_{i,t}^{\gamma+1}\psi_i. \tag{D.70}$$

which is equation (27). □